



# ST7305

---

**Ultra-Low Power Active Matrix 408H x 408V  
TFT Display Driver with Controller**

---

## Datasheet

Sitronix reserves the right to change the contents in this document without prior notice, please contact Sitronix to obtain the latest version of datasheet before placing your order. No responsibility is assumed by Sitronix for any infringement of patent or other rights of third parties which may result from its use.

© 2022 Sitronix Technology Corporation. All rights reserved.

**Version 0.5**

Hazardous Substance Free

RoHS/ REACH Compliant

---

**LIST OF CONTENT**

|                                               |           |
|-----------------------------------------------|-----------|
| <b>1 INTRODUCTION .....</b>                   | <b>8</b>  |
| <b>2 FEATURES .....</b>                       | <b>9</b>  |
| <b>3 PAD ARRANGEMENT .....</b>                | <b>11</b> |
| 3.1 CONFIGURATION OF SIGNAL PADS .....        | 11        |
| 3.2 BUMP .....                                | 12        |
| 3.2.1 Output Pads .....                       | 12        |
| 3.2.2 Input Pads .....                        | 13        |
| 3.2.3 Alignment Marks .....                   | 13        |
| <b>4 PAD CENTER COORDINATES .....</b>         | <b>14</b> |
| <b>5 BLOCK DIAGRAM .....</b>                  | <b>29</b> |
| <b>6 PIN DESCRIPTION .....</b>                | <b>30</b> |
| 6.1 POWER SUPPLY PIN .....                    | 30        |
| 6.2 DIGITAL I/O .....                         | 30        |
| 6.3 CLOCK SYSTEM INPUT .....                  | 32        |
| 6.4 DRIVING OUTPUT PIN .....                  | 32        |
| 6.5 MTP PIN .....                             | 33        |
| 6.6 OTHERS .....                              | 33        |
| 6.7 RECOMMEND RESISTANCE .....                | 33        |
| <b>7 FUNCTION DESCRIPTION .....</b>           | <b>34</b> |
| 7.1 MICROPROCESSOR INTERFACE .....            | 34        |
| 7.1.1 Chip Select Input .....                 | 34        |
| 7.1.2 Interface Selection .....               | 34        |
| 7.1.3 8080 Parallel Interface .....           | 34        |
| 7.1.4 Single 4-Line Serial Interface .....    | 35        |
| 7.1.5 Dual 4-Line Serial Interface .....      | 36        |
| 7.1.6 3-Line Serial Interface .....           | 37        |
| 7.2 DATA COLOR CODING .....                   | 38        |
| 7.2.1 Data Input Mode .....                   | 38        |
| 7.2.2 Data to Display Mapping .....           | 39        |
| 7.2.3 Memory to Display Address Mapping ..... | 40        |
| 7.3 DISPLAY DATA RAM .....                    | 41        |
| 7.3.1 Configuration .....                     | 41        |
| 7.4 ADDRESS CONTROL .....                     | 42        |
| 7.5 TEARING EFFECT .....                      | 43        |
| 7.5.1 Tearing effect line modes .....         | 43        |

|          |                                                          |           |
|----------|----------------------------------------------------------|-----------|
| 7.5.2    | Tearing effect line timings.....                         | 44        |
| 7.5.3    | Example 1: MPU Write is faster than panel read.....      | 45        |
| 7.5.4    | Example 2: MPU write is slower than panel read .....     | 46        |
| 7.6      | OSCILLATION CIRCUIT .....                                | 46        |
| 7.7      | RESET .....                                              | 47        |
| 7.8      | DUAL TFT WAVEFORM .....                                  | 47        |
| 7.9      | POWER ON/OFF SEQUENCE .....                              | 48        |
| 7.10     | REFERENTIAL INITIAL SETUP FLOW .....                     | 49        |
| 7.10.1   | Initial Setting Flow .....                               | 49        |
| 7.10.2   | Referential Initial Code .....                           | 49        |
| 7.11     | SLEEP-IN/OUT SEQUENCE .....                              | 52        |
| 7.12     | POWER MODE (HPM/LPM) SEQUENCE .....                      | 52        |
| <b>8</b> | <b>COMMAND .....</b>                                     | <b>53</b> |
| 8.1      | COMMAND TABLE1 .....                                     | 53        |
| 8.1.1    | NOP (00h) .....                                          | 55        |
| 8.1.2    | SWRESET (01h): Software Reset.....                       | 55        |
| 8.1.3    | RDDID (04h): Read Display ID.....                        | 55        |
| 8.1.4    | RDDST (09h): Read Display Status .....                   | 56        |
| 8.1.5    | RDDPM (0Ah): Read Display Power Mode .....               | 57        |
| 8.1.6    | SLPIN (10h): Sleep in.....                               | 57        |
| 8.1.7    | SLPOUT (11h): Sleep Out.....                             | 58        |
| 8.1.8    | PTLON (12h): Partial On .....                            | 58        |
| 8.1.9    | PTLOFF (13h): Partial Off .....                          | 58        |
| 8.1.10   | INVOFF (20h): Display Inversion Off.....                 | 59        |
| 8.1.11   | INVON (21h): Display Inversion On.....                   | 59        |
| 8.1.12   | DISPOFF (28h): Display Off.....                          | 59        |
| 8.1.13   | DISPON (29h): Display On.....                            | 60        |
| 8.1.14   | CASET (2Ah): Column Address Set.....                     | 60        |
| 8.1.15   | RASET (2Bh): Row Address Set.....                        | 61        |
| 8.1.16   | RAMWR (2Ch): Memory Write .....                          | 62        |
| 8.1.17   | TEOFF (34h): Tearing Effect Line OFF.....                | 62        |
| 8.1.18   | TEON (35h): Tearing Effect Line On .....                 | 62        |
| 8.1.19   | MADCTL (36h): Memory Data Access Control.....            | 63        |
| 8.1.20   | VSCSAD (37h): Vertical Scroll Start Address of RAM ..... | 64        |
| 8.1.21   | HPM (38h): High Power Mode ON .....                      | 65        |
| 8.1.22   | LPM (39h): Low Power Mode ON.....                        | 65        |
| 8.1.23   | DTFORM (3Ah): Data Format Select.....                    | 65        |
| 8.1.24   | WRMEMC (3Ch): Write Memory Continue .....                | 66        |
| 8.1.25   | TESCAN (44h): Set Tear Scanline .....                    | 67        |

|        |                                                            |    |
|--------|------------------------------------------------------------|----|
| 8.1.26 | RDID1 (DAh): Read ID1 .....                                | 67 |
| 8.1.27 | RDID2 (DBh): Read ID2 .....                                | 67 |
| 8.1.28 | RDID3 (DCh): Read ID3 .....                                | 68 |
| 8.2    | COMMAND TABLE2 .....                                       | 69 |
| 8.2.1  | GTCON (62h): Gate Timing Control .....                     | 73 |
| 8.2.2  | GATESET (B0h): Gate Line Setting .....                     | 74 |
| 8.2.3  | FSTCOM (B1h): First Gate Setting .....                     | 74 |
| 8.2.4  | FRCTRL (B2h): Frame Rate Control .....                     | 75 |
| 8.2.5  | GTUPEQH (B3h): Update Period Gate EQ Control in HPM .....  | 76 |
| 8.2.6  | GTUPEQL (B4h): Update Period Gate EQ Control in LPM .....  | 78 |
| 8.2.7  | SOUEQ (B7h): Source EQ Enable .....                        | 79 |
| 8.2.8  | PNLSET (B8h): Panel Setting .....                          | 80 |
| 8.2.9  | CLRAM (BBh): Enable Clear RAM .....                        | 80 |
| 8.2.10 | GCTRL (C0h): Gate Voltage Control .....                    | 81 |
| 8.2.11 | VSHPCTRL (C1h): Source High Positive Voltage Control ..... | 82 |
| 8.2.12 | VSLPCTRL (C2h): Source Low Positive Voltage Control .....  | 83 |
| 8.2.13 | VSHNCTRL (C4h): Source High Negative Voltage Control ..... | 84 |
| 8.2.14 | VSLNCTRL (C5h): Source Low Negative Voltage Control .....  | 85 |
| 8.2.15 | VSHLSEL (C9h): Source Voltage Select .....                 | 86 |
| 8.2.16 | ID1SET (CAh): ID1 Setting .....                            | 86 |
| 8.2.17 | ID2SET (CBh): ID2 Setting .....                            | 86 |
| 8.2.18 | ID3SET (CCh): ID3 Setting .....                            | 87 |
| 8.2.19 | AUTOPWRCTRL (D0h): Enable Auto Power Down .....            | 87 |
| 8.2.20 | BSTEN (D1h): Booster Enable .....                          | 87 |
| 8.2.21 | NVMLOADCTRL (D6h): NVM Load Control .....                  | 88 |
| 8.2.22 | OSCSET (D8h): OSC Setting .....                            | 88 |
| 8.2.23 | NVMRD (E9h): NVM Data Read .....                           | 89 |
| 8.2.24 | EXTBCTRL (ECh): EXTB Control .....                         | 89 |
| 8.2.25 | NVMCTRL1 (F8h): NVM WR/RD Control .....                    | 89 |
| 8.2.26 | NVMCTRL2 (FAh): NVM Program Setting .....                  | 90 |
| 8.2.27 | NVMRDEN (FBh): NVM Read Enable .....                       | 90 |
| 8.2.28 | NVMPROGRAM (FCh): NVM Program Enable .....                 | 90 |
| 9      | NVM PROGRAMMING FLOW .....                                 | 91 |
| 10     | ABSOLUTE MAXIMUM RATING .....                              | 92 |
| 11     | DC CHARACTERISTICS .....                                   | 93 |
| 12     | AC CHARACTERISTICS .....                                   | 95 |
| 12.1   | INTERFACE TIMING .....                                     | 95 |
| 12.1.1 | System Bus Timing for Parallel 8080 MCU Interface .....    | 95 |

|           |                                               |            |
|-----------|-----------------------------------------------|------------|
| 12.1.2    | System Bus Timing for 4SPI MCU Interface..... | 96         |
| 12.1.3    | System Bus Timing for 3SPI MCU Interface..... | 97         |
| 12.1.4    | Reset Timing.....                             | 98         |
| <b>13</b> | <b>APPLICATION NOTE.....</b>                  | <b>100</b> |
| 13.1      | PARALLEL 8080 INTERFACE .....                 | 100        |
| 13.1.1    | VDDA=3.3V.....                                | 100        |
| 13.1.2    | VDDI=VDDA=1.8V .....                          | 101        |
| 13.2      | SINGLE 4-LINE SPI INTERFACE .....             | 102        |
| 13.2.1    | VDDA=3.3V.....                                | 102        |
| 13.2.2    | VDDI=VDDA=1.8V .....                          | 103        |
| 13.3      | DUAL 4-LINE SPI INTERFACE.....                | 104        |
| 13.3.1    | VDDA=3.3V.....                                | 104        |
| 13.3.2    | VDDI=VDDA=1.8V .....                          | 105        |
| 13.4      | 3-LINE SPI INTERFACE.....                     | 106        |
| 13.4.1    | VDDA=3.3V.....                                | 106        |
| 13.4.2    | VDDI=VDDA=1.8V .....                          | 107        |
| <b>14</b> | <b>PANEL LAYOUT .....</b>                     | <b>108</b> |
| 14.1      | TFT STRUCTURE.....                            | 108        |
| 14.2      | GATE LAYOUT .....                             | 108        |
| 14.2.1    | 1-line interlace.....                         | 108        |
| 14.2.2    | 2-line interlace.....                         | 109        |
| 14.2.3    | Non interlace .....                           | 109        |
| <b>15</b> | <b>REVISION HISTORY .....</b>                 | <b>110</b> |

**LIST OF FIGURES**

Figure 3-1 Chip..... 11

Figure 3-2 Output Pads Outline..... 12

Figure 3-3 Input Pads Outline..... 13

Figure 4 Alignment Marks Outline ..... 13

CONFIDENTIAL

**LIST OF TABLES**

Table 1 Parallel/Serial Interface Mode ..... 34

Table 2 Maximum Ratings..... 92

Table 3 Basic DC Characteristics ..... 93

CONFIDENTIAL

## **1 INTRODUCTION**

The ST7305 is a single-chip controller/driver for small and medium TFT LCD display that operate at very low frame frequency to conserve power. This chip is capable of supporting up to 408H x 408V pixels and provides a standard 8-bit 8080 parallel interface, 3-wire SPI serial interface and single/dual 4-wire SPI serial interface to configure the system and update the graphics content. The 408-channel source driver supports 1-bit per pixel (or sub-pixel) control to simplify the driver construction and lower power dissipation.

ST7305 is controlled using a conventional mobile driver SoC interface which supports an interface I/O voltage (VDDI) of 1.8V ~ 3.3V. ST7305 is equipped with all required charge pumps, buffer amplifiers and regulators to run directly from an unregulated coin cell battery source (Normal mode VDDA=2.8V ~ 3.3V; 1.8V mode VDDA=1.8V). All voltages used by the TFT channel drivers are close loop regulated by ST7305 and can be adjusted in fine increments by user controlled digital register settings to optimize power and display characteristics. The built-in timing controller (TCON) supports a wide range of output timing for various pixel organizations and driver waveform generation. The combination of timing and voltage controls allows the ST7305 driver SoC to have wide compatibility with various LCD and TFT types.

CONFIDENTIAL



## **2 FEATURES**

**Design for Low Power Active Matrix a-Si TFT Display**

**Single-chip TFT-LCD Controller/Driver with On-chip Frame Memory**

**Support for 408H x 408V Mono pixels**

**Outputs bi-level 1-bit per pixel source drive**

**Support multiple frame rate (1Hz ~ 32Hz)**

**Supports multiple LCD modes**

- ◆ ECB or MTN
- ◆ Normally black or white

**Built-in 408 x 408 x 1b internal SRAM**

**Dot/Column inversion DC drive of TFTs at low refresh rates**

**Built-in Sequential/Interlace Gate Scan Function**

**Microprocessor Interface**

- ◆ 80 parallel 8bits
- ◆ 3/ 4-line serial interface support write-operation and read- status

**Wide Power Supply Range**

- ◆ Normal Mode  
VDDI(Digital) : 1.8V ~ 3.3V  
VDDA(Analog): 2.8V ~ 3.3V
- ◆ 1.8V Mode  
VDDI(Digital)=VDDA(Analog)=1.8V

**On-chip Build-In Circuits**

- ◆ DC/DC Converter
- ◆ Non-Volatile (NV) Memory to Store Initial Register Setting and Factory Default Value (Module ID, Module Version, etc)
- ◆ Internal Oscillator for Display Clock Generation
- ◆ Timing Controller

**On-chip power management system**

**Built in charge pump circuits**

- ◆ 2x boost generates: 6.2V (Normal Mode)
- ◆ High voltage +16.5V/-15V charge pump

**Built in low power analog circuit**

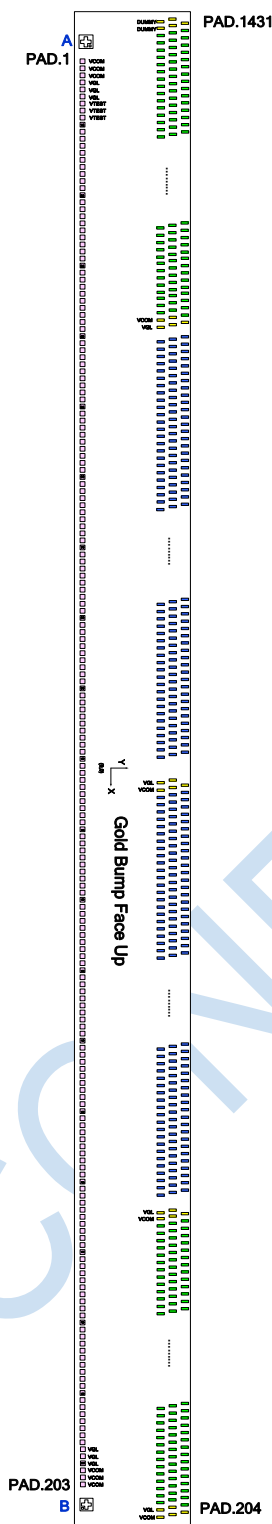
- ◆ On-chip oscillator circuit.
- ◆ Built-in voltage regulator with programmable contrast
- ◆ On-chip power system
  - VCOM level : GND
  - Source Voltage (VSHP/VSHN/VSLP/VSLN ): +6.0V ~ -5.0V

- Gate Driver HIGH Level (VGH to AGND): +8V ~ +16.5V
- Gate Driver LOW Level (VGL to AGND): -6V ~ -15V
- Max. VGH – VGL: 31.5V
- ◆ Built-in NV memory for VSHP/VSLP/VSHN/VSLN amplitude adjustment and ID setting.
- ◆ COG packaging
- ◆ Wide operating temperature range: -30°C ~ +85°C

CONFIDENTIAL

3 PAD ARRANGEMENT

3.1 Configuration of Signal Pads



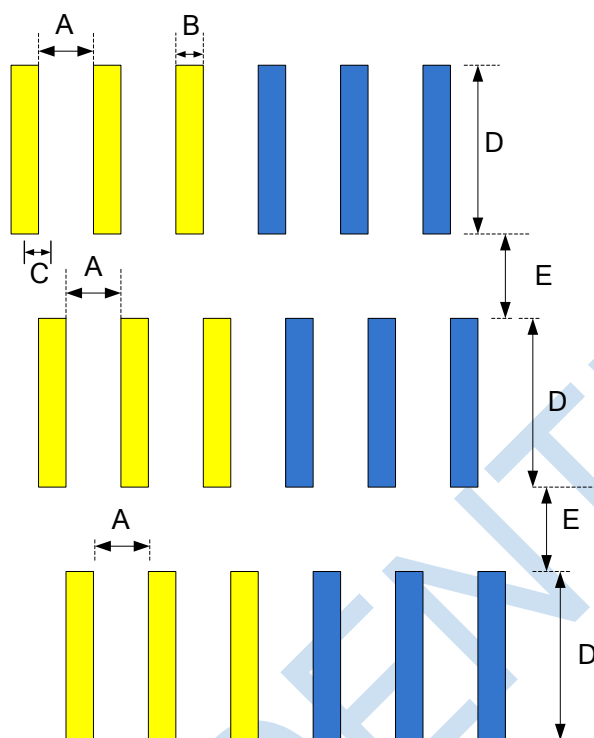
| ST7305         |               |
|----------------|---------------|
| Chip Thickness | 200           |
| Bump Height    | 12            |
| Output Pad     | 13(X) x 95(Y) |
| I/O Pad        | 40(X) x 50(Y) |

Unit: um

Figure 3-1 Chip

## 3.2 Bump

### 3.2.1 Output Pads

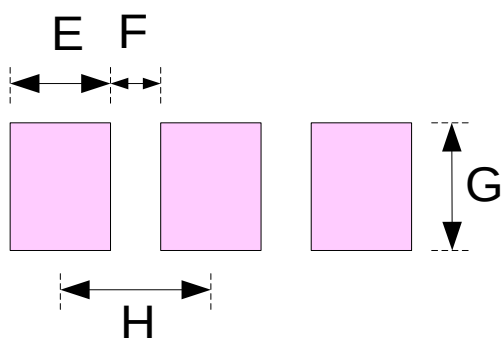


PAD.204~1431

Figure 3-2 Output Pads Outline

| Symbol | Item         | Size (um) |
|--------|--------------|-----------|
| A      | Bump Gap (H) | 17        |
| B      | Bump Width   | 13        |
| C      | Bump Pitch   | 10        |
| D      | Bump Height  | 95        |
| E      | Bump Gap (V) | 20        |

## 3.2.2 Input Pads



| Symbol | Item        | Size (um) |
|--------|-------------|-----------|
| E      | Bump Width  | 40        |
| F      | Bump Gap    | 20        |
| G      | Bump Height | 50        |
| H      | Bump Pitch  | 60        |

Figure 3-3 Input Pads Outline

## 3.2.3 Alignment Marks

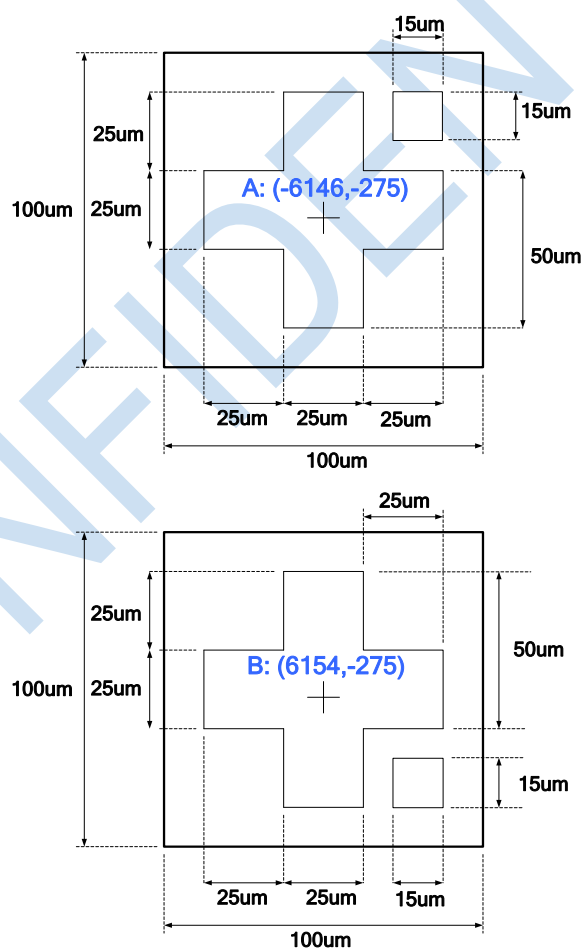


Figure 4 Alignment Marks Outline

## 4 PAD CENTER COORDINATES

| PAD. | Name  | X     | Y       |
|------|-------|-------|---------|
| 1    | VCOM  | -6056 | -292.62 |
| 2    | VCOM  | -5996 | -292.62 |
| 3    | VCOM  | -5936 | -292.62 |
| 4    | VGL   | -5876 | -292.62 |
| 5    | VGL   | -5816 | -292.62 |
| 6    | VGL   | -5756 | -292.62 |
| 7    | VTEST | -5696 | -292.62 |
| 8    | VTEST | -5636 | -292.62 |
| 9    | VTEST | -5576 | -292.62 |
| 10   | VTEST | -5516 | -292.62 |
| 11   | VTEST | -5456 | -292.62 |
| 12   | VTEST | -5396 | -292.62 |
| 13   | VTEST | -5336 | -292.62 |
| 14   | VTEST | -5276 | -292.62 |
| 15   | VCOM  | -5216 | -292.62 |
| 16   | VCOM  | -5156 | -292.62 |
| 17   | VCOM  | -5096 | -292.62 |
| 18   | VCOM  | -5036 | -292.62 |
| 19   | VCOM  | -4976 | -292.62 |
| 20   | VCOM  | -4916 | -292.62 |
| 21   | VDDA  | -4856 | -292.62 |
| 22   | VDDA  | -4796 | -292.62 |
| 23   | VDDA  | -4736 | -292.62 |
| 24   | VAGND | -4676 | -292.62 |
| 25   | VAGND | -4616 | -292.62 |
| 26   | VAGND | -4556 | -292.62 |
| 27   | CA1N  | -4496 | -292.62 |
| 28   | CA1N  | -4436 | -292.62 |
| 29   | CA1N  | -4376 | -292.62 |
| 30   | CA1N  | -4316 | -292.62 |
| 31   | CA1P  | -4256 | -292.62 |
| 32   | CA1P  | -4196 | -292.62 |
| 33   | CA1P  | -4136 | -292.62 |

| PAD. | Name | X     | Y       |
|------|------|-------|---------|
| 34   | CA1P | -4076 | -292.62 |
| 35   | CA1P | -4016 | -292.62 |
| 36   | CA1P | -3956 | -292.62 |
| 37   | VPV  | -3896 | -292.62 |
| 38   | VPV  | -3836 | -292.62 |
| 39   | VPV  | -3776 | -292.62 |
| 40   | VPV  | -3716 | -292.62 |
| 41   | VPV  | -3656 | -292.62 |
| 42   | VPV  | -3596 | -292.62 |
| 43   | CA2P | -3536 | -292.62 |
| 44   | CA2P | -3476 | -292.62 |
| 45   | CA2P | -3416 | -292.62 |
| 46   | CA2P | -3356 | -292.62 |
| 47   | CA2P | -3296 | -292.62 |
| 48   | CA2P | -3236 | -292.62 |
| 49   | CA2N | -3176 | -292.62 |
| 50   | CA2N | -3116 | -292.62 |
| 51   | CA2N | -3056 | -292.62 |
| 52   | CA2N | -2996 | -292.62 |
| 53   | CA2N | -2936 | -292.62 |
| 54   | CA2N | -2876 | -292.62 |
| 55   | VNV  | -2816 | -292.62 |
| 56   | VNV  | -2756 | -292.62 |
| 57   | VNV  | -2696 | -292.62 |
| 58   | VNV  | -2636 | -292.62 |
| 59   | VNV  | -2576 | -292.62 |
| 60   | VNV  | -2516 | -292.62 |
| 61   | CF1N | -2456 | -292.62 |
| 62   | CF1N | -2396 | -292.62 |
| 63   | CF1N | -2336 | -292.62 |
| 64   | CF1N | -2276 | -292.62 |
| 65   | CF1P | -2216 | -292.62 |
| 66   | CF1P | -2156 | -292.62 |

| PAD. | Name  | X     | Y       |
|------|-------|-------|---------|
| 67   | CF1P  | -2096 | -292.62 |
| 68   | CF1P  | -2036 | -292.62 |
| 69   | AVDD  | -1976 | -292.62 |
| 70   | AVDD  | -1916 | -292.62 |
| 71   | AVDD  | -1856 | -292.62 |
| 72   | AVDD  | -1796 | -292.62 |
| 73   | AVDD  | -1736 | -292.62 |
| 74   | AVDD  | -1676 | -292.62 |
| 75   | CB2P  | -1616 | -292.62 |
| 76   | CB2P  | -1556 | -292.62 |
| 77   | CB2P  | -1496 | -292.62 |
| 78   | CB2P  | -1436 | -292.62 |
| 79   | CB2N  | -1376 | -292.62 |
| 80   | CB2N  | -1316 | -292.62 |
| 81   | CB2N  | -1256 | -292.62 |
| 82   | CB2N  | -1196 | -292.62 |
| 83   | CB1P  | -1136 | -292.62 |
| 84   | CB1P  | -1076 | -292.62 |
| 85   | CB1N  | -1016 | -292.62 |
| 86   | CB1N  | -956  | -292.62 |
| 87   | CB1N  | -896  | -292.62 |
| 88   | CB1N  | -836  | -292.62 |
| 89   | V1V   | -776  | -292.62 |
| 90   | V1V   | -716  | -292.62 |
| 91   | CE1P  | -656  | -292.62 |
| 92   | CE1P  | -596  | -292.62 |
| 93   | CE1N  | -536  | -292.62 |
| 94   | CE1N  | -476  | -292.62 |
| 95   | NAVDD | -416  | -292.62 |
| 96   | NAVDD | -356  | -292.62 |
| 97   | CE2N  | -296  | -292.62 |
| 98   | CE2N  | -236  | -292.62 |
| 99   | CE2P  | -176  | -292.62 |

| PAD. | Name  | X    | Y       |
|------|-------|------|---------|
| 100  | CE2P  | -116 | -292.62 |
| 101  | CC1N  | -56  | -292.62 |
| 102  | CC1N  | 4    | -292.62 |
| 103  | CC1P  | 64   | -292.62 |
| 104  | CC1P  | 124  | -292.62 |
| 105  | VGH   | 184  | -292.62 |
| 106  | VGH   | 244  | -292.62 |
| 107  | CD1P  | 304  | -292.62 |
| 108  | CD1P  | 364  | -292.62 |
| 109  | CD1N  | 424  | -292.62 |
| 110  | CD1N  | 484  | -292.62 |
| 111  | VGL   | 544  | -292.62 |
| 112  | VGL   | 604  | -292.62 |
| 113  | VCCO  | 664  | -292.62 |
| 114  | VCCO  | 724  | -292.62 |
| 115  | VCCI  | 784  | -292.62 |
| 116  | VCCI  | 844  | -292.62 |
| 117  | VDDA  | 904  | -292.62 |
| 118  | VDDA  | 964  | -292.62 |
| 119  | VDDA  | 1024 | -292.62 |
| 120  | VDDA  | 1084 | -292.62 |
| 121  | VDDA  | 1144 | -292.62 |
| 122  | VDDA  | 1204 | -292.62 |
| 123  | VDDR  | 1264 | -292.62 |
| 124  | VDDR  | 1324 | -292.62 |
| 125  | VDDI  | 1384 | -292.62 |
| 126  | VDDI  | 1444 | -292.62 |
| 127  | VDDI  | 1504 | -292.62 |
| 128  | VDDI  | 1564 | -292.62 |
| 129  | VDGND | 1624 | -292.62 |
| 130  | VDGND | 1684 | -292.62 |
| 131  | VDGND | 1744 | -292.62 |
| 132  | VDGND | 1804 | -292.62 |
| 133  | VAGND | 1864 | -292.62 |

| PAD. | Name    | X    | Y       |
|------|---------|------|---------|
| 134  | VAGND   | 1924 | -292.62 |
| 135  | VAGND   | 1984 | -292.62 |
| 136  | VAGND   | 2044 | -292.62 |
| 137  | VAGND   | 2104 | -292.62 |
| 138  | VAGND   | 2164 | -292.62 |
| 139  | VRGND   | 2224 | -292.62 |
| 140  | VRGND   | 2284 | -292.62 |
| 141  | VDDI    | 2344 | -292.62 |
| 142  | CSB     | 2404 | -292.62 |
| 143  | A0      | 2464 | -292.62 |
| 144  | ERD     | 2524 | -292.62 |
| 145  | RWR     | 2584 | -292.62 |
| 146  | TESTENB | 2644 | -292.62 |
| 147  | MTest5  | 2704 | -292.62 |
| 148  | MTest4  | 2764 | -292.62 |
| 149  | MTest3  | 2824 | -292.62 |
| 150  | RSTB    | 2884 | -292.62 |
| 151  | VDGND   | 2944 | -292.62 |
| 152  | D0      | 3004 | -292.62 |
| 153  | D1      | 3064 | -292.62 |
| 154  | D2      | 3124 | -292.62 |
| 155  | D3      | 3184 | -292.62 |
| 156  | D4      | 3244 | -292.62 |
| 157  | D5      | 3304 | -292.62 |
| 158  | D6      | 3364 | -292.62 |
| 159  | D7      | 3424 | -292.62 |
| 160  | VDDI    | 3484 | -292.62 |
| 161  | ModeSel | 3544 | -292.62 |
| 162  | IF1     | 3604 | -292.62 |
| 163  | IF0     | 3664 | -292.62 |
| 164  | CLS     | 3724 | -292.62 |
| 165  | CL      | 3784 | -292.62 |
| 166  | TE      | 3844 | -292.62 |
| 167  | EXTB    | 3904 | -292.62 |

| PAD. | Name   | X    | Y       |
|------|--------|------|---------|
| 168  | VPP    | 3964 | -292.62 |
| 169  | VPP    | 4024 | -292.62 |
| 170  | VPP    | 4084 | -292.62 |
| 171  | VPP    | 4144 | -292.62 |
| 172  | VPP    | 4204 | -292.62 |
| 173  | VPP    | 4264 | -292.62 |
| 174  | VDGND  | 4324 | -292.62 |
| 175  | TESTB  | 4384 | -292.62 |
| 176  | TESTA  | 4444 | -292.62 |
| 177  | MTest2 | 4504 | -292.62 |
| 178  | MTest1 | 4564 | -292.62 |
| 179  | MTest0 | 4624 | -292.62 |
| 180  | VDDI   | 4684 | -292.62 |
| 181  | VCOM   | 4744 | -292.62 |
| 182  | VCOM   | 4804 | -292.62 |
| 183  | VCOM   | 4864 | -292.62 |
| 184  | VCOM   | 4924 | -292.62 |
| 185  | VCOM   | 4984 | -292.62 |
| 186  | VCOM   | 5044 | -292.62 |
| 187  | DUMMY  | 5104 | -292.62 |
| 188  | DUMMY  | 5164 | -292.62 |
| 189  | DUMMY  | 5224 | -292.62 |
| 190  | DUMMY  | 5284 | -292.62 |
| 191  | DUMMY  | 5344 | -292.62 |
| 192  | DUMMY  | 5404 | -292.62 |
| 193  | DUMMY  | 5464 | -292.62 |
| 194  | DUMMY  | 5524 | -292.62 |
| 195  | DUMMY  | 5584 | -292.62 |
| 196  | DUMMY  | 5644 | -292.62 |
| 197  | DUMMY  | 5704 | -292.62 |
| 198  | VGL    | 5764 | -292.62 |
| 199  | VGL    | 5824 | -292.62 |
| 200  | VGL    | 5884 | -292.62 |
| 201  | VCOM   | 5944 | -292.62 |

| PAD. | Name  | X    | Y       |
|------|-------|------|---------|
| 202  | VCOM  | 6004 | -292.62 |
| 203  | VCOM  | 6064 | -292.62 |
| 204  | DUMMY | 6239 | 35.58   |
| 205  | DUMMY | 6229 | 150.58  |
| 206  | Gtest | 6219 | 265.58  |
| 207  | DUMMY | 6209 | 35.58   |
| 208  | Gtest | 6199 | 150.58  |
| 209  | G2    | 6189 | 265.58  |
| 210  | G4    | 6179 | 35.58   |
| 211  | G6    | 6169 | 150.58  |
| 212  | G8    | 6159 | 265.58  |
| 213  | G10   | 6149 | 35.58   |
| 214  | G12   | 6139 | 150.58  |
| 215  | G14   | 6129 | 265.58  |
| 216  | G16   | 6119 | 35.58   |
| 217  | G18   | 6109 | 150.58  |
| 218  | G20   | 6099 | 265.58  |
| 219  | G22   | 6089 | 35.58   |
| 220  | G24   | 6079 | 150.58  |
| 221  | G26   | 6069 | 265.58  |
| 222  | G28   | 6059 | 35.58   |
| 223  | G30   | 6049 | 150.58  |
| 224  | G32   | 6039 | 265.58  |
| 225  | G34   | 6029 | 35.58   |
| 226  | G36   | 6019 | 150.58  |
| 227  | G38   | 6009 | 265.58  |
| 228  | G40   | 5999 | 35.58   |
| 229  | G42   | 5989 | 150.58  |
| 230  | G44   | 5979 | 265.58  |
| 231  | G46   | 5969 | 35.58   |
| 232  | G48   | 5959 | 150.58  |
| 233  | G50   | 5949 | 265.58  |
| 234  | G52   | 5939 | 35.58   |
| 235  | G54   | 5929 | 150.58  |

| PAD. | Name | X    | Y      |
|------|------|------|--------|
| 236  | G56  | 5919 | 265.58 |
| 237  | G58  | 5909 | 35.58  |
| 238  | G60  | 5899 | 150.58 |
| 239  | G62  | 5889 | 265.58 |
| 240  | G64  | 5879 | 35.58  |
| 241  | G66  | 5869 | 150.58 |
| 242  | G68  | 5859 | 265.58 |
| 243  | G70  | 5849 | 35.58  |
| 244  | G72  | 5839 | 150.58 |
| 245  | G74  | 5829 | 265.58 |
| 246  | G76  | 5819 | 35.58  |
| 247  | G78  | 5809 | 150.58 |
| 248  | G80  | 5799 | 265.58 |
| 249  | G82  | 5789 | 35.58  |
| 250  | G84  | 5779 | 150.58 |
| 251  | G86  | 5769 | 265.58 |
| 252  | G88  | 5759 | 35.58  |
| 253  | G90  | 5749 | 150.58 |
| 254  | G92  | 5739 | 265.58 |
| 255  | G94  | 5729 | 35.58  |
| 256  | G96  | 5719 | 150.58 |
| 257  | G98  | 5709 | 265.58 |
| 258  | G100 | 5699 | 35.58  |
| 259  | G102 | 5689 | 150.58 |
| 260  | G104 | 5679 | 265.58 |
| 261  | G106 | 5669 | 35.58  |
| 262  | G108 | 5659 | 150.58 |
| 263  | G110 | 5649 | 265.58 |
| 264  | G112 | 5639 | 35.58  |
| 265  | G114 | 5629 | 150.58 |
| 266  | G116 | 5619 | 265.58 |
| 267  | G118 | 5609 | 35.58  |
| 268  | G120 | 5599 | 150.58 |
| 269  | G122 | 5589 | 265.58 |

| PAD. | Name | X    | Y      |
|------|------|------|--------|
| 270  | G124 | 5579 | 35.58  |
| 271  | G126 | 5569 | 150.58 |
| 272  | G128 | 5559 | 265.58 |
| 273  | G130 | 5549 | 35.58  |
| 274  | G132 | 5539 | 150.58 |
| 275  | G134 | 5529 | 265.58 |
| 276  | G136 | 5519 | 35.58  |
| 277  | G138 | 5509 | 150.58 |
| 278  | G140 | 5499 | 265.58 |
| 279  | G142 | 5489 | 35.58  |
| 280  | G144 | 5479 | 150.58 |
| 281  | G146 | 5469 | 265.58 |
| 282  | G148 | 5459 | 35.58  |
| 283  | G150 | 5449 | 150.58 |
| 284  | G152 | 5439 | 265.58 |
| 285  | G154 | 5429 | 35.58  |
| 286  | G156 | 5419 | 150.58 |
| 287  | G158 | 5409 | 265.58 |
| 288  | G160 | 5399 | 35.58  |
| 289  | G162 | 5389 | 150.58 |
| 290  | G164 | 5379 | 265.58 |
| 291  | G166 | 5369 | 35.58  |
| 292  | G168 | 5359 | 150.58 |
| 293  | G170 | 5349 | 265.58 |
| 294  | G172 | 5339 | 35.58  |
| 295  | G174 | 5329 | 150.58 |
| 296  | G176 | 5319 | 265.58 |
| 297  | G178 | 5309 | 35.58  |
| 298  | G180 | 5299 | 150.58 |
| 299  | G182 | 5289 | 265.58 |
| 300  | G184 | 5279 | 35.58  |
| 301  | G186 | 5269 | 150.58 |
| 302  | G188 | 5259 | 265.58 |
| 303  | G190 | 5249 | 35.58  |



| PAD. | Name | X    | Y      |
|------|------|------|--------|
| 304  | G192 | 5239 | 150.58 |
| 305  | G194 | 5229 | 265.58 |
| 306  | G196 | 5219 | 35.58  |
| 307  | G198 | 5209 | 150.58 |
| 308  | G200 | 5199 | 265.58 |
| 309  | G202 | 5189 | 35.58  |
| 310  | G204 | 5179 | 150.58 |
| 311  | G206 | 5169 | 265.58 |
| 312  | G208 | 5159 | 35.58  |
| 313  | G210 | 5149 | 150.58 |
| 314  | G212 | 5139 | 265.58 |
| 315  | G214 | 5129 | 35.58  |
| 316  | G216 | 5119 | 150.58 |
| 317  | G218 | 5109 | 265.58 |
| 318  | G220 | 5099 | 35.58  |
| 319  | G222 | 5089 | 150.58 |
| 320  | G224 | 5079 | 265.58 |
| 321  | G226 | 5069 | 35.58  |
| 322  | G228 | 5059 | 150.58 |
| 323  | G230 | 5049 | 265.58 |
| 324  | G232 | 5039 | 35.58  |
| 325  | G234 | 5029 | 150.58 |
| 326  | G236 | 5019 | 265.58 |
| 327  | G238 | 5009 | 35.58  |
| 328  | G240 | 4999 | 150.58 |
| 329  | G242 | 4989 | 265.58 |
| 330  | G244 | 4979 | 35.58  |
| 331  | G246 | 4969 | 150.58 |
| 332  | G248 | 4959 | 265.58 |
| 333  | G250 | 4949 | 35.58  |
| 334  | G252 | 4939 | 150.58 |
| 335  | G254 | 4929 | 265.58 |
| 336  | G256 | 4919 | 35.58  |
| 337  | G258 | 4909 | 150.58 |

| PAD. | Name | X    | Y      |
|------|------|------|--------|
| 338  | G260 | 4899 | 265.58 |
| 339  | G262 | 4889 | 35.58  |
| 340  | G264 | 4879 | 150.58 |
| 341  | G266 | 4869 | 265.58 |
| 342  | G268 | 4859 | 35.58  |
| 343  | G270 | 4849 | 150.58 |
| 344  | G272 | 4839 | 265.58 |
| 345  | G274 | 4829 | 35.58  |
| 346  | G276 | 4819 | 150.58 |
| 347  | G278 | 4809 | 265.58 |
| 348  | G280 | 4799 | 35.58  |
| 349  | G282 | 4789 | 150.58 |
| 350  | G284 | 4779 | 265.58 |
| 351  | G286 | 4769 | 35.58  |
| 352  | G288 | 4759 | 150.58 |
| 353  | G290 | 4749 | 265.58 |
| 354  | G292 | 4739 | 35.58  |
| 355  | G294 | 4729 | 150.58 |
| 356  | G296 | 4719 | 265.58 |
| 357  | G298 | 4709 | 35.58  |
| 358  | G300 | 4699 | 150.58 |
| 359  | G302 | 4689 | 265.58 |
| 360  | G304 | 4679 | 35.58  |
| 361  | G306 | 4669 | 150.58 |
| 362  | G308 | 4659 | 265.58 |
| 363  | G310 | 4649 | 35.58  |
| 364  | G312 | 4639 | 150.58 |
| 365  | G314 | 4629 | 265.58 |
| 366  | G316 | 4619 | 35.58  |
| 367  | G318 | 4609 | 150.58 |
| 368  | G320 | 4599 | 265.58 |
| 369  | G322 | 4589 | 35.58  |
| 370  | G324 | 4579 | 150.58 |
| 371  | G326 | 4569 | 265.58 |

| PAD. | Name | X    | Y      |
|------|------|------|--------|
| 372  | G328 | 4559 | 35.58  |
| 373  | G330 | 4549 | 150.58 |
| 374  | G332 | 4539 | 265.58 |
| 375  | G334 | 4529 | 35.58  |
| 376  | G336 | 4519 | 150.58 |
| 377  | G338 | 4509 | 265.58 |
| 378  | G340 | 4499 | 35.58  |
| 379  | G342 | 4489 | 150.58 |
| 380  | G344 | 4479 | 265.58 |
| 381  | G346 | 4469 | 35.58  |
| 382  | G348 | 4459 | 150.58 |
| 383  | G350 | 4449 | 265.58 |
| 384  | G352 | 4439 | 35.58  |
| 385  | G354 | 4429 | 150.58 |
| 386  | G356 | 4419 | 265.58 |
| 387  | G358 | 4409 | 35.58  |
| 388  | G360 | 4399 | 150.58 |
| 389  | G362 | 4389 | 265.58 |
| 390  | G364 | 4379 | 35.58  |
| 391  | G366 | 4369 | 150.58 |
| 392  | G368 | 4359 | 265.58 |
| 393  | G370 | 4349 | 35.58  |
| 394  | G372 | 4339 | 150.58 |
| 395  | G374 | 4329 | 265.58 |
| 396  | G376 | 4319 | 35.58  |
| 397  | G378 | 4309 | 150.58 |
| 398  | G380 | 4299 | 265.58 |
| 399  | G382 | 4289 | 35.58  |
| 400  | G384 | 4279 | 150.58 |
| 401  | G386 | 4269 | 265.58 |
| 402  | G388 | 4259 | 35.58  |
| 403  | G390 | 4249 | 150.58 |
| 404  | G392 | 4239 | 265.58 |
| 405  | G394 | 4229 | 35.58  |

| PAD. | Name     | X    | Y      |
|------|----------|------|--------|
| 406  | G396     | 4219 | 150.58 |
| 407  | G398     | 4209 | 265.58 |
| 408  | G400     | 4199 | 35.58  |
| 409  | G402     | 4189 | 150.58 |
| 410  | G404     | 4179 | 265.58 |
| 411  | G406     | 4169 | 35.58  |
| 412  | G408     | 4159 | 150.58 |
| 413  | G410     | 4149 | 265.58 |
| 414  | RESERVED | 4139 | 35.58  |
| 415  | RESERVED | 4129 | 150.58 |
| 416  | RESERVED | 4119 | 265.58 |
| 417  | RESERVED | 4109 | 35.58  |
| 418  | RESERVED | 4099 | 150.58 |
| 419  | RESERVED | 4089 | 265.58 |
| 420  | RESERVED | 4079 | 35.58  |
| 421  | RESERVED | 4069 | 150.58 |
| 422  | RESERVED | 4059 | 265.58 |
| 423  | RESERVED | 4049 | 35.58  |
| 424  | RESERVED | 4039 | 150.58 |
| 425  | RESERVED | 4029 | 265.58 |
| 426  | RESERVED | 4019 | 35.58  |
| 427  | RESERVED | 4009 | 150.58 |
| 428  | RESERVED | 3999 | 265.58 |
| 429  | RESERVED | 3989 | 35.58  |
| 430  | RESERVED | 3979 | 150.58 |
| 431  | RESERVED | 3969 | 265.58 |
| 432  | RESERVED | 3959 | 35.58  |
| 433  | RESERVED | 3949 | 150.58 |
| 434  | RESERVED | 3939 | 265.58 |
| 435  | RESERVED | 3929 | 35.58  |
| 436  | RESERVED | 3919 | 150.58 |
| 437  | RESERVED | 3909 | 265.58 |
| 438  | RESERVED | 3899 | 35.58  |
| 439  | RESERVED | 3889 | 150.58 |

| PAD. | Name     | X    | Y      |
|------|----------|------|--------|
| 440  | RESERVED | 3879 | 265.58 |
| 441  | RESERVED | 3869 | 35.58  |
| 442  | RESERVED | 3859 | 150.58 |
| 443  | RESERVED | 3849 | 265.58 |
| 444  | RESERVED | 3839 | 35.58  |
| 445  | RESERVED | 3829 | 150.58 |
| 446  | RESERVED | 3819 | 265.58 |
| 447  | RESERVED | 3809 | 35.58  |
| 448  | RESERVED | 3799 | 150.58 |
| 449  | RESERVED | 3789 | 265.58 |
| 450  | DUMMY    | 3779 | 35.58  |
| 451  | DUMMY    | 3769 | 150.58 |
| 452  | DUMMY    | 3759 | 265.58 |
| 453  | VGL      | 3749 | 35.58  |
| 454  | DUMMY    | 3739 | 150.58 |
| 455  | DUMMY    | 3699 | 265.58 |
| 456  | RESERVED | 3689 | 35.58  |
| 457  | RESERVED | 3679 | 150.58 |
| 458  | RESERVED | 3669 | 265.58 |
| 459  | RESERVED | 3659 | 35.58  |
| 460  | RESERVED | 3649 | 150.58 |
| 461  | RESERVED | 3639 | 265.58 |
| 462  | RESERVED | 3629 | 35.58  |
| 463  | RESERVED | 3619 | 150.58 |
| 464  | RESERVED | 3609 | 265.58 |
| 465  | RESERVED | 3599 | 35.58  |
| 466  | RESERVED | 3589 | 150.58 |
| 467  | RESERVED | 3579 | 265.58 |
| 468  | RESERVED | 3569 | 35.58  |
| 469  | RESERVED | 3559 | 150.58 |
| 470  | RESERVED | 3549 | 265.58 |
| 471  | RESERVED | 3539 | 35.58  |
| 472  | RESERVED | 3529 | 150.58 |
| 473  | RESERVED | 3519 | 265.58 |

| PAD. | Name     | X    | Y      |
|------|----------|------|--------|
| 474  | RESERVED | 3509 | 35.58  |
| 475  | RESERVED | 3499 | 150.58 |
| 476  | RESERVED | 3489 | 265.58 |
| 477  | RESERVED | 3479 | 35.58  |
| 478  | RESERVED | 3469 | 150.58 |
| 479  | RESERVED | 3459 | 265.58 |
| 480  | RESERVED | 3449 | 35.58  |
| 481  | RESERVED | 3439 | 150.58 |
| 482  | RESERVED | 3429 | 265.58 |
| 483  | RESERVED | 3419 | 35.58  |
| 484  | RESERVED | 3409 | 150.58 |
| 485  | RESERVED | 3399 | 265.58 |
| 486  | RESERVED | 3389 | 35.58  |
| 487  | RESERVED | 3379 | 150.58 |
| 488  | RESERVED | 3369 | 265.58 |
| 489  | RESERVED | 3359 | 35.58  |
| 490  | RESERVED | 3349 | 150.58 |
| 491  | RESERVED | 3339 | 265.58 |
| 492  | RESERVED | 3329 | 35.58  |
| 493  | RESERVED | 3319 | 150.58 |
| 494  | RESERVED | 3309 | 265.58 |
| 495  | RESERVED | 3299 | 35.58  |
| 496  | RESERVED | 3289 | 150.58 |
| 497  | RESERVED | 3279 | 265.58 |
| 498  | RESERVED | 3269 | 35.58  |
| 499  | RESERVED | 3259 | 150.58 |
| 500  | RESERVED | 3249 | 265.58 |
| 501  | RESERVED | 3239 | 35.58  |
| 502  | RESERVED | 3229 | 150.58 |
| 503  | RESERVED | 3219 | 265.58 |
| 504  | RESERVED | 3209 | 35.58  |
| 505  | RESERVED | 3199 | 150.58 |
| 506  | RESERVED | 3189 | 265.58 |
| 507  | RESERVED | 3179 | 35.58  |

| PAD. | Name     | X    | Y      |
|------|----------|------|--------|
| 508  | RESERVED | 3169 | 150.58 |
| 509  | RESERVED | 3159 | 265.58 |
| 510  | RESERVED | 3149 | 35.58  |
| 511  | RESERVED | 3139 | 150.58 |
| 512  | RESERVED | 3129 | 265.58 |
| 513  | RESERVED | 3119 | 35.58  |
| 514  | RESERVED | 3109 | 150.58 |
| 515  | RESERVED | 3099 | 265.58 |
| 516  | RESERVED | 3089 | 35.58  |
| 517  | RESERVED | 3079 | 150.58 |
| 518  | RESERVED | 3069 | 265.58 |
| 519  | RESERVED | 3059 | 35.58  |
| 520  | RESERVED | 3049 | 150.58 |
| 521  | RESERVED | 3039 | 265.58 |
| 522  | RESERVED | 3029 | 35.58  |
| 523  | RESERVED | 3019 | 150.58 |
| 524  | RESERVED | 3009 | 265.58 |
| 525  | RESERVED | 2999 | 35.58  |
| 526  | RESERVED | 2989 | 150.58 |
| 527  | RESERVED | 2979 | 265.58 |
| 528  | RESERVED | 2969 | 35.58  |
| 529  | RESERVED | 2959 | 150.58 |
| 530  | RESERVED | 2949 | 265.58 |
| 531  | RESERVED | 2939 | 35.58  |
| 532  | RESERVED | 2929 | 150.58 |
| 533  | RESERVED | 2919 | 265.58 |
| 534  | RESERVED | 2909 | 35.58  |
| 535  | RESERVED | 2899 | 150.58 |
| 536  | RESERVED | 2889 | 265.58 |
| 537  | RESERVED | 2879 | 35.58  |
| 538  | RESERVED | 2869 | 150.58 |
| 539  | RESERVED | 2859 | 265.58 |
| 540  | RESERVED | 2849 | 35.58  |
| 541  | RESERVED | 2839 | 150.58 |

| PAD. | Name     | X    | Y      |
|------|----------|------|--------|
| 542  | RESERVED | 2829 | 265.58 |
| 543  | RESERVED | 2819 | 35.58  |
| 544  | RESERVED | 2809 | 150.58 |
| 545  | RESERVED | 2799 | 265.58 |
| 546  | RESERVED | 2789 | 35.58  |
| 547  | RESERVED | 2779 | 150.58 |
| 548  | RESERVED | 2769 | 265.58 |
| 549  | RESERVED | 2759 | 35.58  |
| 550  | RESERVED | 2749 | 150.58 |
| 551  | RESERVED | 2739 | 265.58 |
| 552  | RESERVED | 2729 | 35.58  |
| 553  | RESERVED | 2719 | 150.58 |
| 554  | RESERVED | 2709 | 265.58 |
| 555  | RESERVED | 2699 | 35.58  |
| 556  | RESERVED | 2689 | 150.58 |
| 557  | RESERVED | 2679 | 265.58 |
| 558  | RESERVED | 2669 | 35.58  |
| 559  | RESERVED | 2659 | 150.58 |
| 560  | RESERVED | 2649 | 265.58 |
| 561  | RESERVED | 2639 | 35.58  |
| 562  | RESERVED | 2629 | 150.58 |
| 563  | RESERVED | 2619 | 265.58 |
| 564  | RESERVED | 2609 | 35.58  |
| 565  | RESERVED | 2599 | 150.58 |
| 566  | RESERVED | 2589 | 265.58 |
| 567  | RESERVED | 2579 | 35.58  |
| 568  | RESERVED | 2569 | 150.58 |
| 569  | RESERVED | 2559 | 265.58 |
| 570  | RESERVED | 2549 | 35.58  |
| 571  | RESERVED | 2539 | 150.58 |
| 572  | RESERVED | 2529 | 265.58 |
| 573  | RESERVED | 2519 | 35.58  |
| 574  | RESERVED | 2509 | 150.58 |
| 575  | RESERVED | 2499 | 265.58 |

| PAD. | Name     | X    | Y      |
|------|----------|------|--------|
| 576  | RESERVED | 2489 | 35.58  |
| 577  | RESERVED | 2479 | 150.58 |
| 578  | RESERVED | 2469 | 265.58 |
| 579  | RESERVED | 2459 | 35.58  |
| 580  | RESERVED | 2449 | 150.58 |
| 581  | RESERVED | 2439 | 265.58 |
| 582  | RESERVED | 2429 | 35.58  |
| 583  | RESERVED | 2419 | 150.58 |
| 584  | RESERVED | 2409 | 265.58 |
| 585  | RESERVED | 2399 | 35.58  |
| 586  | RESERVED | 2389 | 150.58 |
| 587  | RESERVED | 2379 | 265.58 |
| 588  | RESERVED | 2369 | 35.58  |
| 589  | RESERVED | 2359 | 150.58 |
| 590  | RESERVED | 2349 | 265.58 |
| 591  | RESERVED | 2339 | 35.58  |
| 592  | RESERVED | 2329 | 150.58 |
| 593  | RESERVED | 2319 | 265.58 |
| 594  | RESERVED | 2309 | 35.58  |
| 595  | RESERVED | 2299 | 150.58 |
| 596  | RESERVED | 2289 | 265.58 |
| 597  | RESERVED | 2279 | 35.58  |
| 598  | RESERVED | 2269 | 150.58 |
| 599  | RESERVED | 2259 | 265.58 |
| 600  | RESERVED | 2249 | 35.58  |
| 601  | RESERVED | 2239 | 150.58 |
| 602  | RESERVED | 2229 | 265.58 |
| 603  | RESERVED | 2219 | 35.58  |
| 604  | RESERVED | 2209 | 150.58 |
| 605  | RESERVED | 2199 | 265.58 |
| 606  | RESERVED | 2189 | 35.58  |
| 607  | RESERVED | 2179 | 150.58 |
| 608  | RESERVED | 2169 | 265.58 |
| 609  | RESERVED | 2159 | 35.58  |

| PAD. | Name     | X    | Y      |
|------|----------|------|--------|
| 610  | RESERVED | 2149 | 150.58 |
| 611  | RESERVED | 2139 | 265.58 |
| 612  | S408     | 2129 | 35.58  |
| 613  | S407     | 2119 | 150.58 |
| 614  | S406     | 2109 | 265.58 |
| 615  | S405     | 2099 | 35.58  |
| 616  | S404     | 2089 | 150.58 |
| 617  | S403     | 2079 | 265.58 |
| 618  | S402     | 2069 | 35.58  |
| 619  | S401     | 2059 | 150.58 |
| 620  | S400     | 2049 | 265.58 |
| 621  | S399     | 2039 | 35.58  |
| 622  | S398     | 2029 | 150.58 |
| 623  | S397     | 2019 | 265.58 |
| 624  | S396     | 2009 | 35.58  |
| 625  | S395     | 1999 | 150.58 |
| 626  | S394     | 1989 | 265.58 |
| 627  | S393     | 1979 | 35.58  |
| 628  | S392     | 1969 | 150.58 |
| 629  | S391     | 1959 | 265.58 |
| 630  | S390     | 1949 | 35.58  |
| 631  | S389     | 1939 | 150.58 |
| 632  | S388     | 1929 | 265.58 |
| 633  | S387     | 1919 | 35.58  |
| 634  | S386     | 1909 | 150.58 |
| 635  | S385     | 1899 | 265.58 |
| 636  | S384     | 1889 | 35.58  |
| 637  | S383     | 1879 | 150.58 |
| 638  | S382     | 1869 | 265.58 |
| 639  | S381     | 1859 | 35.58  |
| 640  | S380     | 1849 | 150.58 |
| 641  | S379     | 1839 | 265.58 |
| 642  | S378     | 1829 | 35.58  |
| 643  | S377     | 1819 | 150.58 |

| PAD. | Name | X    | Y      |
|------|------|------|--------|
| 644  | S376 | 1809 | 265.58 |
| 645  | S375 | 1799 | 35.58  |
| 646  | S374 | 1789 | 150.58 |
| 647  | S373 | 1779 | 265.58 |
| 648  | S372 | 1769 | 35.58  |
| 649  | S371 | 1759 | 150.58 |
| 650  | S370 | 1749 | 265.58 |
| 651  | S369 | 1739 | 35.58  |
| 652  | S368 | 1729 | 150.58 |
| 653  | S367 | 1719 | 265.58 |
| 654  | S366 | 1709 | 35.58  |
| 655  | S365 | 1699 | 150.58 |
| 656  | S364 | 1689 | 265.58 |
| 657  | S363 | 1679 | 35.58  |
| 658  | S362 | 1669 | 150.58 |
| 659  | S361 | 1659 | 265.58 |
| 660  | S360 | 1649 | 35.58  |
| 661  | S359 | 1639 | 150.58 |
| 662  | S358 | 1629 | 265.58 |
| 663  | S357 | 1619 | 35.58  |
| 664  | S356 | 1609 | 150.58 |
| 665  | S355 | 1599 | 265.58 |
| 666  | S354 | 1589 | 35.58  |
| 667  | S353 | 1579 | 150.58 |
| 668  | S352 | 1569 | 265.58 |
| 669  | S351 | 1559 | 35.58  |
| 670  | S350 | 1549 | 150.58 |
| 671  | S349 | 1539 | 265.58 |
| 672  | S348 | 1529 | 35.58  |
| 673  | S347 | 1519 | 150.58 |
| 674  | S346 | 1509 | 265.58 |
| 675  | S345 | 1499 | 35.58  |
| 676  | S344 | 1489 | 150.58 |
| 677  | S343 | 1479 | 265.58 |

| PAD. | Name | X    | Y      |
|------|------|------|--------|
| 678  | S342 | 1469 | 35.58  |
| 679  | S341 | 1459 | 150.58 |
| 680  | S340 | 1449 | 265.58 |
| 681  | S339 | 1439 | 35.58  |
| 682  | S338 | 1429 | 150.58 |
| 683  | S337 | 1419 | 265.58 |
| 684  | S336 | 1409 | 35.58  |
| 685  | S335 | 1399 | 150.58 |
| 686  | S334 | 1389 | 265.58 |
| 687  | S333 | 1379 | 35.58  |
| 688  | S332 | 1369 | 150.58 |
| 689  | S331 | 1359 | 265.58 |
| 690  | S330 | 1349 | 35.58  |
| 691  | S329 | 1339 | 150.58 |
| 692  | S328 | 1329 | 265.58 |
| 693  | S327 | 1319 | 35.58  |
| 694  | S326 | 1309 | 150.58 |
| 695  | S325 | 1299 | 265.58 |
| 696  | S324 | 1289 | 35.58  |
| 697  | S323 | 1279 | 150.58 |
| 698  | S322 | 1269 | 265.58 |
| 699  | S321 | 1259 | 35.58  |
| 700  | S320 | 1249 | 150.58 |
| 701  | S319 | 1239 | 265.58 |
| 702  | S318 | 1229 | 35.58  |
| 703  | S317 | 1219 | 150.58 |
| 704  | S316 | 1209 | 265.58 |
| 705  | S315 | 1199 | 35.58  |
| 706  | S314 | 1189 | 150.58 |
| 707  | S313 | 1179 | 265.58 |
| 708  | S312 | 1169 | 35.58  |
| 709  | S311 | 1159 | 150.58 |
| 710  | S310 | 1149 | 265.58 |
| 711  | S309 | 1139 | 35.58  |

# ST7305

| PAD. | Name | X    | Y      |
|------|------|------|--------|
| 712  | S308 | 1129 | 150.58 |
| 713  | S307 | 1119 | 265.58 |
| 714  | S306 | 1109 | 35.58  |
| 715  | S305 | 1099 | 150.58 |
| 716  | S304 | 1089 | 265.58 |
| 717  | S303 | 1079 | 35.58  |
| 718  | S302 | 1069 | 150.58 |
| 719  | S301 | 1059 | 265.58 |
| 720  | S300 | 1049 | 35.58  |
| 721  | S299 | 1039 | 150.58 |
| 722  | S298 | 1029 | 265.58 |
| 723  | S297 | 1019 | 35.58  |
| 724  | S296 | 1009 | 150.58 |
| 725  | S295 | 999  | 265.58 |
| 726  | S294 | 989  | 35.58  |
| 727  | S293 | 979  | 150.58 |
| 728  | S292 | 969  | 265.58 |
| 729  | S291 | 959  | 35.58  |
| 730  | S290 | 949  | 150.58 |
| 731  | S289 | 939  | 265.58 |
| 732  | S288 | 929  | 35.58  |
| 733  | S287 | 919  | 150.58 |
| 734  | S286 | 909  | 265.58 |
| 735  | S285 | 899  | 35.58  |
| 736  | S284 | 889  | 150.58 |
| 737  | S283 | 879  | 265.58 |
| 738  | S282 | 869  | 35.58  |
| 739  | S281 | 859  | 150.58 |
| 740  | S280 | 849  | 265.58 |
| 741  | S279 | 839  | 35.58  |
| 742  | S278 | 829  | 150.58 |
| 743  | S277 | 819  | 265.58 |
| 744  | S276 | 809  | 35.58  |
| 745  | S275 | 799  | 150.58 |

| PAD. | Name | X   | Y      |
|------|------|-----|--------|
| 746  | S274 | 789 | 265.58 |
| 747  | S273 | 779 | 35.58  |
| 748  | S272 | 769 | 150.58 |
| 749  | S271 | 759 | 265.58 |
| 750  | S270 | 749 | 35.58  |
| 751  | S269 | 739 | 150.58 |
| 752  | S268 | 729 | 265.58 |
| 753  | S267 | 719 | 35.58  |
| 754  | S266 | 709 | 150.58 |
| 755  | S265 | 699 | 265.58 |
| 756  | S264 | 689 | 35.58  |
| 757  | S263 | 679 | 150.58 |
| 758  | S262 | 669 | 265.58 |
| 759  | S261 | 659 | 35.58  |
| 760  | S260 | 649 | 150.58 |
| 761  | S259 | 639 | 265.58 |
| 762  | S258 | 629 | 35.58  |
| 763  | S257 | 619 | 150.58 |
| 764  | S256 | 609 | 265.58 |
| 765  | S255 | 599 | 35.58  |
| 766  | S254 | 589 | 150.58 |
| 767  | S253 | 579 | 265.58 |
| 768  | S252 | 569 | 35.58  |
| 769  | S251 | 559 | 150.58 |
| 770  | S250 | 549 | 265.58 |
| 771  | S249 | 539 | 35.58  |
| 772  | S248 | 529 | 150.58 |
| 773  | S247 | 519 | 265.58 |
| 774  | S246 | 509 | 35.58  |
| 775  | S245 | 499 | 150.58 |
| 776  | S244 | 489 | 265.58 |
| 777  | S243 | 479 | 35.58  |
| 778  | S242 | 469 | 150.58 |
| 779  | S241 | 459 | 265.58 |

| PAD. | Name | X   | Y      |
|------|------|-----|--------|
| 780  | S240 | 449 | 35.58  |
| 781  | S239 | 439 | 150.58 |
| 782  | S238 | 429 | 265.58 |
| 783  | S237 | 419 | 35.58  |
| 784  | S236 | 409 | 150.58 |
| 785  | S235 | 399 | 265.58 |
| 786  | S234 | 389 | 35.58  |
| 787  | S233 | 379 | 150.58 |
| 788  | S232 | 369 | 265.58 |
| 789  | S231 | 359 | 35.58  |
| 790  | S230 | 349 | 150.58 |
| 791  | S229 | 339 | 265.58 |
| 792  | S228 | 329 | 35.58  |
| 793  | S227 | 319 | 150.58 |
| 794  | S226 | 309 | 265.58 |
| 795  | S225 | 299 | 35.58  |
| 796  | S224 | 289 | 150.58 |
| 797  | S223 | 279 | 265.58 |
| 798  | S222 | 269 | 35.58  |
| 799  | S221 | 259 | 150.58 |
| 800  | S220 | 249 | 265.58 |
| 801  | S219 | 239 | 35.58  |
| 802  | S218 | 229 | 150.58 |
| 803  | S217 | 219 | 265.58 |
| 804  | S216 | 209 | 35.58  |
| 805  | S215 | 199 | 150.58 |
| 806  | S214 | 189 | 265.58 |
| 807  | S213 | 179 | 35.58  |
| 808  | S212 | 169 | 150.58 |
| 809  | S211 | 159 | 265.58 |
| 810  | S210 | 149 | 35.58  |
| 811  | S209 | 139 | 150.58 |
| 812  | S208 | 129 | 265.58 |
| 813  | S207 | 119 | 35.58  |

| PAD. | Name  | X    | Y      |
|------|-------|------|--------|
| 814  | S206  | 109  | 150.58 |
| 815  | S205  | 99   | 265.58 |
| 816  | DUMMY | 89   | 35.58  |
| 817  | DUMMY | 79   | 150.58 |
| 818  | GTest | 69   | 265.58 |
| 819  | DUMMY | 59   | 35.58  |
| 820  | GTest | 49   | 150.58 |
| 821  | S204  | -99  | 35.58  |
| 822  | S203  | -109 | 150.58 |
| 823  | S202  | -119 | 265.58 |
| 824  | S201  | -129 | 35.58  |
| 825  | S200  | -139 | 150.58 |
| 826  | S199  | -149 | 265.58 |
| 827  | S198  | -159 | 35.58  |
| 828  | S197  | -169 | 150.58 |
| 829  | S196  | -179 | 265.58 |
| 830  | S195  | -189 | 35.58  |
| 831  | S194  | -199 | 150.58 |
| 832  | S193  | -209 | 265.58 |
| 833  | S192  | -219 | 35.58  |
| 834  | S191  | -229 | 150.58 |
| 835  | S190  | -239 | 265.58 |
| 836  | S189  | -249 | 35.58  |
| 837  | S188  | -259 | 150.58 |
| 838  | S187  | -269 | 265.58 |
| 839  | S186  | -279 | 35.58  |
| 840  | S185  | -289 | 150.58 |
| 841  | S184  | -299 | 265.58 |
| 842  | S183  | -309 | 35.58  |
| 843  | S182  | -319 | 150.58 |
| 844  | S181  | -329 | 265.58 |
| 845  | S180  | -339 | 35.58  |
| 846  | S179  | -349 | 150.58 |
| 847  | S178  | -359 | 265.58 |

| PAD. | Name | X    | Y      |
|------|------|------|--------|
| 848  | S177 | -369 | 35.58  |
| 849  | S176 | -379 | 150.58 |
| 850  | S175 | -389 | 265.58 |
| 851  | S174 | -399 | 35.58  |
| 852  | S173 | -409 | 150.58 |
| 853  | S172 | -419 | 265.58 |
| 854  | S171 | -429 | 35.58  |
| 855  | S170 | -439 | 150.58 |
| 856  | S169 | -449 | 265.58 |
| 857  | S168 | -459 | 35.58  |
| 858  | S167 | -469 | 150.58 |
| 859  | S166 | -479 | 265.58 |
| 860  | S165 | -489 | 35.58  |
| 861  | S164 | -499 | 150.58 |
| 862  | S163 | -509 | 265.58 |
| 863  | S162 | -519 | 35.58  |
| 864  | S161 | -529 | 150.58 |
| 865  | S160 | -539 | 265.58 |
| 866  | S159 | -549 | 35.58  |
| 867  | S158 | -559 | 150.58 |
| 868  | S157 | -569 | 265.58 |
| 869  | S156 | -579 | 35.58  |
| 870  | S155 | -589 | 150.58 |
| 871  | S154 | -599 | 265.58 |
| 872  | S153 | -609 | 35.58  |
| 873  | S152 | -619 | 150.58 |
| 874  | S151 | -629 | 265.58 |
| 875  | S150 | -639 | 35.58  |
| 876  | S149 | -649 | 150.58 |
| 877  | S148 | -659 | 265.58 |
| 878  | S147 | -669 | 35.58  |
| 879  | S146 | -679 | 150.58 |
| 880  | S145 | -689 | 265.58 |
| 881  | S144 | -699 | 35.58  |

| PAD. | Name | X     | Y      |
|------|------|-------|--------|
| 882  | S143 | -709  | 150.58 |
| 883  | S142 | -719  | 265.58 |
| 884  | S141 | -729  | 35.58  |
| 885  | S140 | -739  | 150.58 |
| 886  | S139 | -749  | 265.58 |
| 887  | S138 | -759  | 35.58  |
| 888  | S137 | -769  | 150.58 |
| 889  | S136 | -779  | 265.58 |
| 890  | S135 | -789  | 35.58  |
| 891  | S134 | -799  | 150.58 |
| 892  | S133 | -809  | 265.58 |
| 893  | S132 | -819  | 35.58  |
| 894  | S131 | -829  | 150.58 |
| 895  | S130 | -839  | 265.58 |
| 896  | S129 | -849  | 35.58  |
| 897  | S128 | -859  | 150.58 |
| 898  | S127 | -869  | 265.58 |
| 899  | S126 | -879  | 35.58  |
| 900  | S125 | -889  | 150.58 |
| 901  | S124 | -899  | 265.58 |
| 902  | S123 | -909  | 35.58  |
| 903  | S122 | -919  | 150.58 |
| 904  | S121 | -929  | 265.58 |
| 905  | S120 | -939  | 35.58  |
| 906  | S119 | -949  | 150.58 |
| 907  | S118 | -959  | 265.58 |
| 908  | S117 | -969  | 35.58  |
| 909  | S116 | -979  | 150.58 |
| 910  | S115 | -989  | 265.58 |
| 911  | S114 | -999  | 35.58  |
| 912  | S113 | -1009 | 150.58 |
| 913  | S112 | -1019 | 265.58 |
| 914  | S111 | -1029 | 35.58  |
| 915  | S110 | -1039 | 150.58 |



# ST7305

| PAD. | Name | X     | Y      |
|------|------|-------|--------|
| 916  | S109 | -1049 | 265.58 |
| 917  | S108 | -1059 | 35.58  |
| 918  | S107 | -1069 | 150.58 |
| 919  | S106 | -1079 | 265.58 |
| 920  | S105 | -1089 | 35.58  |
| 921  | S104 | -1099 | 150.58 |
| 922  | S103 | -1109 | 265.58 |
| 923  | S102 | -1119 | 35.58  |
| 924  | S101 | -1129 | 150.58 |
| 925  | S100 | -1139 | 265.58 |
| 926  | S99  | -1149 | 35.58  |
| 927  | S98  | -1159 | 150.58 |
| 928  | S97  | -1169 | 265.58 |
| 929  | S96  | -1179 | 35.58  |
| 930  | S95  | -1189 | 150.58 |
| 931  | S94  | -1199 | 265.58 |
| 932  | S93  | -1209 | 35.58  |
| 933  | S92  | -1219 | 150.58 |
| 934  | S91  | -1229 | 265.58 |
| 935  | S90  | -1239 | 35.58  |
| 936  | S89  | -1249 | 150.58 |
| 937  | S88  | -1259 | 265.58 |
| 938  | S87  | -1269 | 35.58  |
| 939  | S86  | -1279 | 150.58 |
| 940  | S85  | -1289 | 265.58 |
| 941  | S84  | -1299 | 35.58  |
| 942  | S83  | -1309 | 150.58 |
| 943  | S82  | -1319 | 265.58 |
| 944  | S81  | -1329 | 35.58  |
| 945  | S80  | -1339 | 150.58 |
| 946  | S79  | -1349 | 265.58 |
| 947  | S78  | -1359 | 35.58  |
| 948  | S77  | -1369 | 150.58 |
| 949  | S76  | -1379 | 265.58 |

| PAD. | Name | X     | Y      |
|------|------|-------|--------|
| 950  | S75  | -1389 | 35.58  |
| 951  | S74  | -1399 | 150.58 |
| 952  | S73  | -1409 | 265.58 |
| 953  | S72  | -1419 | 35.58  |
| 954  | S71  | -1429 | 150.58 |
| 955  | S70  | -1439 | 265.58 |
| 956  | S69  | -1449 | 35.58  |
| 957  | S68  | -1459 | 150.58 |
| 958  | S67  | -1469 | 265.58 |
| 959  | S66  | -1479 | 35.58  |
| 960  | S65  | -1489 | 150.58 |
| 961  | S64  | -1499 | 265.58 |
| 962  | S63  | -1509 | 35.58  |
| 963  | S62  | -1519 | 150.58 |
| 964  | S61  | -1529 | 265.58 |
| 965  | S60  | -1539 | 35.58  |
| 966  | S59  | -1549 | 150.58 |
| 967  | S58  | -1559 | 265.58 |
| 968  | S57  | -1569 | 35.58  |
| 969  | S56  | -1579 | 150.58 |
| 970  | S55  | -1589 | 265.58 |
| 971  | S54  | -1599 | 35.58  |
| 972  | S53  | -1609 | 150.58 |
| 973  | S52  | -1619 | 265.58 |
| 974  | S51  | -1629 | 35.58  |
| 975  | S50  | -1639 | 150.58 |
| 976  | S49  | -1649 | 265.58 |
| 977  | S48  | -1659 | 35.58  |
| 978  | S47  | -1669 | 150.58 |
| 979  | S46  | -1679 | 265.58 |
| 980  | S45  | -1689 | 35.58  |
| 981  | S44  | -1699 | 150.58 |
| 982  | S43  | -1709 | 265.58 |
| 983  | S42  | -1719 | 35.58  |

| PAD. | Name | X     | Y      |
|------|------|-------|--------|
| 984  | S41  | -1729 | 150.58 |
| 985  | S40  | -1739 | 265.58 |
| 986  | S39  | -1749 | 35.58  |
| 987  | S38  | -1759 | 150.58 |
| 988  | S37  | -1769 | 265.58 |
| 989  | S36  | -1779 | 35.58  |
| 990  | S35  | -1789 | 150.58 |
| 991  | S34  | -1799 | 265.58 |
| 992  | S33  | -1809 | 35.58  |
| 993  | S32  | -1819 | 150.58 |
| 994  | S31  | -1829 | 265.58 |
| 995  | S30  | -1839 | 35.58  |
| 996  | S29  | -1849 | 150.58 |
| 997  | S28  | -1859 | 265.58 |
| 998  | S27  | -1869 | 35.58  |
| 999  | S26  | -1879 | 150.58 |
| 1000 | S25  | -1889 | 265.58 |
| 1001 | S24  | -1899 | 35.58  |
| 1002 | S23  | -1909 | 150.58 |
| 1003 | S22  | -1919 | 265.58 |
| 1004 | S21  | -1929 | 35.58  |
| 1005 | S20  | -1939 | 150.58 |
| 1006 | S19  | -1949 | 265.58 |
| 1007 | S18  | -1959 | 35.58  |
| 1008 | S17  | -1969 | 150.58 |
| 1009 | S16  | -1979 | 265.58 |
| 1010 | S15  | -1989 | 35.58  |
| 1011 | S14  | -1999 | 150.58 |
| 1012 | S13  | -2009 | 265.58 |
| 1013 | S12  | -2019 | 35.58  |
| 1014 | S11  | -2029 | 150.58 |
| 1015 | S10  | -2039 | 265.58 |
| 1016 | S9   | -2049 | 35.58  |
| 1017 | S8   | -2059 | 150.58 |

| PAD. | Name     | X     | Y      |
|------|----------|-------|--------|
| 1018 | S7       | -2069 | 265.58 |
| 1019 | S6       | -2079 | 35.58  |
| 1020 | S5       | -2089 | 150.58 |
| 1021 | S4       | -2099 | 265.58 |
| 1022 | S3       | -2109 | 35.58  |
| 1023 | S2       | -2119 | 150.58 |
| 1024 | S1       | -2129 | 265.58 |
| 1025 | RESERVED | -2139 | 35.58  |
| 1026 | RESERVED | -2149 | 150.58 |
| 1027 | RESERVED | -2159 | 265.58 |
| 1028 | RESERVED | -2169 | 35.58  |
| 1029 | RESERVED | -2179 | 150.58 |
| 1030 | RESERVED | -2189 | 265.58 |
| 1031 | RESERVED | -2199 | 35.58  |
| 1032 | RESERVED | -2209 | 150.58 |
| 1033 | RESERVED | -2219 | 265.58 |
| 1034 | RESERVED | -2229 | 35.58  |
| 1035 | RESERVED | -2239 | 150.58 |
| 1036 | RESERVED | -2249 | 265.58 |
| 1037 | RESERVED | -2259 | 35.58  |
| 1038 | RESERVED | -2269 | 150.58 |
| 1039 | RESERVED | -2279 | 265.58 |
| 1040 | RESERVED | -2289 | 35.58  |
| 1041 | RESERVED | -2299 | 150.58 |
| 1042 | RESERVED | -2309 | 265.58 |
| 1043 | RESERVED | -2319 | 35.58  |
| 1044 | RESERVED | -2329 | 150.58 |
| 1045 | RESERVED | -2339 | 265.58 |
| 1046 | RESERVED | -2349 | 35.58  |
| 1047 | RESERVED | -2359 | 150.58 |
| 1048 | RESERVED | -2369 | 265.58 |
| 1049 | RESERVED | -2379 | 35.58  |
| 1050 | RESERVED | -2389 | 150.58 |
| 1051 | RESERVED | -2399 | 265.58 |

| PAD. | Name     | X     | Y      |
|------|----------|-------|--------|
| 1052 | RESERVED | -2409 | 35.58  |
| 1053 | RESERVED | -2419 | 150.58 |
| 1054 | RESERVED | -2429 | 265.58 |
| 1055 | RESERVED | -2439 | 35.58  |
| 1056 | RESERVED | -2449 | 150.58 |
| 1057 | RESERVED | -2459 | 265.58 |
| 1058 | RESERVED | -2469 | 35.58  |
| 1059 | RESERVED | -2479 | 150.58 |
| 1060 | RESERVED | -2489 | 265.58 |
| 1061 | RESERVED | -2499 | 35.58  |
| 1062 | RESERVED | -2509 | 150.58 |
| 1063 | RESERVED | -2519 | 265.58 |
| 1064 | RESERVED | -2529 | 35.58  |
| 1065 | RESERVED | -2539 | 150.58 |
| 1066 | RESERVED | -2549 | 265.58 |
| 1067 | RESERVED | -2559 | 35.58  |
| 1068 | RESERVED | -2569 | 150.58 |
| 1069 | RESERVED | -2579 | 265.58 |
| 1070 | RESERVED | -2589 | 35.58  |
| 1071 | RESERVED | -2599 | 150.58 |
| 1072 | RESERVED | -2609 | 265.58 |
| 1073 | RESERVED | -2619 | 35.58  |
| 1074 | RESERVED | -2629 | 150.58 |
| 1075 | RESERVED | -2639 | 265.58 |
| 1076 | RESERVED | -2649 | 35.58  |
| 1077 | RESERVED | -2659 | 150.58 |
| 1078 | RESERVED | -2669 | 265.58 |
| 1079 | RESERVED | -2679 | 35.58  |
| 1080 | RESERVED | -2689 | 150.58 |
| 1081 | RESERVED | -2699 | 265.58 |
| 1082 | RESERVED | -2709 | 35.58  |
| 1083 | RESERVED | -2719 | 150.58 |
| 1084 | RESERVED | -2729 | 265.58 |
| 1085 | RESERVED | -2739 | 35.58  |

| PAD. | Name     | X     | Y      |
|------|----------|-------|--------|
| 1086 | RESERVED | -2749 | 150.58 |
| 1087 | RESERVED | -2759 | 265.58 |
| 1088 | RESERVED | -2769 | 35.58  |
| 1089 | RESERVED | -2779 | 150.58 |
| 1090 | RESERVED | -2789 | 265.58 |
| 1091 | RESERVED | -2799 | 35.58  |
| 1092 | RESERVED | -2809 | 150.58 |
| 1093 | RESERVED | -2819 | 265.58 |
| 1094 | RESERVED | -2829 | 35.58  |
| 1095 | RESERVED | -2839 | 150.58 |
| 1096 | RESERVED | -2849 | 265.58 |
| 1097 | RESERVED | -2859 | 35.58  |
| 1098 | RESERVED | -2869 | 150.58 |
| 1099 | RESERVED | -2879 | 265.58 |
| 1100 | RESERVED | -2889 | 35.58  |
| 1101 | RESERVED | -2899 | 150.58 |
| 1102 | RESERVED | -2909 | 265.58 |
| 1103 | RESERVED | -2919 | 35.58  |
| 1104 | RESERVED | -2929 | 150.58 |
| 1105 | RESERVED | -2939 | 265.58 |
| 1106 | RESERVED | -2949 | 35.58  |
| 1107 | RESERVED | -2959 | 150.58 |
| 1108 | RESERVED | -2969 | 265.58 |
| 1109 | RESERVED | -2979 | 35.58  |
| 1110 | RESERVED | -2989 | 150.58 |
| 1111 | RESERVED | -2999 | 265.58 |
| 1112 | RESERVED | -3009 | 35.58  |
| 1113 | RESERVED | -3019 | 150.58 |
| 1114 | RESERVED | -3029 | 265.58 |
| 1115 | RESERVED | -3039 | 35.58  |
| 1116 | RESERVED | -3049 | 150.58 |
| 1117 | RESERVED | -3059 | 265.58 |
| 1118 | RESERVED | -3069 | 35.58  |
| 1119 | RESERVED | -3079 | 150.58 |



| PAD. | Name     | X     | Y      |
|------|----------|-------|--------|
| 1120 | RESERVED | -3089 | 265.58 |
| 1121 | RESERVED | -3099 | 35.58  |
| 1122 | RESERVED | -3109 | 150.58 |
| 1123 | RESERVED | -3119 | 265.58 |
| 1124 | RESERVED | -3129 | 35.58  |
| 1125 | RESERVED | -3139 | 150.58 |
| 1126 | RESERVED | -3149 | 265.58 |
| 1127 | RESERVED | -3159 | 35.58  |
| 1128 | RESERVED | -3169 | 150.58 |
| 1129 | RESERVED | -3179 | 265.58 |
| 1130 | RESERVED | -3189 | 35.58  |
| 1131 | RESERVED | -3199 | 150.58 |
| 1132 | RESERVED | -3209 | 265.58 |
| 1133 | RESERVED | -3219 | 35.58  |
| 1134 | RESERVED | -3229 | 150.58 |
| 1135 | RESERVED | -3239 | 265.58 |
| 1136 | RESERVED | -3249 | 35.58  |
| 1137 | RESERVED | -3259 | 150.58 |
| 1138 | RESERVED | -3269 | 265.58 |
| 1139 | RESERVED | -3279 | 35.58  |
| 1140 | RESERVED | -3289 | 150.58 |
| 1141 | RESERVED | -3299 | 265.58 |
| 1142 | RESERVED | -3309 | 35.58  |
| 1143 | RESERVED | -3319 | 150.58 |
| 1144 | RESERVED | -3329 | 265.58 |
| 1145 | RESERVED | -3339 | 35.58  |
| 1146 | RESERVED | -3349 | 150.58 |
| 1147 | RESERVED | -3359 | 265.58 |
| 1148 | RESERVED | -3369 | 35.58  |
| 1149 | RESERVED | -3379 | 150.58 |
| 1150 | RESERVED | -3389 | 265.58 |
| 1151 | RESERVED | -3399 | 35.58  |
| 1152 | RESERVED | -3409 | 150.58 |
| 1153 | RESERVED | -3419 | 265.58 |

| PAD. | Name     | X     | Y      |
|------|----------|-------|--------|
| 1154 | RESERVED | -3429 | 35.58  |
| 1155 | RESERVED | -3439 | 150.58 |
| 1156 | RESERVED | -3449 | 265.58 |
| 1157 | RESERVED | -3459 | 35.58  |
| 1158 | RESERVED | -3469 | 150.58 |
| 1159 | RESERVED | -3479 | 265.58 |
| 1160 | RESERVED | -3489 | 35.58  |
| 1161 | RESERVED | -3499 | 150.58 |
| 1162 | RESERVED | -3509 | 265.58 |
| 1163 | RESERVED | -3519 | 35.58  |
| 1164 | RESERVED | -3529 | 150.58 |
| 1165 | RESERVED | -3539 | 265.58 |
| 1166 | RESERVED | -3549 | 35.58  |
| 1167 | RESERVED | -3559 | 150.58 |
| 1168 | RESERVED | -3569 | 265.58 |
| 1169 | RESERVED | -3579 | 35.58  |
| 1170 | RESERVED | -3589 | 150.58 |
| 1171 | RESERVED | -3599 | 265.58 |
| 1172 | RESERVED | -3609 | 35.58  |
| 1173 | RESERVED | -3619 | 150.58 |
| 1174 | RESERVED | -3629 | 265.58 |
| 1175 | RESERVED | -3639 | 35.58  |
| 1176 | RESERVED | -3649 | 150.58 |
| 1177 | RESERVED | -3659 | 265.58 |
| 1178 | RESERVED | -3669 | 35.58  |
| 1179 | RESERVED | -3679 | 150.58 |
| 1180 | RESERVED | -3689 | 265.58 |
| 1181 | VGL      | -3739 | 35.58  |
| 1182 | DUMMY    | -3749 | 150.58 |
| 1183 | DUMMY    | -3759 | 265.58 |
| 1184 | DUMMY    | -3769 | 35.58  |
| 1185 | DUMMY    | -3779 | 150.58 |
| 1186 | RESERVED | -3789 | 265.58 |
| 1187 | RESERVED | -3799 | 35.58  |

| PAD. | Name     | X     | Y      |
|------|----------|-------|--------|
| 1188 | RESERVED | -3809 | 150.58 |
| 1189 | RESERVED | -3819 | 265.58 |
| 1190 | RESERVED | -3829 | 35.58  |
| 1191 | RESERVED | -3839 | 150.58 |
| 1192 | RESERVED | -3849 | 265.58 |
| 1193 | RESERVED | -3859 | 35.58  |
| 1194 | RESERVED | -3869 | 150.58 |
| 1195 | RESERVED | -3879 | 265.58 |
| 1196 | RESERVED | -3889 | 35.58  |
| 1197 | RESERVED | -3899 | 150.58 |
| 1198 | RESERVED | -3909 | 265.58 |
| 1199 | RESERVED | -3919 | 35.58  |
| 1200 | RESERVED | -3929 | 150.58 |
| 1201 | RESERVED | -3939 | 265.58 |
| 1202 | RESERVED | -3949 | 35.58  |
| 1203 | RESERVED | -3959 | 150.58 |
| 1204 | RESERVED | -3969 | 265.58 |
| 1205 | RESERVED | -3979 | 35.58  |
| 1206 | RESERVED | -3989 | 150.58 |
| 1207 | RESERVED | -3999 | 265.58 |
| 1208 | RESERVED | -4009 | 35.58  |
| 1209 | RESERVED | -4019 | 150.58 |
| 1210 | RESERVED | -4029 | 265.58 |
| 1211 | RESERVED | -4039 | 35.58  |
| 1212 | RESERVED | -4049 | 150.58 |
| 1213 | RESERVED | -4059 | 265.58 |
| 1214 | RESERVED | -4069 | 35.58  |
| 1215 | RESERVED | -4079 | 150.58 |
| 1216 | RESERVED | -4089 | 265.58 |
| 1217 | RESERVED | -4099 | 35.58  |
| 1218 | RESERVED | -4109 | 150.58 |
| 1219 | RESERVED | -4119 | 265.58 |
| 1220 | RESERVED | -4129 | 35.58  |
| 1221 | RESERVED | -4139 | 150.58 |

| PAD. | Name | X     | Y      |
|------|------|-------|--------|
| 1222 | G409 | -4149 | 265.58 |
| 1223 | G407 | -4159 | 35.58  |
| 1224 | G405 | -4169 | 150.58 |
| 1225 | G403 | -4179 | 265.58 |
| 1226 | G401 | -4189 | 35.58  |
| 1227 | G399 | -4199 | 150.58 |
| 1228 | G397 | -4209 | 265.58 |
| 1229 | G395 | -4219 | 35.58  |
| 1230 | G393 | -4229 | 150.58 |
| 1231 | G391 | -4239 | 265.58 |
| 1232 | G389 | -4249 | 35.58  |
| 1233 | G387 | -4259 | 150.58 |
| 1234 | G385 | -4269 | 265.58 |
| 1235 | G383 | -4279 | 35.58  |
| 1236 | G381 | -4289 | 150.58 |
| 1237 | G379 | -4299 | 265.58 |
| 1238 | G377 | -4309 | 35.58  |
| 1239 | G375 | -4319 | 150.58 |
| 1240 | G373 | -4329 | 265.58 |
| 1241 | G371 | -4339 | 35.58  |
| 1242 | G369 | -4349 | 150.58 |
| 1243 | G367 | -4359 | 265.58 |
| 1244 | G365 | -4369 | 35.58  |
| 1245 | G363 | -4379 | 150.58 |
| 1246 | G361 | -4389 | 265.58 |
| 1247 | G359 | -4399 | 35.58  |
| 1248 | G357 | -4409 | 150.58 |
| 1249 | G355 | -4419 | 265.58 |
| 1250 | G353 | -4429 | 35.58  |
| 1251 | G351 | -4439 | 150.58 |
| 1252 | G349 | -4449 | 265.58 |
| 1253 | G347 | -4459 | 35.58  |
| 1254 | G345 | -4469 | 150.58 |
| 1255 | G343 | -4479 | 265.58 |

| PAD. | Name | X     | Y      |
|------|------|-------|--------|
| 1256 | G341 | -4489 | 35.58  |
| 1257 | G339 | -4499 | 150.58 |
| 1258 | G337 | -4509 | 265.58 |
| 1259 | G335 | -4519 | 35.58  |
| 1260 | G333 | -4529 | 150.58 |
| 1261 | G331 | -4539 | 265.58 |
| 1262 | G329 | -4549 | 35.58  |
| 1263 | G327 | -4559 | 150.58 |
| 1264 | G325 | -4569 | 265.58 |
| 1265 | G323 | -4579 | 35.58  |
| 1266 | G321 | -4589 | 150.58 |
| 1267 | G319 | -4599 | 265.58 |
| 1268 | G317 | -4609 | 35.58  |
| 1269 | G315 | -4619 | 150.58 |
| 1270 | G313 | -4629 | 265.58 |
| 1271 | G311 | -4639 | 35.58  |
| 1272 | G309 | -4649 | 150.58 |
| 1273 | G307 | -4659 | 265.58 |
| 1274 | G305 | -4669 | 35.58  |
| 1275 | G303 | -4679 | 150.58 |
| 1276 | G301 | -4689 | 265.58 |
| 1277 | G299 | -4699 | 35.58  |
| 1278 | G297 | -4709 | 150.58 |
| 1279 | G295 | -4719 | 265.58 |
| 1280 | G293 | -4729 | 35.58  |
| 1281 | G291 | -4739 | 150.58 |
| 1282 | G289 | -4749 | 265.58 |
| 1283 | G287 | -4759 | 35.58  |
| 1284 | G285 | -4769 | 150.58 |
| 1285 | G283 | -4779 | 265.58 |
| 1286 | G281 | -4789 | 35.58  |
| 1287 | G279 | -4799 | 150.58 |
| 1288 | G277 | -4809 | 265.58 |
| 1289 | G275 | -4819 | 35.58  |

| PAD. | Name | X     | Y      |
|------|------|-------|--------|
| 1290 | G273 | -4829 | 150.58 |
| 1291 | G271 | -4839 | 265.58 |
| 1292 | G269 | -4849 | 35.58  |
| 1293 | G267 | -4859 | 150.58 |
| 1294 | G265 | -4869 | 265.58 |
| 1295 | G263 | -4879 | 35.58  |
| 1296 | G261 | -4889 | 150.58 |
| 1297 | G259 | -4899 | 265.58 |
| 1298 | G257 | -4909 | 35.58  |
| 1299 | G255 | -4919 | 150.58 |
| 1300 | G253 | -4929 | 265.58 |
| 1301 | G251 | -4939 | 35.58  |
| 1302 | G249 | -4949 | 150.58 |
| 1303 | G247 | -4959 | 265.58 |
| 1304 | G245 | -4969 | 35.58  |
| 1305 | G243 | -4979 | 150.58 |
| 1306 | G241 | -4989 | 265.58 |
| 1307 | G239 | -4999 | 35.58  |
| 1308 | G237 | -5009 | 150.58 |
| 1309 | G235 | -5019 | 265.58 |
| 1310 | G233 | -5029 | 35.58  |
| 1311 | G231 | -5039 | 150.58 |
| 1312 | G229 | -5049 | 265.58 |
| 1313 | G227 | -5059 | 35.58  |
| 1314 | G225 | -5069 | 150.58 |
| 1315 | G223 | -5079 | 265.58 |
| 1316 | G221 | -5089 | 35.58  |
| 1317 | G219 | -5099 | 150.58 |
| 1318 | G217 | -5109 | 265.58 |
| 1319 | G215 | -5119 | 35.58  |
| 1320 | G213 | -5129 | 150.58 |
| 1321 | G211 | -5139 | 265.58 |
| 1322 | G209 | -5149 | 35.58  |
| 1323 | G207 | -5159 | 150.58 |

| PAD. | Name | X     | Y      |
|------|------|-------|--------|
| 1324 | G205 | -5169 | 265.58 |
| 1325 | G203 | -5179 | 35.58  |
| 1326 | G201 | -5189 | 150.58 |
| 1327 | G199 | -5199 | 265.58 |
| 1328 | G197 | -5209 | 35.58  |
| 1329 | G195 | -5219 | 150.58 |
| 1330 | G193 | -5229 | 265.58 |
| 1331 | G191 | -5239 | 35.58  |
| 1332 | G189 | -5249 | 150.58 |
| 1333 | G187 | -5259 | 265.58 |
| 1334 | G185 | -5269 | 35.58  |
| 1335 | G183 | -5279 | 150.58 |
| 1336 | G181 | -5289 | 265.58 |
| 1337 | G179 | -5299 | 35.58  |
| 1338 | G177 | -5309 | 150.58 |
| 1339 | G175 | -5319 | 265.58 |
| 1340 | G173 | -5329 | 35.58  |
| 1341 | G171 | -5339 | 150.58 |
| 1342 | G169 | -5349 | 265.58 |
| 1343 | G167 | -5359 | 35.58  |
| 1344 | G165 | -5369 | 150.58 |
| 1345 | G163 | -5379 | 265.58 |
| 1346 | G161 | -5389 | 35.58  |
| 1347 | G159 | -5399 | 150.58 |
| 1348 | G157 | -5409 | 265.58 |
| 1349 | G155 | -5419 | 35.58  |
| 1350 | G153 | -5429 | 150.58 |
| 1351 | G151 | -5439 | 265.58 |
| 1352 | G149 | -5449 | 35.58  |
| 1353 | G147 | -5459 | 150.58 |
| 1354 | G145 | -5469 | 265.58 |
| 1355 | G143 | -5479 | 35.58  |
| 1356 | G141 | -5489 | 150.58 |
| 1357 | G139 | -5499 | 265.58 |

| PAD. | Name | X     | Y      |
|------|------|-------|--------|
| 1358 | G137 | -5509 | 35.58  |
| 1359 | G135 | -5519 | 150.58 |
| 1360 | G133 | -5529 | 265.58 |
| 1361 | G131 | -5539 | 35.58  |
| 1362 | G129 | -5549 | 150.58 |
| 1363 | G127 | -5559 | 265.58 |
| 1364 | G125 | -5569 | 35.58  |
| 1365 | G123 | -5579 | 150.58 |
| 1366 | G121 | -5589 | 265.58 |
| 1367 | G119 | -5599 | 35.58  |
| 1368 | G117 | -5609 | 150.58 |
| 1369 | G115 | -5619 | 265.58 |
| 1370 | G113 | -5629 | 35.58  |
| 1371 | G111 | -5639 | 150.58 |
| 1372 | G109 | -5649 | 265.58 |
| 1373 | G107 | -5659 | 35.58  |
| 1374 | G105 | -5669 | 150.58 |
| 1375 | G103 | -5679 | 265.58 |
| 1376 | G101 | -5689 | 35.58  |
| 1377 | G99  | -5699 | 150.58 |
| 1378 | G97  | -5709 | 265.58 |
| 1379 | G95  | -5719 | 35.58  |
| 1380 | G93  | -5729 | 150.58 |
| 1381 | G91  | -5739 | 265.58 |
| 1382 | G89  | -5749 | 35.58  |
| 1383 | G87  | -5759 | 150.58 |
| 1384 | G85  | -5769 | 265.58 |
| 1385 | G83  | -5779 | 35.58  |
| 1386 | G81  | -5789 | 150.58 |
| 1387 | G79  | -5799 | 265.58 |
| 1388 | G77  | -5809 | 35.58  |
| 1389 | G75  | -5819 | 150.58 |
| 1390 | G73  | -5829 | 265.58 |
| 1391 | G71  | -5839 | 35.58  |

| PAD. | Name | X     | Y      |
|------|------|-------|--------|
| 1392 | G69  | -5849 | 150.58 |
| 1393 | G67  | -5859 | 265.58 |
| 1394 | G65  | -5869 | 35.58  |
| 1395 | G63  | -5879 | 150.58 |
| 1396 | G61  | -5889 | 265.58 |
| 1397 | G59  | -5899 | 35.58  |
| 1398 | G57  | -5909 | 150.58 |
| 1399 | G55  | -5919 | 265.58 |
| 1400 | G53  | -5929 | 35.58  |
| 1401 | G51  | -5939 | 150.58 |
| 1402 | G49  | -5949 | 265.58 |
| 1403 | G47  | -5959 | 35.58  |
| 1404 | G45  | -5969 | 150.58 |
| 1405 | G43  | -5979 | 265.58 |
| 1406 | G41  | -5989 | 35.58  |
| 1407 | G39  | -5999 | 150.58 |
| 1408 | G37  | -6009 | 265.58 |
| 1409 | G35  | -6019 | 35.58  |
| 1410 | G33  | -6029 | 150.58 |
| 1411 | G31  | -6039 | 265.58 |
| 1412 | G29  | -6049 | 35.58  |
| 1413 | G27  | -6059 | 150.58 |
| 1414 | G25  | -6069 | 265.58 |
| 1415 | G23  | -6079 | 35.58  |
| 1416 | G21  | -6089 | 150.58 |
| 1417 | G19  | -6099 | 265.58 |
| 1418 | G17  | -6109 | 35.58  |
| 1419 | G15  | -6119 | 150.58 |
| 1420 | G13  | -6129 | 265.58 |
| 1421 | G11  | -6139 | 35.58  |
| 1422 | G9   | -6149 | 150.58 |
| 1423 | G7   | -6159 | 265.58 |
| 1424 | G5   | -6169 | 35.58  |
| 1425 | G3   | -6179 | 150.58 |

| PAD. | Name  | X     | Y      |
|------|-------|-------|--------|
| 1426 | G1    | -6189 | 265.58 |
| 1427 | DUMMY | -6199 | 35.58  |
| 1428 | GTest | -6209 | 150.58 |

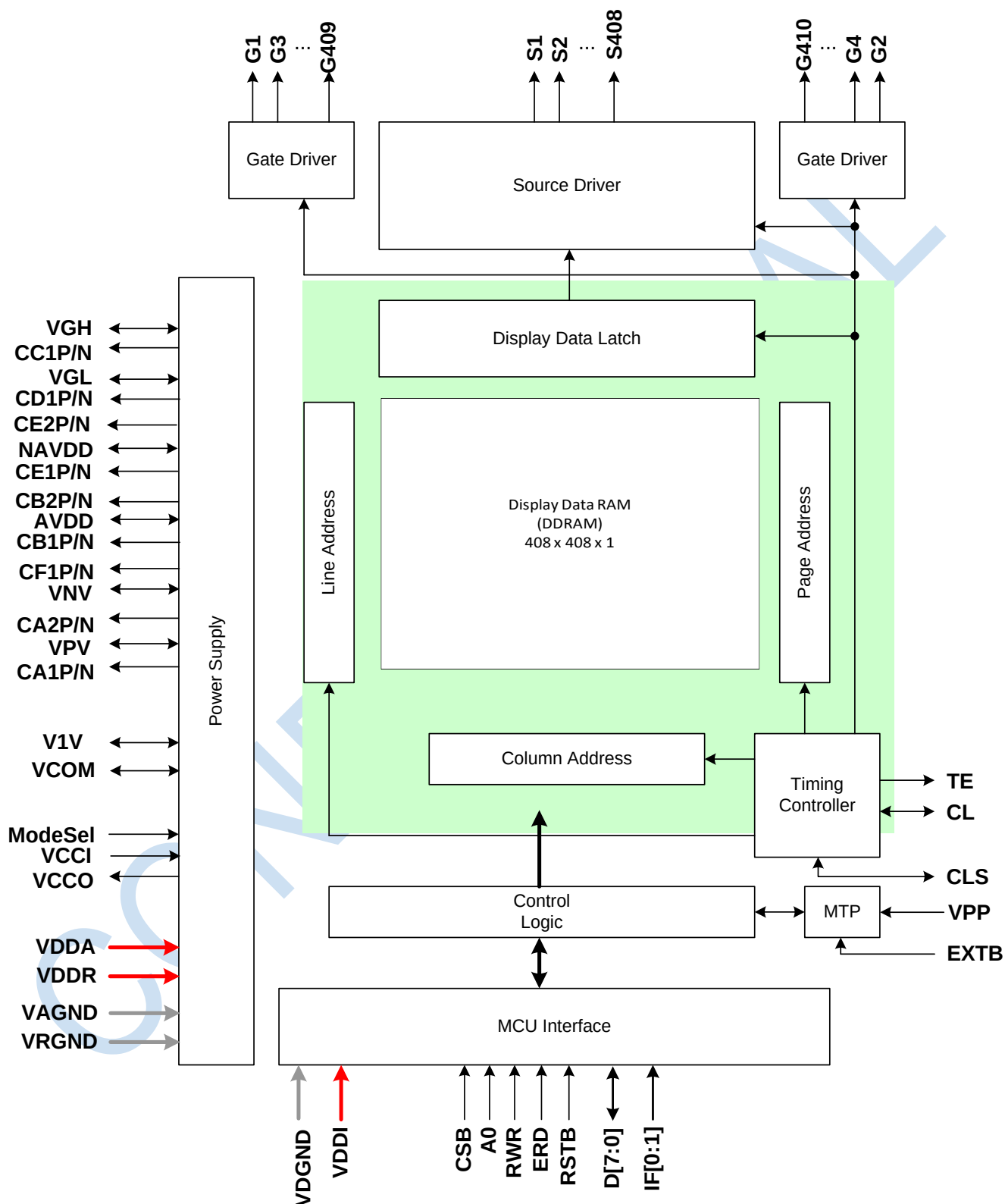
| PAD. | Name  | X     | Y      |
|------|-------|-------|--------|
| 1429 | GTest | -6219 | 265.58 |
| 1430 | DUMMY | -6229 | 35.58  |
| 1431 | DUMMY | -6239 | 150.58 |

| PAD. | Name    | X     | Y    |
|------|---------|-------|------|
| Mark | ALIGN_L | -6146 | -275 |
| Mark | ALIGN_R | 6154  | -275 |

Unit: um

CONFIDENTIAL

## 5 BLOCK DIAGRAM



## 6 PIN DESCRIPTION

### 6.1 Power supply Pin

| Name      | Type  | Description                                                                                                                               |
|-----------|-------|-------------------------------------------------------------------------------------------------------------------------------------------|
| VDDI      | Power | Power Supply (Digital)                                                                                                                    |
| VDDA      | Power | Power Supply (Analog)                                                                                                                     |
| VDDR      | Power | Power Supply (Reference)                                                                                                                  |
| VCCI/VCCO | Power | VCCI is the power source of digital circuits.<br>VCCO is the VCC output. VCCI and VCCO should be connected together.                      |
| ModeSel   | I     | Power Mode Selection.<br>ModeSel = "VAGND": 1.8V mode (VDDI=VDDA=1.8V)<br>ModeSel = "VDDA": Normal mode (VDDI=1.8V~3.6V ; VDDA=2.55~3.6V) |
| VDGND     | Power | Ground (Digital)                                                                                                                          |
| VAGND     | Power | Ground (Analog)                                                                                                                           |
| VRGND     | Power | Ground (Reference)                                                                                                                        |

### 6.2 Digital I/O

| Name    | Type | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |     |                    |   |   |                               |   |   |                                |   |   |                              |   |   |                         |
|---------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|--------------------|---|---|-------------------------------|---|---|--------------------------------|---|---|------------------------------|---|---|-------------------------|
| RSTB    | I    | Reset input pin. When RSTB is "L", internal initialization procedure is executed.                                                                                                                                                                                                                                                                                                                                                                            |     |     |                    |   |   |                               |   |   |                                |   |   |                              |   |   |                         |
| IF[1:0] | I    | <p>These pins select interface operation mode.</p> <table border="1"> <thead> <tr> <th>IF1</th><th>IF0</th><th>MPU interface type</th></tr> </thead> <tbody> <tr> <td>L</td><td>L</td><td>8-bit 8080 parallel interface</td></tr> <tr> <td>L</td><td>H</td><td>Single 4-line serial interface</td></tr> <tr> <td>H</td><td>L</td><td>Dual 4-line serial interface</td></tr> <tr> <td>H</td><td>H</td><td>3-line serial interface</td></tr> </tbody> </table> | IF1 | IF0 | MPU interface type | L | L | 8-bit 8080 parallel interface | L | H | Single 4-line serial interface | H | L | Dual 4-line serial interface | H | H | 3-line serial interface |
| IF1     | IF0  | MPU interface type                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |     |                    |   |   |                               |   |   |                                |   |   |                              |   |   |                         |
| L       | L    | 8-bit 8080 parallel interface                                                                                                                                                                                                                                                                                                                                                                                                                                |     |     |                    |   |   |                               |   |   |                                |   |   |                              |   |   |                         |
| L       | H    | Single 4-line serial interface                                                                                                                                                                                                                                                                                                                                                                                                                               |     |     |                    |   |   |                               |   |   |                                |   |   |                              |   |   |                         |
| H       | L    | Dual 4-line serial interface                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |     |                    |   |   |                               |   |   |                                |   |   |                              |   |   |                         |
| H       | H    | 3-line serial interface                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |     |                    |   |   |                               |   |   |                                |   |   |                              |   |   |                         |
| CSB     | I    | <p>Chip select input pin.</p> <p>CSB="L": This chip is selected and the MPU interface is active.</p> <p>CSB="H": This chip is not selected and the MPU interface is disabled (D[7:0] are high impedance).</p>                                                                                                                                                                                                                                                |     |     |                    |   |   |                               |   |   |                                |   |   |                              |   |   |                         |
| A0      | I    | <p><b>When using 8080, Single 4SPI and Dual 4SPI mode.</b></p> <p>It determines whether the access is related to data or command.</p> <p>A0 = "H": Indicates that D[7:0] are display data;</p> <p>A0 = "L": Indicates that D[7:0] are control data.</p> <p><b>When using 3SPI.</b></p> <p>There is no A0 pin in 3-Line. A0 should be fixed to "H" by VDDI.</p>                                                                                               |     |     |                    |   |   |                               |   |   |                                |   |   |                              |   |   |                         |

| Name   | Type | Description                                                                                                                                                                                                                                                                                                                                        |
|--------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RWR    | I    | <p><b>When using 8080</b><br/>Write enable in 8080 parallel interface.</p> <p><b>When using Dual 4SPI</b><br/>Second data lane in Dual 4SPI interface.<br/>This pin is not used in 3-Line and Single 4-Line SPI. RWR should be connected to VDDI.</p>                                                                                              |
| ERD    | I    | <p>Read enable in 8080 interface.<br/>This pin is not used in serial interfaces and should be connected to VDDI.</p>                                                                                                                                                                                                                               |
| D[7:0] | I/O  | <p><b>When using 8-bit parallel interface: 8080 mode</b><br/>8 bit bi-directional data bus. Connect to the data bus of 8-bit microprocessor.<br/>When CSB is "H", D[7:0] are high impedance.</p>                                                                                                                                                   |
|        | I/O  | <p><b>When using serial interface: Single/Dual 4-line SPI, 3-line SPI</b><br/>D[7:4] : fix to "H" by VDDI.<br/>D[3:2] : serial output data (SDA_OUT).<br/>D[1] : serial input data (SDA_IN).<br/>D[0] : serial input clock (SCLK).<br/><b><u>D1 to D3 must be connected together (SDA)</u></b><br/>When CSB is "H", D[7:0] are high impedance.</p> |
| TE     | O    | <p>Tearing effect signal is used to synchronize MCU to frame memory writing.<br/>If not used, please let this pin open.</p>                                                                                                                                                                                                                        |

Note:

1. After VDDI is turned ON, all MPU interface pins should not be left OPEN.

### 6.3 Clock System Input

| Name | Type | Description                                                                                                               |
|------|------|---------------------------------------------------------------------------------------------------------------------------|
| CLS  | I    | Clock source selection pin.<br>CLS="H": enable internal clock.<br>CLS="L": disable internal clock and use external clock. |
| CL   | I/O  | For external clock.<br>If CLS="H": this pin is open.<br>If CLS="L": this pin is the input of oscillator.                  |

### 6.4 Driving Output Pin

| Name                               | Type | Description                                                                                              |
|------------------------------------|------|----------------------------------------------------------------------------------------------------------|
| S1 to S408                         | O    | Source driver output pins.                                                                               |
| G1 to G410                         | O    | Gate driver output pins.                                                                                 |
| VPV<br>VNV<br>V1V<br>AVDD<br>NAVDD | O    | Power output pin for analog circuit.                                                                     |
| VGH                                | O    | Power output (positive) pin for gate driver.                                                             |
| VGL                                | O    | Power output (negative) pin for gate driver.                                                             |
| VCOM                               | O    | COM outputs. A power supply for the TFT-LCD common electrode.<br><b>Must be connected to GND of FPC.</b> |
| CA1P/N<br>CA2P/N                   | O    | Capacitor connecting pins for step-up circuit. (for VPV)                                                 |
| CB1P/N<br>CB2P/N                   | O    | Capacitor connecting pins for step-up circuit. (for AVDD)                                                |
| CC1P/N                             | O    | Capacitor connecting pins for step-up circuit. (for VGH)                                                 |
| CD1P/N                             | O    | Capacitor connecting pins for step-up circuit. (for VGL)                                                 |
| CE1P/N<br>CE2P/N                   | O    | Capacitor connecting pins for step-up circuit. (for NAVDD)                                               |
| CF1P/N                             | O    | Capacitor connecting pins for step-up circuit. (for VNV)                                                 |



## 6.5 MTP Pin

| Name | Type  | Description                                                                                                                                                                                                   |
|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VPP  | Power | The programming power supply of the built-in NVM. Apply external power 7.5V here when programming (> 8mA for successful programming).<br>If not used, left this pin open.                                     |
| EXTB | I     | EXTB="L": Enable the extension operation mode.<br>When programming MTP, connect EXTB to VDGND externally.<br>This pin has an internal pull-high resistor. Please leave this pin OPEN after special operation. |

## 6.6 Others

| Name                                                                           | Type | Description                                              |
|--------------------------------------------------------------------------------|------|----------------------------------------------------------|
| VTEST<br>TESTENB<br>TESTA<br>TESTB<br>GTest<br>DUMMY<br>RESERVED<br>MTest[5:0] | O    | Reserved for test only.<br>Please leave these pins open. |

## 6.7 Recommend Resistance

| Item           | Pin name                                          | Resistance |
|----------------|---------------------------------------------------|------------|
| Powers         | VDDA, VAGND                                       | < 3 ohm    |
|                | VCCO, VCCI                                        | < 20 ohm   |
|                | VDDI, VDGND, VDDR, VRGND                          | < 50 ohm   |
| Driving Output | CAXN/P, VPV, VNV, AVDD                            | < 3 ohm    |
|                | CBxN/P, CF1N/P                                    | < 5 ohm    |
|                | V1V, CExP/N, NAVDD                                | < 10 ohm   |
|                | CC1N/P, VGH, CD1N/P, VGL                          | < 20 ohm   |
|                | VCOM                                              | < 50 ohm   |
| Digital        | D[7:0], CSB, RSTB, A0, RWR, ERD                   | < 100 ohm  |
|                | IF[1:0], CLS, CL, TE, EXTB, TESTB, TESTA, ModeSel | < 200 ohm  |
| MTP            | VPP                                               | < 10 ohm   |

## 7 FUNCTION DESCRIPTION

ST7305 supports 8-bit parallel data bus for 8080 series CPU, 3-line and single/dual 4-line interface.

### 7.1 Microprocessor Interface

#### 7.1.1 Chip Select Input

CSB pin is used for chip selection. This driver can interface with an MPU when CSB is "L". If CSB is "H", the inputs of A0, ERD and RWR with any combination will be ignored and D[7:0] are high impedance. In 4-Line and 3-Line serial interface, the internal shift register and serial counter are reset when CSB is "H".

#### 7.1.2 Interface Selection

The interface selection is controlled by IF[1:0] pins. Please refer to the table below:

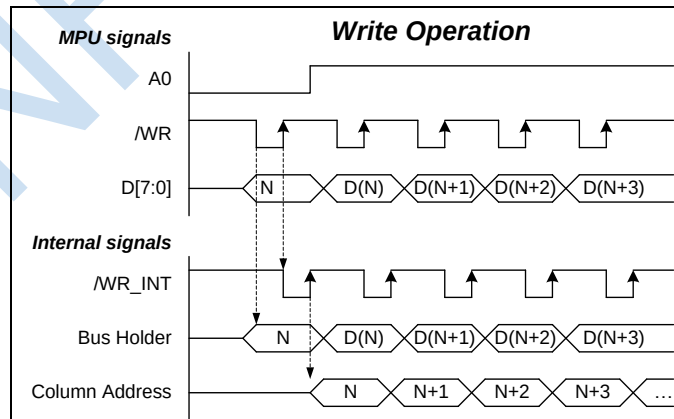
**Table 1 Parallel/Serial Interface Mode**

| Setting |     | MCU Type                       | Interface Pin Function |    |      |     |                                 |
|---------|-----|--------------------------------|------------------------|----|------|-----|---------------------------------|
| IF1     | IF0 |                                | CSB                    | A0 | RWR  | ERD | D[7:0]                          |
| L       | L   | 8-bit 8080 Parallel interface  | CSB                    | A0 | /WR  | /RD | D[7:0]                          |
| L       | H   | Single 4-line serial interface |                        |    | -    | -   | D[0]=SCL, D[1:3]=SDA, D[4:7]= - |
| H       | L   | Dual 4-line serial interface   |                        |    | SDA2 | -   |                                 |
| H       | H   | 3-line serial interface        |                        | -  | -    | -   |                                 |

Note: The un-used pins are marked as "-" and should be connected to "H" by VDDI.

#### 7.1.3 8080 Parallel Interface

ST7305 uses bus holder and internal data bus for data transfer with MPU. When writing data from the MPU to on-chip RAM, data is automatically transferred from the bus holder to the RAM as shown in Fig1.

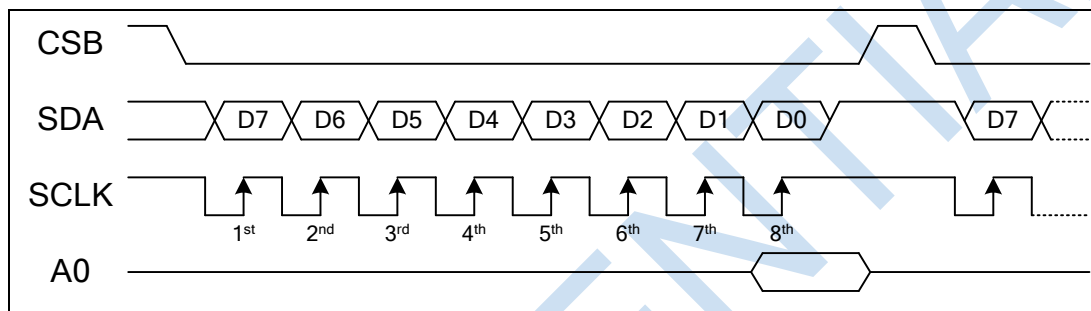


**Fig1. Data Transfer: Write**

## 7.1.4 Single 4-Line Serial Interface

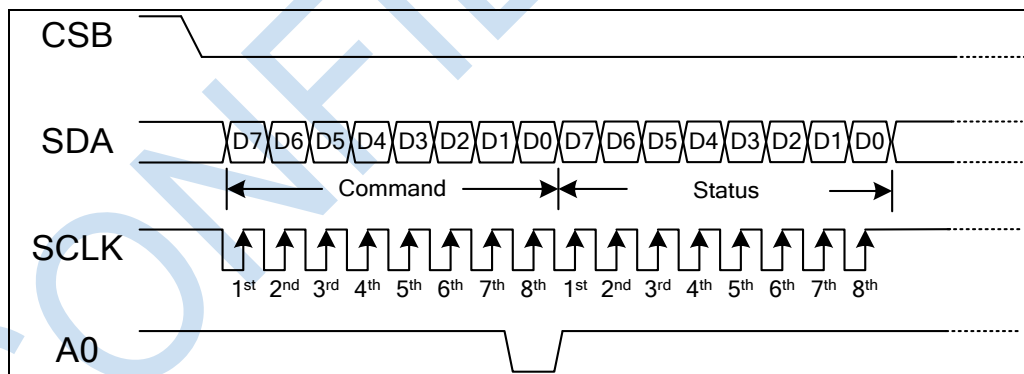
ST7305 is active when CSB is “L”, serial data (SDA) and serial clock (SCLK) inputs are enabled. When CSB is “H”, ST7305 is not active, the internal 8-bit shift register and 3-bit counter are reset. The DDRAM column address pointer will be increased by one automatically after writing each byte of DDRAM.

The display data/command indication is controlled by the register selection pin (A0). The signals transferred on data bus will be display data when A0 is high and will be instruction when A0 is low. Serial data (SDA) is latched at the rising edge of serial clock (SCLK). After the 8<sup>th</sup> serial clock, the serial data will be processed as 8-bit parallel data. The DDRAM column address pointer will be increased by one automatically after each byte of DDRAM access.



**Write Operation of 4-Line SPI**

After entering the “Read Status” instruction to read IC status, the information is shifted out as shown below. CSB signal must be kept at “L” during this period. All read out data will be 8 bits.



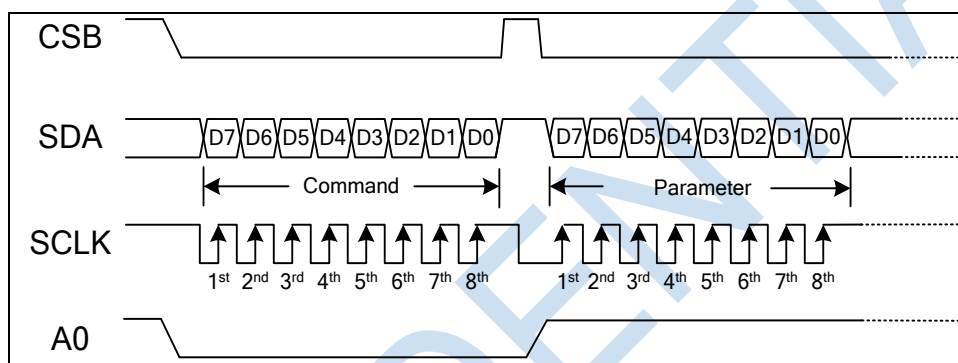
**Read Status Operation of 4-Line SPI**

### 7.1.5 Dual 4-Line Serial Interface

Dual 4-line serial interface use: CSB (chip enable), SCLK (serial clock), SDA (serial data input/output 1), RWR (serial data input 2) and A0.

#### Command write mode:

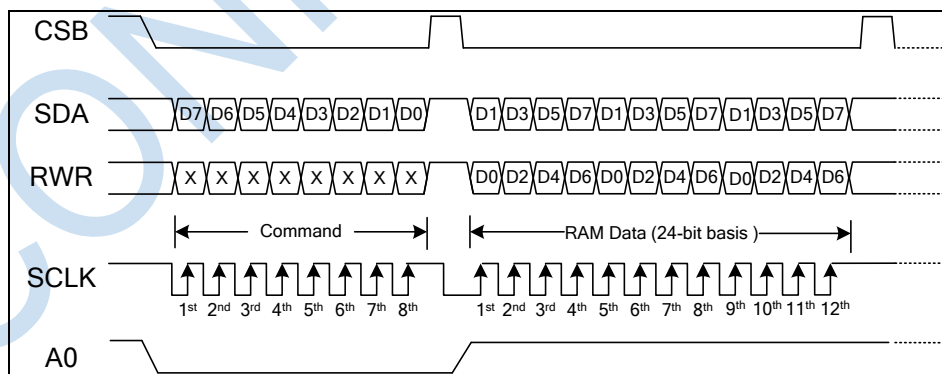
The command write protocol of dual 4-line serial interface is the same with the single 4-line serial interface. The display data/command indication is controlled by the register selection pin (A0). Any instruction can be sent in any order to the driver. The MSB is transmitted first. The serial interface is initialized when CSB is high. In this state, SCLK clock pulse or SDA data have no effect. A falling edge on CSB enables the serial interface and indicates the start of data transmission.



Command Write Operation of Dual 4-Line SPI

#### RAM write mode:

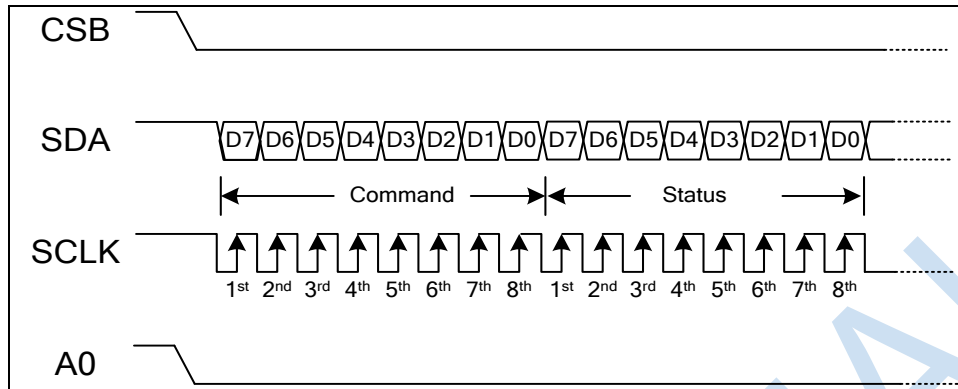
The RAM write mode of dual 4-line serial interface need use SDA pin and RWR pin to be data input pins. The LSB is transmitted first and the data bit will be exchanged between SDA and RWR.



RAM Write Operation of Dual 4-Line SPI

**Read protocol for dual 4-line serial interface:**

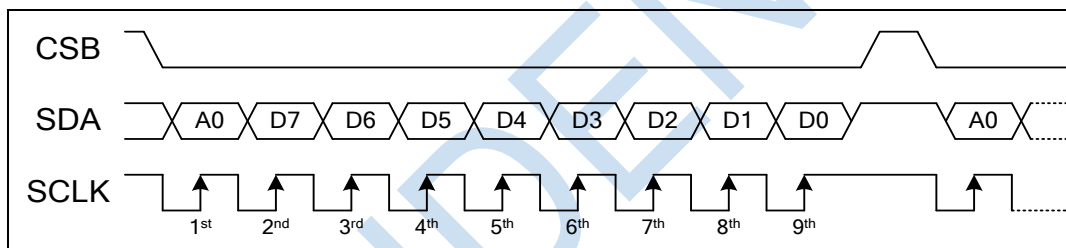
Read protocol (for RDID1/RDID2/RDID3 command: 8-bit read):



**Read Status Operation of Dual 4-Line SPI Interface**

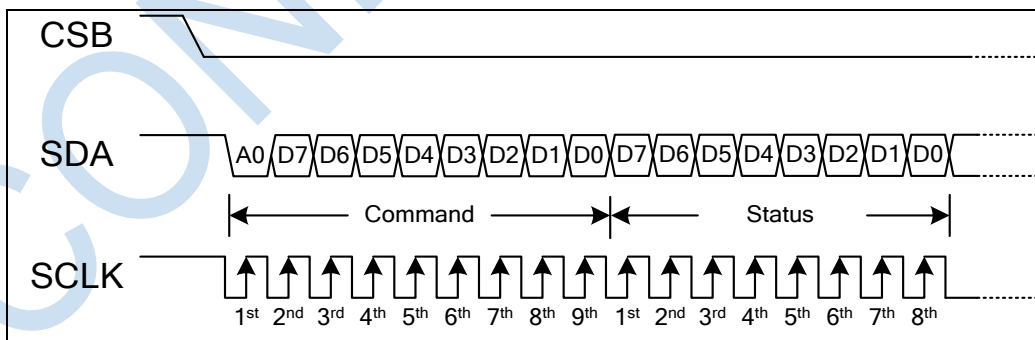
**7.1.6 3-Line Serial Interface**

In 3-Line interface, A0 pin is not available. The 1<sup>st</sup> output bit defines command byte or parameter byte.



**Write Operation of 3-Line SPI**

After entering the “Read Status” instruction to read IC status, the information is shifted out as shown below. CSB signal must be kept at “L” during this period. All read out data will be 8 bits.



**Read Status Operation of 3-Line SPI**

## 7.2 Data Color Coding

Use specific command for switching between data input modes.

### 7.2.1 Data Input Mode

#### (1) 8080/Single 4SPI/3SPI Interface

TYPE1: There are 4 write operations for 24-bit data. (Set by "BPS=0" of command 0x3Ah)

| Command     | A0 | D7  | D6  | D5  | D4  | D3  | D2  | D1 | D0 |
|-------------|----|-----|-----|-----|-----|-----|-----|----|----|
| DDRAM write | 0  | 0   | 0   | 1   | 0   | 1   | 1   | 0  | 0  |
| 1st write   | 1  | P1  | P2  | P3  | P4  | P5  | P6  | -  | -  |
| 2nd write   | 1  | P7  | P8  | P9  | P10 | P11 | P12 | -  | -  |
| 3rd write   | 1  | P13 | P14 | P15 | P16 | P17 | P18 | -  | -  |
| 4th write   | 1  | P19 | P20 | P21 | P22 | P23 | P24 | -  | -  |

Note: - don't care

TYPE2: There are 3 write operations for 24-bit data. (Set by "BPS=1" of command 0x3Ah)

| Command     | A0 | D7  | D6  | D5  | D4  | D3  | D2  | D1  | D0  |
|-------------|----|-----|-----|-----|-----|-----|-----|-----|-----|
| DDRAM write | 0  | 0   | 0   | 1   | 0   | 1   | 1   | 0   | 0   |
| 1st write   | 1  | P1  | P2  | P3  | P4  | P5  | P6  | P7  | P8  |
| 2nd write   | 1  | P9  | P10 | P11 | P12 | P13 | P14 | P15 | P16 |
| 3rd write   | 1  | P17 | P18 | P19 | P20 | P21 | P22 | P23 | P24 |

#### (2) Dual 4SP Interface

There are 3 write operations for 24-bit data at the same time.

| Command     | A0 | D7  | D6  | D5  | D4  | D3  | D2  | D1  | D0  |
|-------------|----|-----|-----|-----|-----|-----|-----|-----|-----|
| DDRAM write | 0  | 0   | 0   | 1   | 0   | 1   | 1   | 0   | 0   |
| 1st write   | 1  | P1  | P2  | P3  | P4  | P5  | P6  | P7  | P8  |
| 2nd write   | 1  | P9  | P10 | P11 | P12 | P13 | P14 | P15 | P16 |
| 3rd write   | 1  | P17 | P18 | P19 | P20 | P21 | P22 | P23 | P24 |

### 7.2.2 Data to Display Mapping

The data mapping of mono display is as below.

|    |    |    |    |     |     |     |     |     |     |     |     |
|----|----|----|----|-----|-----|-----|-----|-----|-----|-----|-----|
| P1 | P3 | P5 | P7 | P9  | P11 | P13 | P15 | P17 | P19 | P21 | P23 |
| P2 | P4 | P6 | P8 | P10 | P12 | P14 | P16 | P18 | P20 | P22 | P24 |

|      |      |      |      |       |       |       |       |       |       |       |       |
|------|------|------|------|-------|-------|-------|-------|-------|-------|-------|-------|
| Pix1 | Pix3 | Pix5 | Pix7 | Pix9  | Pix11 | Pix13 | Pix15 | Pix17 | Pix19 | Pix21 | Pix23 |
| Pix2 | Pix4 | Pix6 | Pix8 | Pix10 | Pix12 | Pix14 | Pix16 | Pix18 | Pix20 | Pix22 | Pix24 |

## 7.2.3 Memory to Display Address Mapping

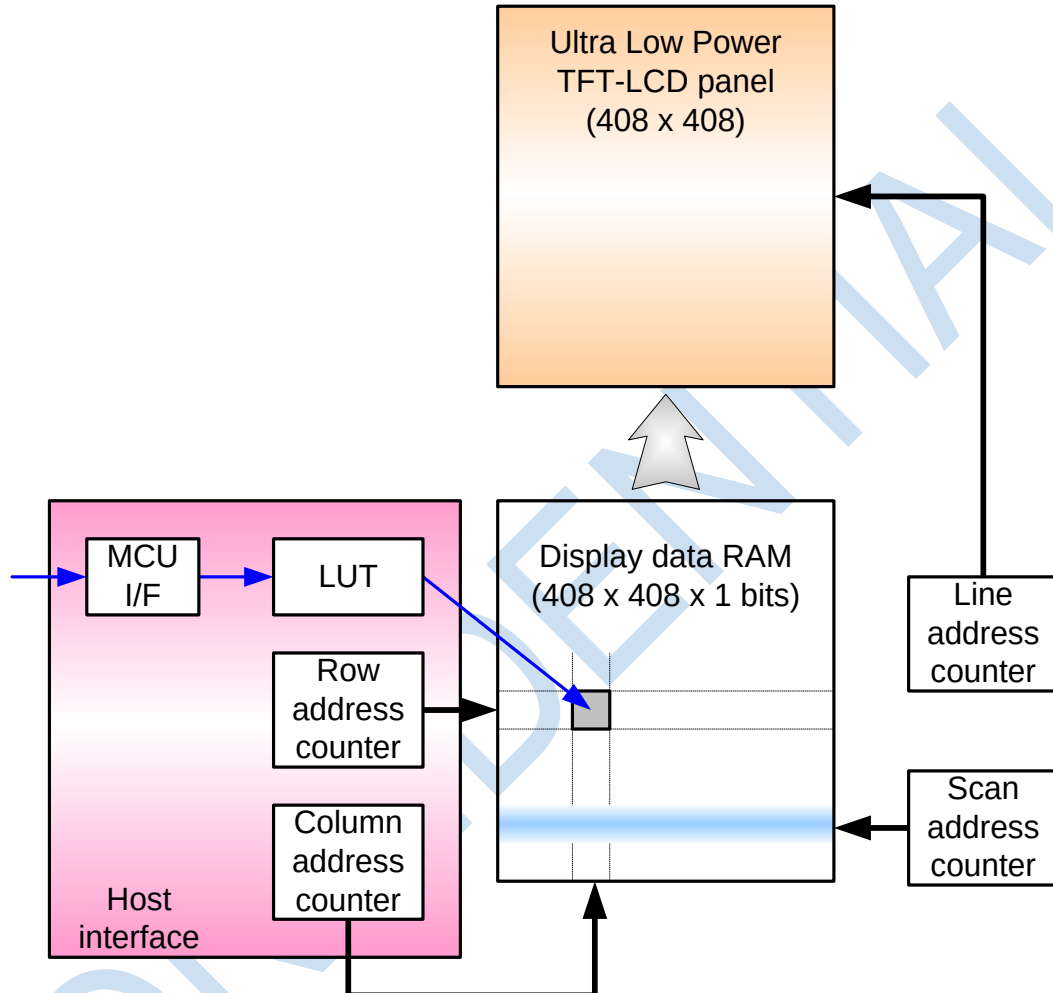
| PA     |     | 13 |    |    |    |     |     |     |     |     |     |     |     |   | ∴    | 46   |      |      |      |      |      |      |      |      |      |      |      |  | NOR  | CA |
|--------|-----|----|----|----|----|-----|-----|-----|-----|-----|-----|-----|-----|---|------|------|------|------|------|------|------|------|------|------|------|------|------|--|------|----|
| NOR    | INV | 46 |    |    |    |     |     |     |     |     |     |     |     |   | ∴    | 13   |      |      |      |      |      |      |      |      |      |      |      |  | INV  |    |
| 0      | 203 | P1 | P3 | P5 | P7 | P9  | P11 | P13 | P15 | P17 | P19 | P21 | P23 | ∴ |      |      |      |      |      |      |      |      |      |      |      |      |      |  | L1   |    |
|        |     | P2 | P4 | P6 | P8 | P10 | P12 | P14 | P16 | P18 | P20 | P22 | P24 | ∴ |      |      |      |      |      |      |      |      |      |      |      |      |      |  | L2   |    |
| 1      | 202 |    |    |    |    |     |     |     |     |     |     |     |     |   |      |      |      |      |      |      |      |      |      |      |      |      |      |  | L3   |    |
|        |     |    |    |    |    |     |     |     |     |     |     |     |     |   |      |      |      |      |      |      |      |      |      |      |      |      |      |  | L4   |    |
| ∴      | ∴   | ∴  | ∴  | ∴  |    |     |     |     |     |     |     |     |     |   |      |      |      |      | ∴    | ∴    | ∴    |      |      |      |      |      |      |  | ∴    |    |
| 202    | 1   |    |    |    |    |     |     |     |     |     |     |     |     |   |      |      |      |      |      |      |      |      |      |      |      |      |      |  | L405 |    |
|        |     |    |    |    |    |     |     |     |     |     |     |     |     |   |      |      |      |      |      |      |      |      |      |      |      |      |      |  | L406 |    |
| 203    | 0   |    |    |    |    |     |     |     |     |     |     |     |     |   |      |      |      |      |      |      |      |      |      |      |      |      |      |  | L407 |    |
|        |     |    |    |    |    |     |     |     |     |     |     |     |     |   |      |      |      |      |      |      |      |      |      |      |      |      |      |  | L408 |    |
| Source |     | S1 | S2 | S3 | S4 | S5  | S6  | S7  | S8  | S9  | S10 | S11 | S12 | ∴ | S397 | S398 | S399 | S400 | S401 | S402 | S403 | S404 | S405 | S406 | S407 | S408 | Line |  |      |    |



### 7.3 Display Data RAM

#### 7.3.1 Configuration

The display module has an integrated 408 x 408 x 1b graphic type static RAM. There will be no abnormal visible effect on the display when there is a simultaneous Panel Read and Write to the same location of the Frame Memory.



## 7.4 Address Control

The address counter sets the addresses of the display data RAM for writing and reading. Data is written pixel-wise into the RAM matrix of DRIVER. The locations of RAM are addressed by the address pointers. The address ranges are X=13(0Dh) to X=46 (2Eh) and Y=0 to Y=203 (CBh). Addresses outside these ranges are not allowed. Before writing to the RAM, a window must be defined that will be written. The window is programmable via the command registers XS, YS designating the start address and XE, YE designating the end address.

For example the whole display contents will be written, the window is defined by the following values: XS=13 (0Dh) YS=0 (0h) and XE=46 (2Eh), YE=203 (CBh).

In vertical addressing mode (MV=1), the Y-address increments after each byte, after the last Y-address (Y=YE), Y wraps around to YS and X increments to address the next column. In horizontal addressing mode (V=0), the X-address increments after each byte, after the last X-address (X=XE), X wraps around to XS and Y increments to address the next row. After every last address (X=XE and Y=YE) the address pointers wrap around to address (X=XS and Y=YS).

For flexibility in handling a wide variety of display architectures, the commands “CASET, RASET and MADCTL”, define flags MX and MY, which allows mirroring of the X-address and Y-address. All combinations of flags are allowed. Section 7.4 show the available combinations of writing to the display RAM. When MX, MY and MV will be changed the data must be rewritten to the display RAM.

For each image condition, the controls for the column and row counters apply as below

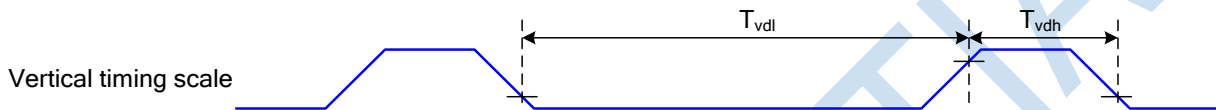
| Condition                                                                                                         | Column Counter                | Row Counter                |
|-------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------|
| When RAMWR/RAMRD command is accepted                                                                              | Return to “Start Column (XS)” | Return to “Start Row (YS)” |
| Complete Pixel Read / Write action                                                                                | Increment by 1                | No change                  |
| The Column counter value is larger than “End Column (XE)”                                                         | Return to “Start Column (XS)” | Increment by 1             |
| The Column counter value is larger than “End Column (XE)” and the Row counter value is larger than “End Row (YE)” | Return to “Start Column (XS)” | Return to “Start Row (YS)” |

## 7.5 Tearing Effect

The Tearing Effect output line supplies to the MPU a Panel synchronization signal. This signal can be enabled or disabled by the Tearing Effect Line Off & On commands. The mode of the Tearing Effect signal is defined by the parameter of the Tearing Effect Line On command. The signal can be used by the MPU to synchronize Frame Memory Writing when displaying video images.

### 7.5.1 Tearing effect line modes

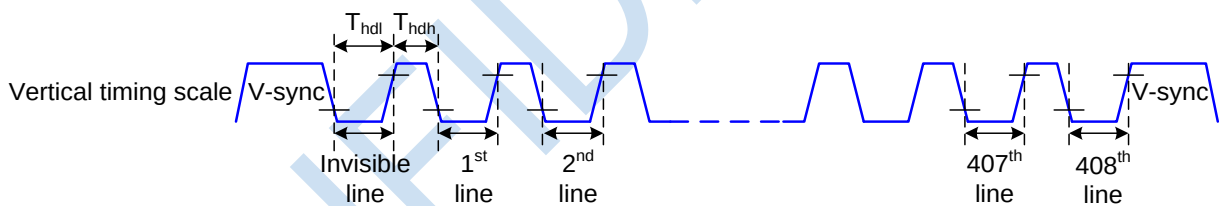
Mode 1, the Tearing Effect Output signal consists of V-Blanking Information only:



$t_{vdh}$ = The LCD display is not updated from the Frame Memory

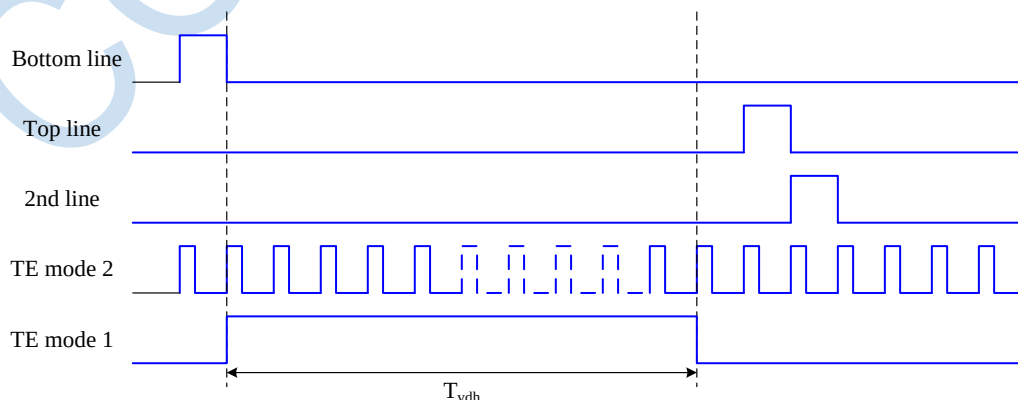
$t_{vdl}$ = The LCD display is updated from the Frame Memory (except Invisible Line – see above)

Mode 2, the Tearing Effect Output signal consists of V-Blanking and H-Blanking Information, there is one V-sync and 408 H-sync pulses per field.



$t_{hdh}$ = The LCD display is not updated from the Frame Memory

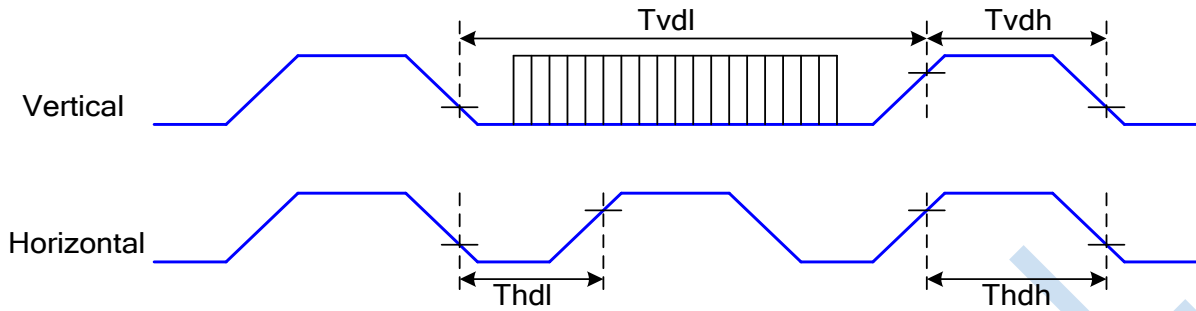
$t_{hdl}$ = The LCD display is updated from the Frame Memory (except Invisible Line – see above)



Note: During Sleep In Mode, the Tearing Output Pin is active Low.

### 7.5.2 Tearing effect line timings

The Tearing Effect signal is described below:



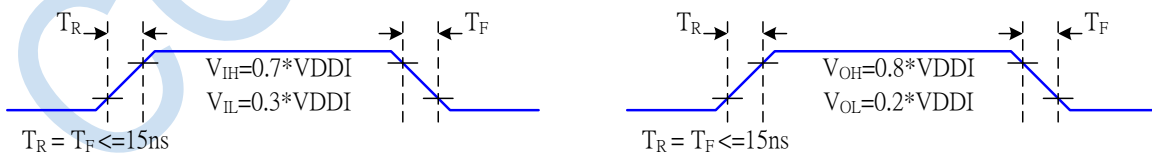
| Symbol | Parameter                      | min  | max | unit | description |
|--------|--------------------------------|------|-----|------|-------------|
| tvdl   | Vertical Timing Low Duration   | 30   | -   | ms   |             |
| tvdh   | Vertical Timing High Duration  | 1.2  | -   | ms   |             |
| thdl   | Horizontal Timing Low Duration | 31.2 | -   | μs   |             |
| thdh   | Horizontal Timing Low Duration | 31.2 | -   | μs   |             |

AC characteristics of Tearing Effect Signal HPM Mode (Frame Rate = 32 Hz, Ta=25°C)

| Symbol | Parameter                      | min   | max | unit | description |
|--------|--------------------------------|-------|-----|------|-------------|
| tvdl   | Vertical Timing Low Duration   | 990.5 | -   | ms   |             |
| tvdh   | Vertical Timing High Duration  | 9.5   | -   | ms   |             |
| thdl   | Horizontal Timing Low Duration | 250   | -   | μs   |             |
| thdh   | Horizontal Timing Low Duration | 250   | -   | μs   |             |

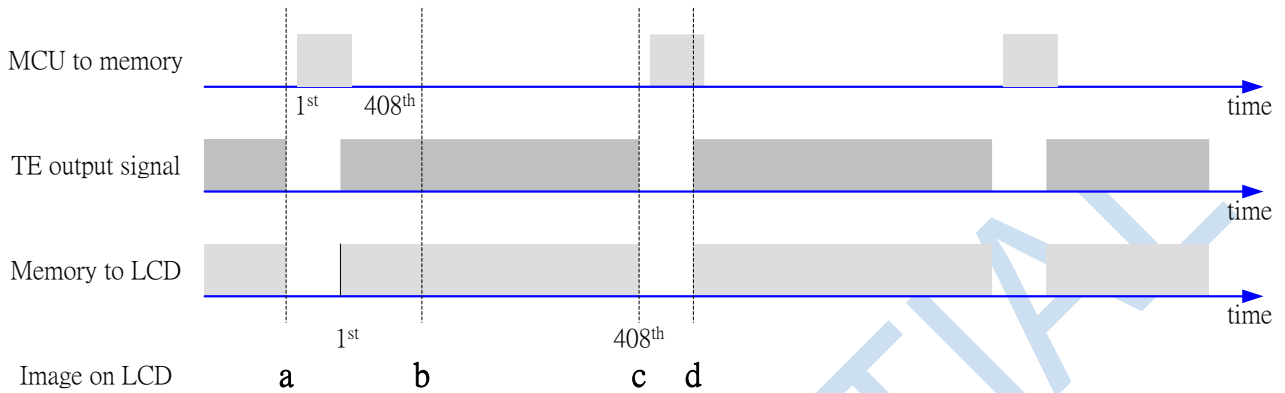
AC characteristics of Tearing Effect Signal LPM Mode (Frame Rate = 1 Hz, Ta=25°C)

The signal's rise and fall times ( $t_f$ ,  $t_r$ ) are stipulated to be equal to or less than 15ns.

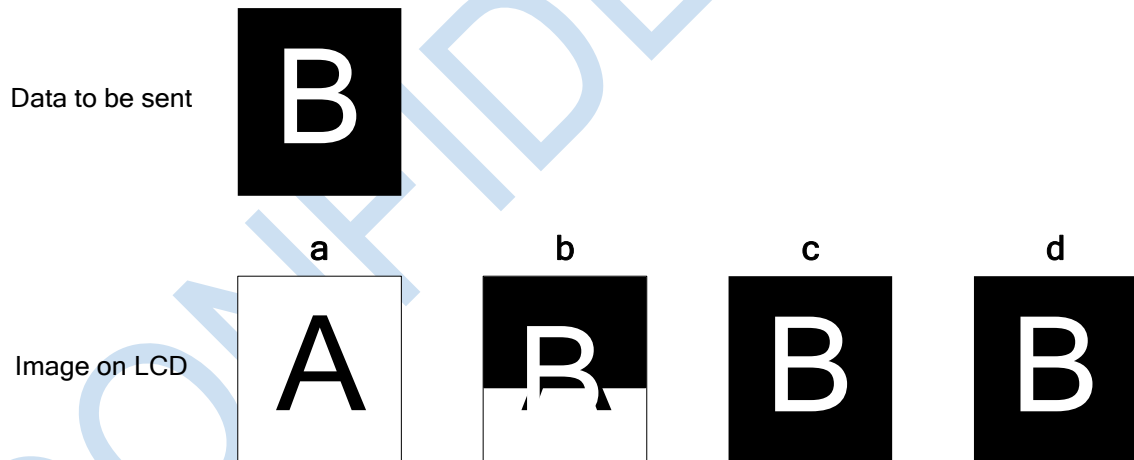


The Tearing Effect Output Line is fed back to the MPU and should be used as shown below to avoid Tearing Effect:

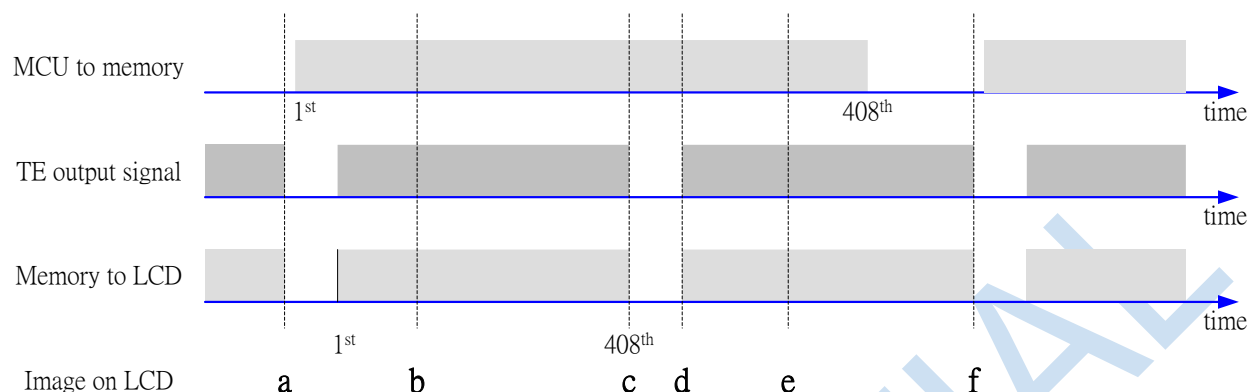
### 7.5.3 Example 1: MPU Write is faster than panel read



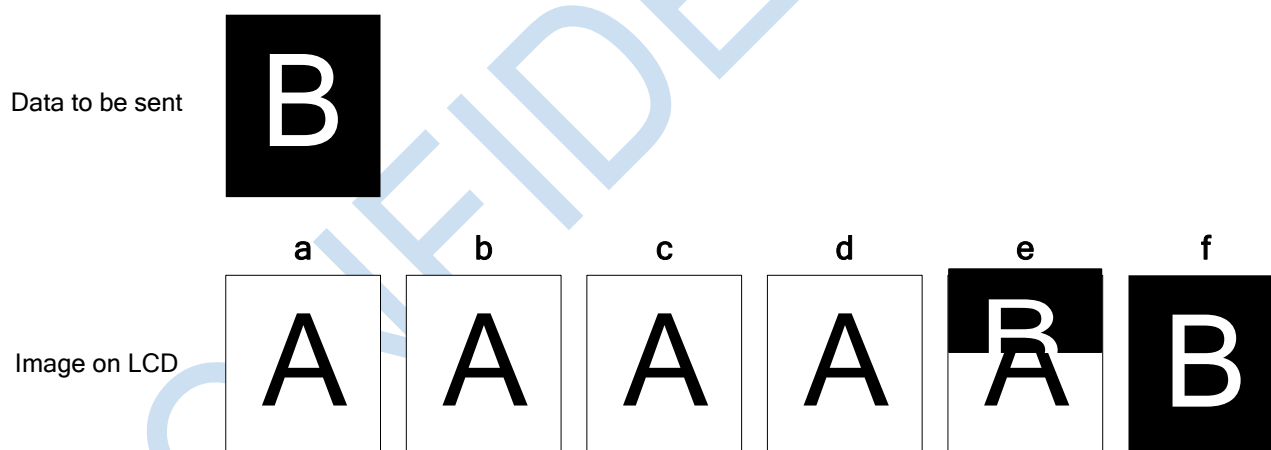
Data write to Frame Memory is now synchronized to the Panel Scan. It should be written during the vertical sync pulse of the Tearing Effect Output Line. This ensures that data is always written ahead of the panel scan and each Panel Frame refresh has a complete new image:



### 7.5.4 Example 2: MPU write is slower than panel read



The MPU to Frame Memory write begins just after Panel Read has commenced i.e. after one horizontal sync pulse of the Tearing Effect Output Line. This allows time for the image to download behind the Panel Read pointer and finishing download during the subsequent Frame before the Read Pointer “catches” the MPU to Frame memory write position.



## 7.6 Oscillation Circuit

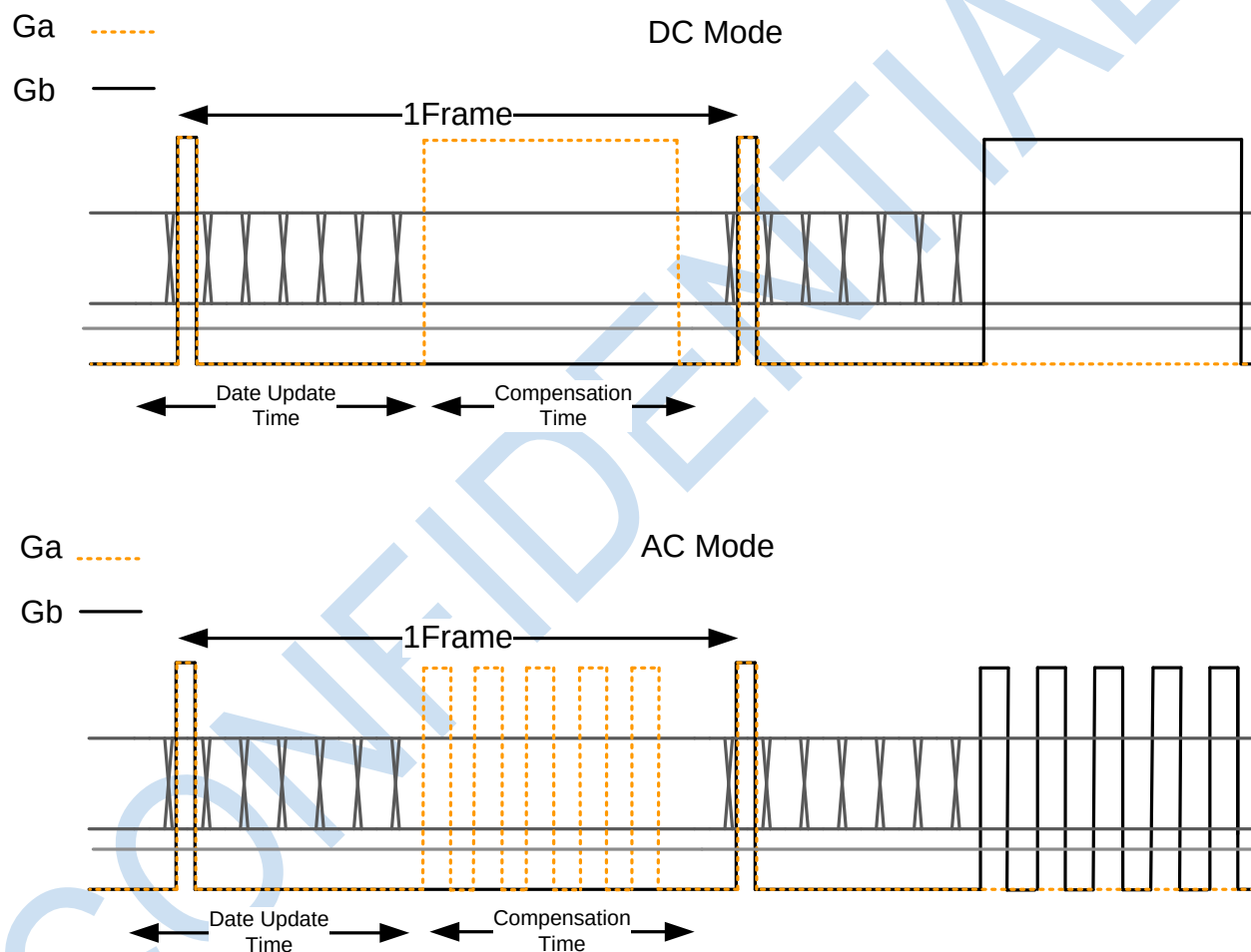
This is on-chip Oscillator without external resistor. When the internal oscillator is used, connect CLS to VDDI; when the external oscillator is used, connect CLS to VSS and input external clock to CL pin. This oscillator signal is used by the voltage booster and display timing generation circuit.

## 7.7 Reset

Setting RSTB pin to “L” (hardware reset) can initialize internal function. Generally, VDDI is not stable at the time that the system power is just turned ON. The hardware reset or software reset is required to initialize internal registers after VDDI is stable. Initialization by RSTB pin or command SWRESET is essential before operating.

## 7.8 Dual TFT Waveform

The driving waveform of two transistors are as below, including data updating and compensation time.



## 7.9 Power ON/OFF Sequence

VDDI and VDDA can be applied in any order.

VDDA and VDDI can be power down in any order.

During power off, if LCD is in the Sleep Out mode, VDDA and VDDI must be powered down minimum 120msec after RSTB has been released.

During power off, if LCD is in the Sleep In mode, VDDI or VDDA can be powered down minimum 0msec after RSTB has been released.

CSB can be applied at any timing or can be permanently grounded. RSTB has priority over CSB.

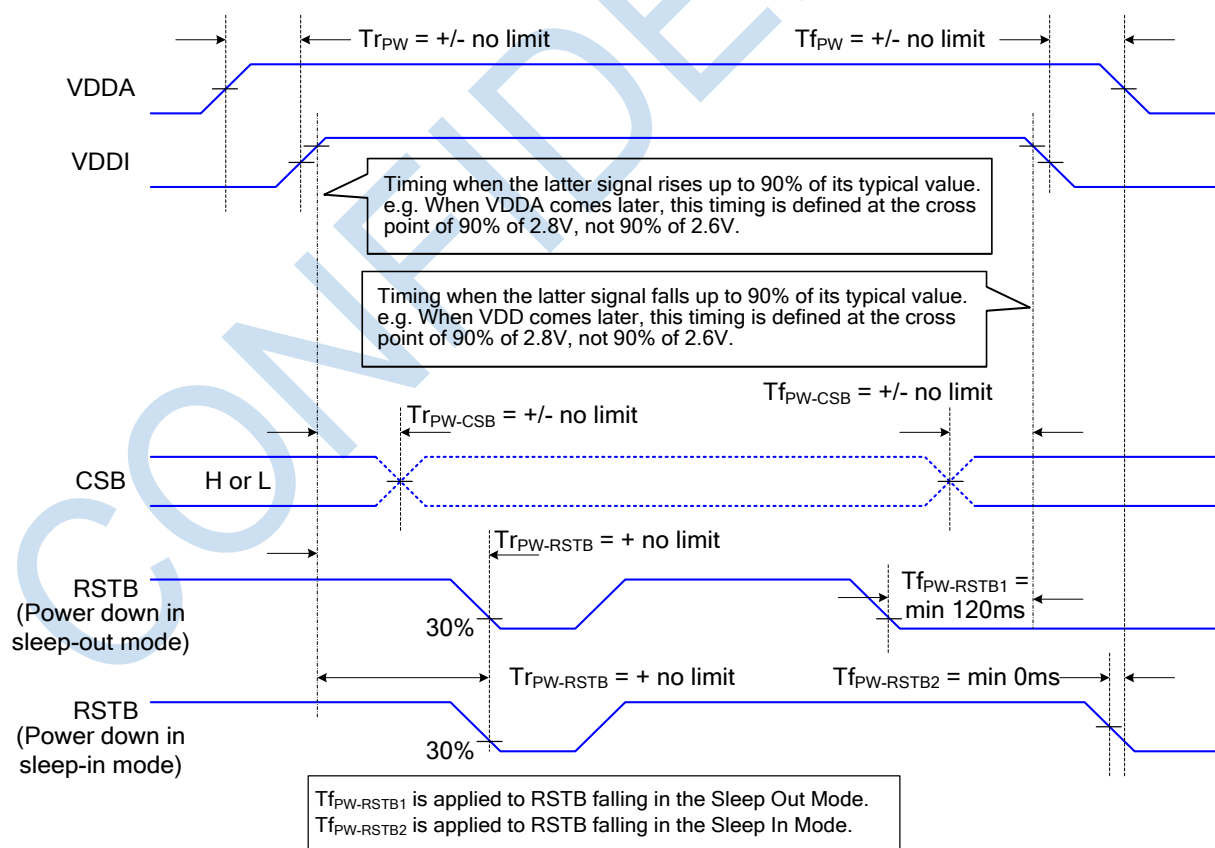
*Note 1: There will be no damage to the display module if the power sequences are not met.*

*Note 2: There will be no abnormal visible effects on the display panel during the Power On/Off Sequences.*

*Note 3: There will be no abnormal visible effects on the display between end of Power On Sequence and before receiving Sleep Out command. Also between receiving Sleep In command and Power Off Sequence.*

*Note 4: If RSTB line is not held stable by host during Power On Sequence as defined in the sequence below, then it will be necessary to apply a Hardware Reset (RSTB) after Host Power On Sequence is complete to ensure correct operation. Otherwise function is not guaranteed.*

The power on/off sequence is illustrated below

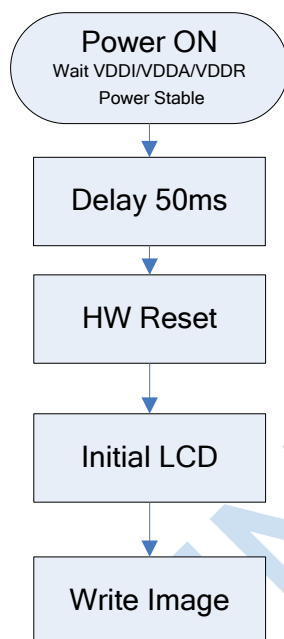




## 7.10 Referential Initial Setup Flow

### 7.10.1 Initial Setting Flow

#### Referential Initial Flow



### 7.10.2 Referential Initial Code

Void Initialization\_ ST7305 (void)

```

{
    Write(Command, 0xD6);    // NVM Load Control
    Write(Data, 0x17);
    Write(Data, 0x02);
    Write(Command, 0xD1);    // Booster Enable
    Write(Data, 0x01);
    Write(Command, 0xC0);    // Gate Voltage Control
    Write(Data, 0x0E)        // VGH=15V
    Write(Data, 0x0A)        // VGL=-10V
    Write(Command, 0xC1);    // VSHP Setting
    Write(Data, 0x41);       // VSHP1=5V
    Write(Data, 0x41);       // VSHP2=5V
    Write(Data, 0x41);       // VSHP3=5V
    Write(Data, 0x41);       // VSHP4=5V
    Write(Command, 0xC2);    // VSLP Setting
    Write(Data, 0x32);       // VSLP1=1V
    Write(Data, 0x32);       // VSLP2=1V
  }
  
```

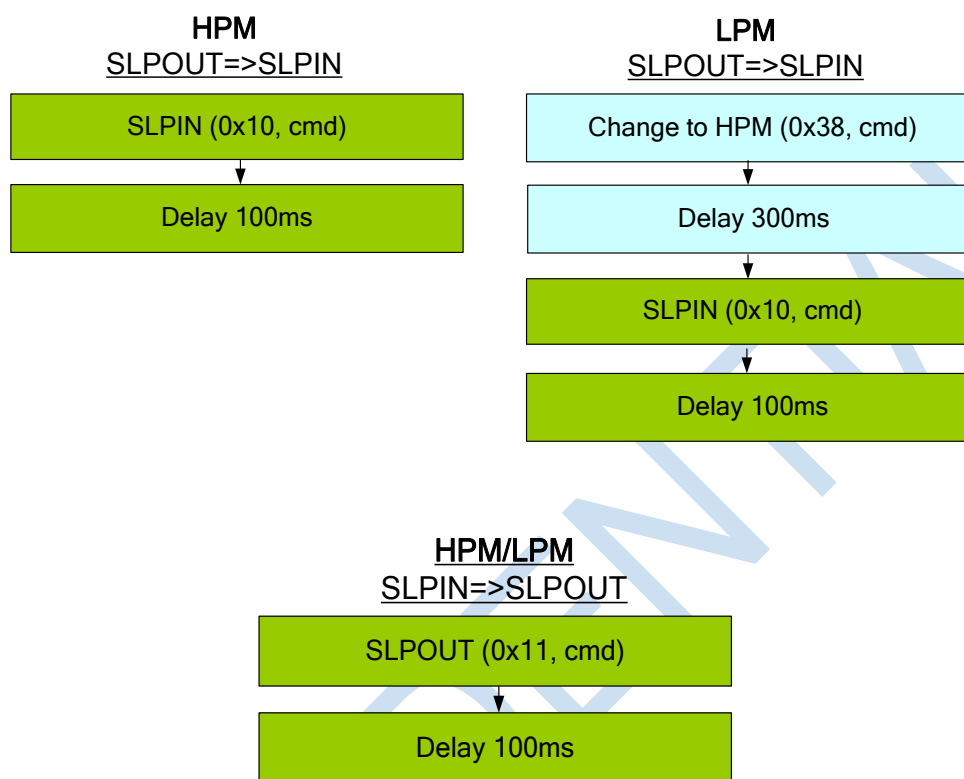
```
Write(Data, 0x32);           //VSLP3=1V
Write(Data, 0x32);           //VSLP4=1V
Write(Command, 0xC4);         //VSHN Setting
Write(Data, 0x46);           //VSHN1=-3.9V
Write(Data, 0x46);           //VSHN2=-3.9V
Write(Data, 0x46);           //VSHN3=-3.9V
Write(Data, 0x46);           //VSHN4=-3.9V
Write(Command, 0xC5);         //VSLN Setting
Write(Data, 0x46);           //VSLN1=-0.4V
Write(Data, 0x46);           //VSLN2=-0.4V
Write(Data, 0x46);           //VSLN3=-0.4V
Write(Data, 0x46);           //VSLN4=-0.4V
Write(Command, 0xB2);         //Frame Rate Control
Write(Data, 0x02);           //HPM=16Hz ; LPM=1Hz
Write(Command, 0xB3);         //Update Period Gate EQ Control in HPM
Write(Data, 0xE5);
Write(Data, 0xF6);
Write(Data, 0x05);           //HPM EQ Control
Write(Data, 0x46);
Write(Data, 0x77);
Write(Data, 0x77);
Write(Data, 0x77);
Write(Data, 0x77);
Write(Data, 0x76);
Write(Data, 0x45);
Write(Command, 0xB4);         //Update Period Gate EQ Control in LPM
Write(Data, 0x05);           //LPM EQ Control
Write(Data, 0x46);
Write(Data, 0x77);
Write(Data, 0x77);
Write(Data, 0x77);
Write(Data, 0x77);
Write(Data, 0x76);
Write(Data, 0x45);
Write(Command, 0x62);         //Gate Timing Control
Write(Data, 0x32);           //Enable/Disable Compensation
Write(Data, 0x03);
```

```
Write(Data, 0x1F);
Write(Command, 0xB7);      //Source EQ Enable
Write(Data, 0x13);
Write(Command, 0xB0);      //Gate Line Setting
Write(Data, 0x66);        //408 line
Write(Command, 0x11);      //Sleep-out
Delay_ms (120);

Write(Command, 0xD8);      //OSC Setting
Write(Data, 0xA6);
Write(Data, 0xE9);
Write(Command, 0xC9);      //Source Voltage Select
Write(Data, 0x00);        //VSH1; VSL1 ; VSHN1 ; VSLN1
Write(Command, 0x36);      //Memory Data Access Control
Write(Data, 0x48);
Write(Command, 0x3A);      //Data Format Select
Write(Data, 0x10);
Write(Command, 0xB8);      //Panel Setting
Write(Data, 0x00);        // 2line interlace and 2line panel layout
Write(Command, 0x2A);      //Column Address Setting
Write(Data, 0x0D);
Write(Data, 0x2E);
Write(Command, 0x2B);      //Row Address Setting
Write(Data, 0x00);
Write(Data, 0xCB);
Write(Command, 0x35);      //TE Setting
Write(Data, 0x00);
Write(Command, 0xD0);      //Enable Auto Power down
Write(Data, 0xFF);
Write(Command, 0x39);      //LPM ON
Write(Command, 0x29);      //Display ON
}
```

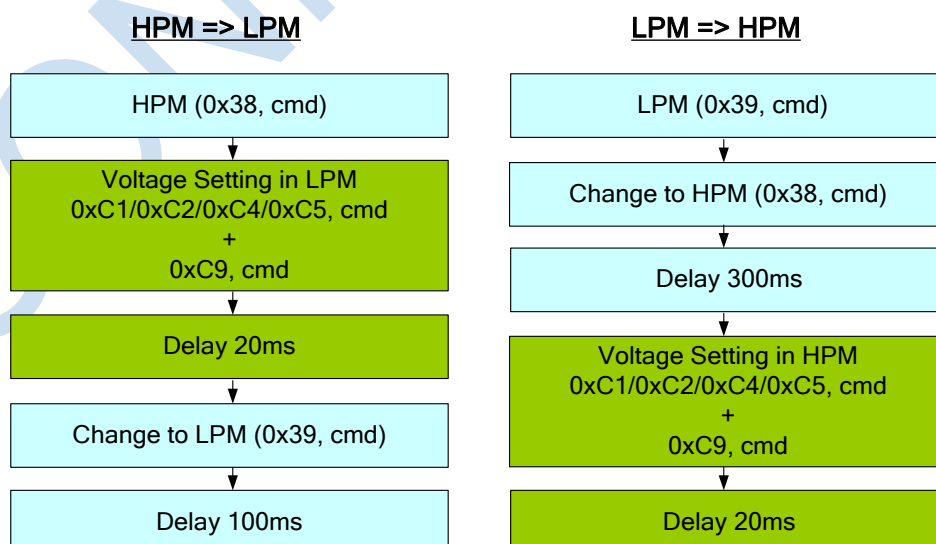
### 7.11 Sleep-In/Out Sequence

Sleep-in/out mode can be used to save power when display function is not required. To enter or exit from Sleep-in/out mode, refer to the following sequence.



### 7.12 Power Mode (HPM/LPM) Sequence

To switch between HPM and LPM, follow the sequences below.



## 8 COMMAND

## 8.1 Command Table1

| Instruction | A0 | R/W | ERD | D7    | D6    | D5    | D4     | D3    | D2    | D1    | D0    | Hex   | Function                             |
|-------------|----|-----|-----|-------|-------|-------|--------|-------|-------|-------|-------|-------|--------------------------------------|
| 1. NOP      | 0  | ↑   | 1   | 0     | 0     | 0     | 0      | 0     | 0     | 0     | 0     | (00h) | No operation                         |
| 2.SWRESET   | 0  | ↑   | 1   | 0     | 0     | 0     | 0      | 0     | 0     | 0     | 1     | (01h) | Software reset                       |
| 3.RDDID     | 0  | ↑   | 1   | 0     | 0     | 0     | 0      | 0     | 1     | 0     | 0     | (04h) | Read display ID                      |
|             | 1  | 1   | ↑   | ID17  | ID16  | ID15  | ID14   | ID13  | ID12  | ID11  | ID10  |       | ID1 read                             |
|             | 1  | 1   | ↑   | ID27  | ID26  | ID25  | ID24   | ID23  | ID22  | ID21  | ID20  |       | ID2 read                             |
|             | 1  | 1   | ↑   | ID37  | ID36  | ID35  | ID34   | ID33  | ID32  | ID31  | ID30  |       | ID3 read                             |
| 4.RDDST     | 0  | ↑   | 1   | 0     | 0     | 0     | 0      | 1     | 0     | 0     | 1     | (09h) | Read display status                  |
|             | 1  | 1   | ↑   | BSTON | MY    | MX    | MV     | 0     | GS    | DO    | 0     |       | Display status1                      |
|             | 1  | 1   | ↑   | 0     | XDE   | 0     | BPS    | LPM   | PTLON | SLOUT | NORON |       | Display status2                      |
|             | 1  | 1   | ↑   | VSON  | 0     | INVON | 0      | 0     | DISON | TEON  | 0     |       | Display status3                      |
| 5.RDDPM     | 0  | ↑   | 1   | 0     | 0     | 0     | 0      | 1     | 0     | 1     | 0     | (0Ah) | Read display Power Mode              |
|             | 1  | 1   | ↑   | BSTON | LPM   | PTLON | SLPOUT | NORON | DISON | 0     | 0     |       | Display Power Status                 |
| 6.SLPIN     | 0  | ↑   | 1   | 0     | 0     | 0     | 1      | 0     | 0     | 0     | 0     | (10h) | Sleep in                             |
| 7.SLPOUT    | 0  | ↑   | 1   | 0     | 0     | 0     | 1      | 0     | 0     | 0     | 1     | (11h) | Sleep out                            |
| 8.PTLON     | 0  | ↑   | 1   | 0     | 0     | 0     | 1      | 0     | 0     | 1     | 0     | (12h) | Partial on (240 duty set)            |
| 9.PTLOFF    | 0  | ↑   | 1   | 0     | 0     | 0     | 1      | 0     | 0     | 1     | 1     | (13h) | Partial off (408 duty set)           |
| 10.INVOFF   | 0  | ↑   | 1   | 0     | 0     | 1     | 0      | 0     | 0     | 0     | 0     | (20h) | Display inversion off                |
| 11.INVON    | 0  | ↑   | 1   | 0     | 0     | 1     | 0      | 0     | 0     | 0     | 1     | (21h) | Display inversion on                 |
| 12.DISPOFF  | 0  | ↑   | 1   | 0     | 0     | 1     | 0      | 1     | 0     | 0     | 0     | (28h) | Display off                          |
| 13.DISPON   | 0  | ↑   | 1   | 0     | 0     | 1     | 0      | 1     | 0     | 0     | 1     | (29h) | Display on                           |
| 14.CASET    | 0  | ↑   | 1   | 0     | 0     | 1     | 0      | 1     | 0     | 1     | 0     | (2Ah) | Column address set                   |
|             | 1  | ↑   | 1   | 0     | 0     | XS5   | XS4    | XS3   | XS2   | XS1   | XS0   |       | X address start: $13 \leq XS \leq X$ |
|             | 1  | ↑   | 1   | 0     | 0     | XE5   | XE4    | XE3   | XE2   | XE1   | XE0   |       | X address end: $S \leq XE \leq X$    |
| 15.RASET    | 0  | ↑   | 1   | 0     | 0     | 1     | 0      | 1     | 0     | 1     | 1     | (2Bh) | Row address set                      |
|             | 1  | ↑   | 1   | YS7   | YS6   | YS5   | YS4    | YS3   | YS2   | YS1   | YS0   |       | Y address start: $0 \leq YS \leq Y$  |
|             | 1  | ↑   | 1   | YE7   | YE6   | YE5   | YE4    | YE3   | YE2   | YE1   | YE0   |       | Y address end: $S \leq YE \leq Y$    |
| 16.RAMWR    | 0  | ↑   | 1   | 0     | 0     | 1     | 0      | 1     | 1     | 0     | 0     | (2Ch) | Memory write                         |
|             | 1  | ↑   | 1   | D1[7] | D1[6] | D1[5] | D1[4]  | D1[3] | D1[2] | D1[1] | D1[0] |       | Write data                           |
| 17.TEOFF    | 0  | ↑   | 1   | 0     | 0     | 1     | 1      | 0     | 1     | 0     | 0     | (34h) | Tearing effect line off              |
| 18.TEON     | 0  | ↑   | 1   | 0     | 0     | 1     | 1      | 0     | 1     | 0     | 1     | (35h) | Tearing effect line on               |
|             | 1  | ↑   | 1   | 0     | 0     | 0     | 0      | 0     | 0     | 0     | TEM   |       |                                      |
| 19.MADCTL   | 0  | ↑   | 1   | 0     | 0     | 1     | 1      | 0     | 1     | 1     | 0     | (36h) | Memory data access control           |

| Instruction | A0 | R/W | ERD | D7    | D6    | D5    | D4    | D3    | D2    | D1    | D0    | Hex   | Function                         |
|-------------|----|-----|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------------------------------|
|             | 1  | ↑   | 1   | MY    | MX    | MV    | 0     | DO    | GS    | 0     | 0     |       |                                  |
| 20.VSCSAD   | 0  | ↑   | 1   | 0     | 0     | 1     | 1     | 0     | 1     | 1     | 1     | (37h) | Vertical scrolling start address |
|             | 1  | ↑   | 1   | 0     | 0     | 0     | 0     | 0     | 0     | 0     | VSP8  |       |                                  |
|             | 1  | ↑   | 1   | VSP7  | VSP6  | VSP5  | VSP4  | VSP3  | VSP2  | VSP1  | VSP0  |       |                                  |
| 21.HPM      | 0  | ↑   | 1   | 0     | 0     | 1     | 1     | 1     | 0     | 0     | 0     | (38h) | High power mode                  |
| 22.LPM      | 0  | ↑   | 1   | 0     | 0     | 1     | 1     | 1     | 0     | 0     | 1     | (39h) | Low power mode                   |
| 23.DTFORM   | 0  | ↑   | 1   | 0     | 0     | 1     | 1     | 1     | 0     | 1     | 0     | (3Ah) | Data format select               |
|             | 1  | ↑   | 1   | 0     | 0     | 0     | XDE   | 0     | 0     | 0     | BPS   |       |                                  |
| 24.RAMWRC   | 0  | ↑   | 1   | 0     | 0     | 1     | 1     | 1     | 1     | 0     | 0     | (3Ch) | Write memory continue            |
|             | 1  | ↑   | 1   | D1[7] | D1[6] | D1[5] | D1[4] | D1[3] | D1[2] | D1[1] | D1[0] |       | Write data                       |
|             | 1  | ↑   | 1   | Dx[7] | Dx[6] | Dx[5] | Dx[4] | Dx[3] | Dx[2] | Dx[1] | Dx[0] |       |                                  |
|             | 1  | ↑   | 1   | Dn[7] | Dn[6] | Dn[5] | Dn[4] | Dn[3] | Dn[2] | Dn[1] | Dn[0] |       |                                  |
|             | 1  | 1   | ↑   | Dn[7] | Dn[6] | Dn[5] | Dn[4] | Dn[3] | Dn[2] | Dn[1] | Dn[0] |       |                                  |
| 25.TESCAN   | 0  | ↑   | 1   | 0     | 1     | 0     | 0     | 0     | 1     | 0     | 0     | (44h) | Set tear scanline                |
|             | 1  | ↑   | 1   | 0     | 0     | 0     | 0     | 0     | 0     | 0     | N8    |       |                                  |
|             | 1  | ↑   | 1   | N7    | N6    | N5    | N4    | N3    | N2    | N1    | N0    |       |                                  |
| 26.RDID1    | 0  | ↑   | 1   | 1     | 1     | 0     | 1     | 1     | 0     | 1     | 0     | (DAh) | Read ID1                         |
|             | 1  | 1   | ↑   | ID17  | ID16  | ID15  | ID14  | ID13  | ID12  | ID11  | ID10  |       | Read parameter                   |
| 27.RDID2    | 0  | ↑   | 1   | 1     | 1     | 0     | 1     | 1     | 0     | 1     | 1     | (DBh) | Read ID2                         |
|             | 1  | 1   | ↑   | ID27  | ID26  | ID25  | ID24  | ID23  | ID22  | ID21  | ID20  |       | Read parameter                   |
| 28.RDID3    | 0  | ↑   | 1   | 1     | 1     | 0     | 1     | 1     | 1     | 0     | 0     | (DCh) | Read ID3                         |
|             | 1  | 1   | ↑   | ID37  | ID36  | ID35  | ID34  | ID33  | ID32  | ID31  | ID30  |       | Read parameter                   |

“-”: Don't care

## 8.1.1 NOP (00h)

| 00H         | NOP (No Operation)             |     |     |    |    |    |    |    |    |    |    |       |
|-------------|--------------------------------|-----|-----|----|----|----|----|----|----|----|----|-------|
| Inst / Para | A0                             | RWR | ERD | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| NOP         | 0                              | ↑   | 1   | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | (00h) |
| Parameter   | No Parameter                   |     |     |    |    |    |    |    |    |    |    |       |
| Description | This command is empty command. |     |     |    |    |    |    |    |    |    |    |       |

## 8.1.2 SWRESET (01h): Software Reset

| 01H         | SWRESET (Software Reset)                                                                                                                                                                                                                                                                                                                                                                            |     |     |    |    |    |    |    |    |    |    |       |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----|----|----|----|----|----|----|----|-------|
| Inst / Para | A0                                                                                                                                                                                                                                                                                                                                                                                                  | RWR | ERD | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| SWRESET     | 0                                                                                                                                                                                                                                                                                                                                                                                                   | ↑   | 1   | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | (01h) |
| Parameter   | No Parameter-                                                                                                                                                                                                                                                                                                                                                                                       |     |     |    |    |    |    |    |    |    |    |       |
| Description | -The display module performs a software reset, registers are written with their SW reset default values.<br>-Frame memory contents are unaffected by this command.                                                                                                                                                                                                                                  |     |     |    |    |    |    |    |    |    |    |       |
| Restriction | It will be necessary to wait 5msec before sending new command following software reset.<br>The display module loads all display suppliers' factory default values to the registers during this 5msec.<br>If software reset is sent during sleep in mode, it will be necessary to wait 120msec before sending sleep out command.<br>Software reset command cannot be sent during sleep out sequence. |     |     |    |    |    |    |    |    |    |    |       |

## 8.1.3 RDDID (04h): Read Display ID

| 04H                         | RDDID (Read Display ID)                                                                                                                                                                                                                                                                                                                                                                                                        |     |     |      |      |      |      |      |      |      |      |       |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|------|------|------|------|------|------|------|------|-------|
| Inst / Para                 | A0                                                                                                                                                                                                                                                                                                                                                                                                                             | RWR | ERD | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX   |
| RDDID                       | 0                                                                                                                                                                                                                                                                                                                                                                                                                              | ↑   | 1   | 0    | 0    | 0    | 0    | 0    | 1    | 0    | 0    | (04h) |
| 1 <sup>st</sup> parameter   | 1                                                                                                                                                                                                                                                                                                                                                                                                                              | 1   | ↑   | ID17 | ID16 | ID15 | ID14 | ID13 | ID12 | ID11 | ID10 |       |
| 2 <sup>nd</sup> parameter   | 1                                                                                                                                                                                                                                                                                                                                                                                                                              | 1   | ↑   | ID27 | ID26 | ID25 | ID24 | ID23 | ID22 | ID21 | ID20 |       |
| 3 <sup>rd</sup> parameter   | 1                                                                                                                                                                                                                                                                                                                                                                                                                              | 1   | ↑   | ID37 | ID36 | ID35 | ID34 | ID33 | ID32 | ID31 | ID30 |       |
| Description                 | -This read byte returns 24-bit display identification information.<br>-The 1 <sup>st</sup> parameter (ID17 to ID10): LCD module's manufacturer ID.<br>-The 2 <sup>nd</sup> parameter (ID26 to ID20): LCD module/driver version ID<br>-The 3 <sup>rd</sup> parameter (ID37 to UD30): LCD module/driver ID.<br>-Commands RDID1/2/3(DAh, DBh, DCh) read data correspond to the parameters 1,2,3 of the command 04h, respectively. |     |     |      |      |      |      |      |      |      |      |       |
| Default Value<br>(D7 to D0) | 1 <sup>st</sup> parameter= 0x00h<br>2 <sup>nd</sup> parameter =0x00h<br>3 <sup>rd</sup> parameter = 0x00h                                                                                                                                                                                                                                                                                                                      |     |     |      |      |      |      |      |      |      |      |       |

## 8.1.4 RDDST (09h): Read Display Status

| 09H                       | RDDST(Read Display Status)                                                                 |                                  |     |       |                                                                                                                                                        |       |     |     |       |       |       |       |  |
|---------------------------|--------------------------------------------------------------------------------------------|----------------------------------|-----|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-------|-------|-------|-------|--|
| Inst / Para               | A0                                                                                         | RWR                              | ERD | D7    | D6                                                                                                                                                     | D5    | D4  | D3  | D2    | D1    | D0    | HEX   |  |
| RDDST                     | 0                                                                                          | ↑                                | 1   | 0     | 0                                                                                                                                                      | 0     | 0   | 1   | 0     | 0     | 1     | (09h) |  |
| 1 <sup>st</sup> parameter | 1                                                                                          | 1                                | ↑   | BSTON | MY                                                                                                                                                     | MX    | MV  | 0   | GS    | DO    | 0     |       |  |
| 2 <sup>nd</sup> parameter | 1                                                                                          | 1                                | ↑   | 0     | XDE                                                                                                                                                    | 0     | BPS | LPM | PTLON | SLOUT | NORON |       |  |
| 3 <sup>rd</sup> parameter | 1                                                                                          | 1                                | ↑   | VSON  | 0                                                                                                                                                      | INVON | 0   | 0   | DISON | TEON  | 0     |       |  |
| Description               | - This command indicates the current status of the display as described in the table below |                                  |     |       |                                                                                                                                                        |       |     |     |       |       |       |       |  |
|                           | Bit                                                                                        | Description                      |     |       | Value                                                                                                                                                  |       |     |     |       |       |       |       |  |
|                           | BSTON                                                                                      | Booster Voltage Status           |     |       | '1' =Booster on, (when BSTEN (D1h) D0='1')<br>'0' =Booster off, (when BSTEN (D1h) D0='0')                                                              |       |     |     |       |       |       |       |  |
|                           | MY                                                                                         | Row Address Order                |     |       | '1' =Decrement, (Bottom to Top, when MADCTL (36h) D7='1')<br>'0' =Increment, (Top to Bottom, when MADCTL (36h) D7='0')                                 |       |     |     |       |       |       |       |  |
|                           | MX                                                                                         | Column Address Order             |     |       | '1' =Decrement, (Right to Left, when MADCTL (36h) D6='1')<br>'0' =Increment, (Left to Right, when MADCTL (36h) D6='0')                                 |       |     |     |       |       |       |       |  |
|                           | MV                                                                                         | Row/Column Exchange              |     |       | '1' = Row/column exchange, (when MADCTL (36h) D5='1')<br>'0' = Normal, (when MADCTL (36h) D5='0')                                                      |       |     |     |       |       |       |       |  |
|                           | DO                                                                                         | Data Order                       |     |       | '0' =Left to right (When MADCTL (36h) D3='1')<br>'1' = Right to left (When MADCTL (36h) D3='0', using in MX=1)                                         |       |     |     |       |       |       |       |  |
|                           | GS                                                                                         | Gate Scan Order                  |     |       | '0' = Data refresh Top to Bottom (when MADCTL (36h) D2='0')<br>'1' = Data refresh Bottom to Top (when MADCTL (36h) D2='1')                             |       |     |     |       |       |       |       |  |
|                           | XDE                                                                                        | Data Up Down Switch              |     |       | '0' = Switch ON(when DTFORM (3Ah) D4='1', Using with MY=1)<br>'1' = Switch OFF (when DTFORM (3Ah) D4='0')                                              |       |     |     |       |       |       |       |  |
|                           | BPS                                                                                        | Bytes Per Pixel Select           |     |       | '0' = 4 writes operations for 24-bit data<br>(when DTFORM (3Ah) D0='0')<br><br>'1' = 3 writes operations for 24-bit data<br>(when DTFORM (3Ah) D0='1') |       |     |     |       |       |       |       |  |
|                           | LPM                                                                                        | Power Mode Select                |     |       | '0' = HPM, '1' = LPM                                                                                                                                   |       |     |     |       |       |       |       |  |
|                           | PTLON                                                                                      | Partial Mode ON/OFF              |     |       | '0' = OFF, '1' = ON                                                                                                                                    |       |     |     |       |       |       |       |  |
|                           | SLPOUT                                                                                     | Sleep In/Out                     |     |       | '0' = In, '1' = Out                                                                                                                                    |       |     |     |       |       |       |       |  |
|                           | NORON                                                                                      | Display Normal Mode On/Off       |     |       | '0' = Scroll or Partial ON, '1' = Normal,                                                                                                              |       |     |     |       |       |       |       |  |
|                           | VSON                                                                                       | Scroll Mode On/Off               |     |       | '0' = Off, '1' = On                                                                                                                                    |       |     |     |       |       |       |       |  |
|                           | INVON                                                                                      | Inversion Status                 |     |       | '0' = Normal, '1' = Inverse                                                                                                                            |       |     |     |       |       |       |       |  |
|                           | DISON                                                                                      | Display On/Off                   |     |       | '1' = On, "0" = Off                                                                                                                                    |       |     |     |       |       |       |       |  |
|                           | TEON                                                                                       | Tearing effect line on/off       |     |       | '1' = On, "0" = Off                                                                                                                                    |       |     |     |       |       |       |       |  |
|                           | Default Value                                                                              | 1 <sup>st</sup> parameter= 0x00h |     |       |                                                                                                                                                        |       |     |     |       |       |       |       |  |
|                           | (D7 to D0)                                                                                 | 2 <sup>nd</sup> parameter =0x40h |     |       |                                                                                                                                                        |       |     |     |       |       |       |       |  |



3<sup>rd</sup> parameter = 0x00h**8.1.5 RDDPM (0Ah): Read Display Power Mode**

| 0AH                         | RDDPM (Read Display Power Mode)                                                             |     |                            |       |     |       |                                                                                           |       |       |    |    |       |
|-----------------------------|---------------------------------------------------------------------------------------------|-----|----------------------------|-------|-----|-------|-------------------------------------------------------------------------------------------|-------|-------|----|----|-------|
| Inst / Para                 | A0                                                                                          | RWR | ERD                        | D7    | D6  | D5    | D4                                                                                        | D3    | D2    | D1 | D0 | HEX   |
| RDDPM                       | 0                                                                                           | ↑   | 1                          | 0     | 0   | 0     | 0                                                                                         | 1     | 0     | 1  | 0  | (0Ah) |
| Parameter                   | 1                                                                                           | 1   | ↑                          | BSTON | LPM | PTLON | SLPOUT                                                                                    | NORON | DISON | 0  | 0  |       |
| Description                 | - This command indicates the current status of the display as described in the table below: |     |                            |       |     |       |                                                                                           |       |       |    |    |       |
|                             | Bit                                                                                         |     | Description                |       |     |       | Value                                                                                     |       |       |    |    |       |
|                             | BSTON                                                                                       |     | Booster Voltage Status     |       |     |       | '1' =Booster on, (when BSTEN (D1h) D0='1')<br>'0' =Booster off, (when BSTEN (D1h) D0='0') |       |       |    |    |       |
|                             | LPM                                                                                         |     | Power Mode Select          |       |     |       | '0' = High Power Mode, '1' = Low Power Mode                                               |       |       |    |    |       |
|                             | PTLON                                                                                       |     | Partial Mode ON/OFF        |       |     |       | '0' = OFF, '1' = ON                                                                       |       |       |    |    |       |
|                             | SLPOUT                                                                                      |     | Sleep In/Out               |       |     |       | '0' = In, '1' = Out                                                                       |       |       |    |    |       |
|                             | NORON                                                                                       |     | Display Normal Mode On/Off |       |     |       | '0' = Scroll or Partial ON, '1' = Normal,                                                 |       |       |    |    |       |
|                             | DISON                                                                                       |     | Display On/Off             |       |     |       | '1' = On, "0" = Off                                                                       |       |       |    |    |       |
| Default Value<br>(D7 to D0) | 0x08h                                                                                       |     |                            |       |     |       |                                                                                           |       |       |    |    |       |

**8.1.6 SLPIN (10h): Sleep in**

| 10H         | SLPIN (Sleep In)                                                                                                                                                                      |     |     |    |    |    |    |    |    |    |    |       |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----|----|----|----|----|----|----|----|-------|
| Inst / Para | A0                                                                                                                                                                                    | RWR | ERD | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| SLPIN       | 0                                                                                                                                                                                     | ↑   | 1   | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 0  | (10h) |
| Parameter   | No Parameter                                                                                                                                                                          |     |     |    |    |    |    |    |    |    |    |       |
| Description | -This command causes the LCD module to enter the minimum power consumption mode.                                                                                                      |     |     |    |    |    |    |    |    |    |    |       |
|             | -In this mode the DC/DC converter is stopped, internal oscillator is stopped, and panel scanning is stopped.                                                                          |     |     |    |    |    |    |    |    |    |    |       |
| Restriction | -MCU interface and memory are still working and the memory keeps its contents.                                                                                                        |     |     |    |    |    |    |    |    |    |    |       |
|             | -This command has no effect when module is already in sleep in mode. Sleep in mode can only be left by the sleep out command (11h).                                                   |     |     |    |    |    |    |    |    |    |    |       |
|             | -It will be necessary to wait 5msec before sending any new commands to a display module following this command to allow time for the supply voltages and clock circuits to stabilize. |     |     |    |    |    |    |    |    |    |    |       |
| Restriction | -It will be necessary to wait 100msec after sending sleep out command (when in sleep in mode) before sending an sleep in command.                                                     |     |     |    |    |    |    |    |    |    |    |       |
|             |                                                                                                                                                                                       |     |     |    |    |    |    |    |    |    |    |       |

**8.1.7 SLPOUT (11h): Sleep Out**

| 11H         | SLPOUT (Sleep Out)                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |     |    |    |    |    |    |    |    |    |       |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----|----|----|----|----|----|----|----|-------|
| Inst / Para | A0                                                                                                                                                                                                                                                                                                                                                                                                                                                                | RWR | ERD | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| SLPOUT      | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ↑   | 1   | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 1  | (11h) |
| Parameter   | No Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |     |    |    |    |    |    |    |    |    |       |
| Description | -This command turn off sleep mode.<br>-In this mode the DC/DC converter is enable, internal display oscillator is started, and panel scanning is started.                                                                                                                                                                                                                                                                                                         |     |     |    |    |    |    |    |    |    |    |       |
| Restriction | -This command has no effect when module is already in sleep in mode. Sleep in mode can only be left by the sleep out command (11h).<br>-It will be necessary to wait 5msec before sending any new commands to a display module following this command to allow time for the supply voltages and clock circuits to stabilize.<br>-It will be necessary to wait 100msec after sending sleep out command (when in sleep in mode) before sending an sleep in command. |     |     |    |    |    |    |    |    |    |    |       |

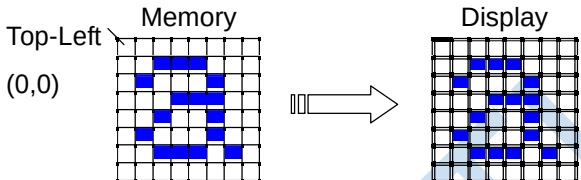
**8.1.8 PTLON (12h): Partial On**

| 12H         | PTLON (Partial On)                                                                                                                                               |     |     |    |    |    |    |    |    |    |    |       |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----|----|----|----|----|----|----|----|-------|
| Inst / Para | A0                                                                                                                                                               | RWR | ERD | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| PTLON       | 0                                                                                                                                                                | ↑   | 1   | 0  | 0  | 0  | 1  | 0  | 0  | 1  | 0  | (12h) |
| Parameter   | No Parameter                                                                                                                                                     |     |     |    |    |    |    |    |    |    |    |       |
| Description | -This command turns on Partial mode. When partial on, display window become 240duty.<br>-To leave Partial mode, the Partial off command (13h) should be written. |     |     |    |    |    |    |    |    |    |    |       |

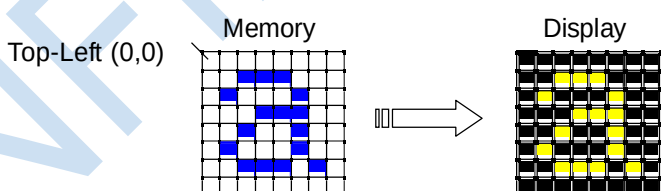
**8.1.9 PTLOFF (13h): Partial Off**

| 13H         | PTLOFF (Partial Off)                                                                                                                                      |     |     |    |    |    |    |    |    |    |    |       |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----|----|----|----|----|----|----|----|-------|
| Inst / Para | A0                                                                                                                                                        | RWR | ERD | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| PTLOFF      | 0                                                                                                                                                         | ↑   | 1   | 0  | 0  | 0  | 1  | 0  | 0  | 1  | 1  | (13h) |
| Parameter   | No Parameter                                                                                                                                              |     |     |    |    |    |    |    |    |    |    |       |
| Description | -This command turns the display to partial off mode. When partial off, display window become 408duty.<br>-Exit from NORON by the partial mode on command. |     |     |    |    |    |    |    |    |    |    |       |

**8.1.10 INVOFF (20h): Display Inversion Off**

| 20H         | INVOFF (Display Inversion Off)                                                                                                                                           |     |     |    |    |    |    |    |    |    |    |       |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----|----|----|----|----|----|----|----|-------|
| Inst / Para | A0                                                                                                                                                                       | RWR | ERD | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| INVOFF      | 0                                                                                                                                                                        | ↑   | 1   | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 0  | (20h) |
| Parameter   | No Parameter                                                                                                                                                             |     |     |    |    |    |    |    |    |    |    |       |
| Description | <p>-This command is used to recover from display inversion mode.</p> <p>(Example)</p>  |     |     |    |    |    |    |    |    |    |    |       |

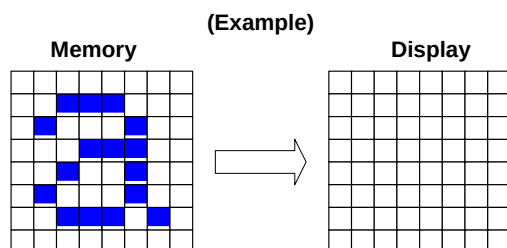
**8.1.11 INVON (21h): Display Inversion On**

| 21H         | INVON (Display Inversion On)                                                                                                                                             |     |     |    |    |    |    |    |    |    |    |       |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----|----|----|----|----|----|----|----|-------|
| Inst / Para | A0                                                                                                                                                                       | RWR | ERD | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| INVON       | 0                                                                                                                                                                        | ↑   | 1   | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 1  | (21h) |
| Parameter   | No Parameter                                                                                                                                                             |     |     |    |    |    |    |    |    |    |    |       |
| Description | <p>-This command is used to enter into display inversion mode.</p> <p>(Example)</p>  |     |     |    |    |    |    |    |    |    |    |       |

**8.1.12 DISPOFF (28h): Display Off**

| 28H         | DISPOFF (Display Off)                                                                                                                                                                                                                                                    |     |     |    |    |    |    |    |    |    |    |       |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----|----|----|----|----|----|----|----|-------|
| Inst / Para | A0                                                                                                                                                                                                                                                                       | RWR | ERD | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| DISPOFF     | 0                                                                                                                                                                                                                                                                        | ↑   | 1   | 0  | 0  | 1  | 0  | 1  | 0  | 0  | 0  | (28h) |
| Parameter   | No Parameter                                                                                                                                                                                                                                                             |     |     |    |    |    |    |    |    |    |    |       |
| Description | <p>- This command is used to enter into DISPLAY OFF mode. In this mode, the output from Frame Memory is disabled and blank page inserted.</p> <p>- This command makes no change of contents of frame memory.</p> <p>- This command does not change any other status.</p> |     |     |    |    |    |    |    |    |    |    |       |

- There will be no abnormal visible effect on the display.
- Exit from this command by Display On (29h)



### 8.1.13 DISPON (29h): Display On

| 29H         | DISPON (Display On)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |     |    |    |    |    |    |    |    |    |       |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----|----|----|----|----|----|----|----|-------|
| Inst / Para | A0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | RWR | ERD | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| DISPON      | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | ↑   | 1   | 0  | 0  | 1  | 0  | 1  | 0  | 0  | 1  | (29h) |
| Parameter   | No Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |     |    |    |    |    |    |    |    |    |       |
| Description | <ul style="list-style-type: none"> <li>- This command is used to recover from DISPLAY OFF mode.</li> <li>- Output from the Frame Memory is enabled.</li> <li>- This command makes no change of contents of frame memory.</li> <li>- This command does not change any other status.</li> </ul> <p style="text-align: center;"><b>(Example)</b></p> <div style="display: flex; justify-content: space-around; align-items: center;"> <div style="text-align: center;"> <p><b>Memory</b></p> </div> <div style="font-size: 2em;">→</div> <div style="text-align: center;"> <p><b>Display</b></p> </div> </div> |     |     |    |    |    |    |    |    |    |    |       |

### 8.1.14 CASET (2Ah): Column Address Set

| 2AH                       | CASET (Column Address Set)                                                                                                                                                                        |     |     |    |    |     |     |     |     |     |     |       |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----|----|-----|-----|-----|-----|-----|-----|-------|
| Inst / Para               | A0                                                                                                                                                                                                | RWR | ERD | D7 | D6 | D5  | D4  | D3  | D2  | D1  | D0  | HEX   |
| CASET                     | 0                                                                                                                                                                                                 | ↑   | 1   | 0  | 0  | 1   | 0   | 1   | 0   | 1   | 0   | (2Ah) |
| 1 <sup>st</sup> parameter | 1                                                                                                                                                                                                 | ↑   | 1   | 0  | 0  | XS5 | XS4 | XS3 | XS2 | XS1 | XS0 |       |
| 2 <sup>th</sup> parameter | 1                                                                                                                                                                                                 | ↑   | 1   | 0  | 0  | XE5 | XE4 | XE3 | XE2 | XE1 | XE0 |       |
| Description               | <ul style="list-style-type: none"> <li>-The value of XS [5:0] and XE [5:0] are referred when RAMWR command comes.</li> <li>-Each value represents one column line in the Frame Memory.</li> </ul> |     |     |    |    |     |     |     |     |     |     |       |

|                             |                                                                                                                                                                                                                                               |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             |                                                                                                                                                                                                                                               |
| Restriction                 | <p>XS [5:0] always must be equal to or less than XE [5:0]</p> <p>When XS [5:0] or XE [5:0] is greater than maximum address like below, data of out of range will be ignored.(Parameter range: 13(0Dh) ≤ XS [5:0] &lt; XE [5:0] ≤ 46 (2Eh)</p> |
| Default Value<br>(D7 to D0) | <p>1<sup>st</sup> parameter= 0x00h</p> <p>2<sup>nd</sup> parameter =0x00h</p>                                                                                                                                                                 |

### 8.1.15 RASET (2Bh): Row Address Set

| 2BH                         | RASET (Row Address Set)                                                                                                                                                                                                                   |     |     |     |     |     |     |     |     |     |     |       |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-------|
| Inst / Para                 | A0                                                                                                                                                                                                                                        | RWR | ERD | D7  | D6  | D5  | D4  | D3  | D2  | D1  | D0  | HEX   |
| RASET                       | 0                                                                                                                                                                                                                                         | ↑   | 1   | 0   | 0   | 1   | 0   | 1   | 0   | 1   | 1   | (2Bh) |
| 1 <sup>nd</sup> parameter   | 1                                                                                                                                                                                                                                         | ↑   | 1   | YS7 | YS6 | YS5 | YS4 | YS3 | YS2 | YS1 | YS0 |       |
| 2 <sup>th</sup> parameter   | 1                                                                                                                                                                                                                                         | ↑   | 1   | YE7 | YE6 | YE5 | YE4 | YE3 | YE2 | YE1 | YE0 |       |
| Description                 | <p>-This command is used to defined area of frame memory where MCU can access.</p> <p>-The value of YS [7:0] and YE [7:0] are referred when RAMWR command comes.</p> <p>-Each value represents one page line in the Frame Memory.</p>     |     |     |     |     |     |     |     |     |     |     |       |
| Restriction                 | <p>YS [7:0] always must be equal to or less than YE [15:0]</p> <p>When YS [7:0] or YE [7:0] is greater than maximum address like below, data of out of range will be ignored.(Parameter range: 0 ≤ YS [7:0] &lt; YE [7:0] ≤ 203 (CBh)</p> |     |     |     |     |     |     |     |     |     |     |       |
| Default Value<br>(D7 to D0) | <p>1<sup>st</sup> parameter= 0x00h</p> <p>2<sup>nd</sup> parameter =0x00h</p>                                                                                                                                                             |     |     |     |     |     |     |     |     |     |     |       |

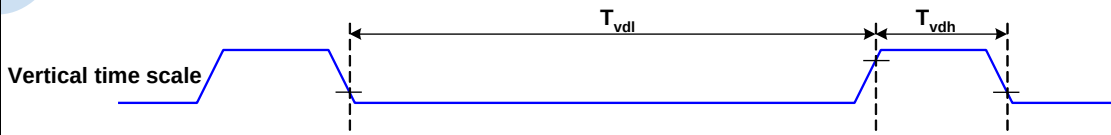
## 8.1.16 RAMWR (2Ch): Memory Write

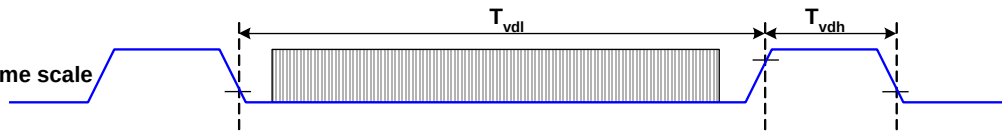
| 2CH                       | RAMWR (Memory Write)                                                                                                                                                                                                                                                                                                                                                |     |     |       |       |       |       |       |       |       |       |       |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Inst / Para               | A0                                                                                                                                                                                                                                                                                                                                                                  | RWR | ERD | D7    | D6    | D5    | D4    | D3    | D2    | D1    | D0    | HEX   |
| RAMWR                     | 0                                                                                                                                                                                                                                                                                                                                                                   | ↑   | 1   | 0     | 0     | 1     | 0     | 1     | 1     | 0     | 0     | (2Ch) |
| 1 <sup>st</sup> parameter | 1                                                                                                                                                                                                                                                                                                                                                                   | ↑   | 1   | D1[7] | D1[6] | D1[5] | D1[4] | D1[3] | D1[2] | D1[1] | D1[0] |       |
| ...                       | 1                                                                                                                                                                                                                                                                                                                                                                   | ↑   | 1   | Dx[7] | Dx[6] | Dx[5] | Dx[4] | Dx[3] | Dx[2] | Dx[1] | Dx[0] |       |
| N parameter               | 1                                                                                                                                                                                                                                                                                                                                                                   | ↑   | 1   | Dn[7] | Dn[6] | Dn[5] | Dn[4] | Dn[3] | Dn[2] | Dn[1] | Dn[0] |       |
| Description               | <p>-This command is used to transfer data from MCU to frame memory.</p> <p>-When this command is accepted, the column register and the page register are reset to the start column/start page positions.</p> <p>-The start column/start page positions are different in accordance with MADCTL setting.</p> <p>-Sending any other command can stop frame write.</p> |     |     |       |       |       |       |       |       |       |       |       |

## 8.1.17 TEOFF (34h): Tearing Effect Line OFF

| 34H         | TEOFF (Tearing Effect Line OFF)                                                                          |     |     |    |    |    |    |    |    |    |    |       |
|-------------|----------------------------------------------------------------------------------------------------------|-----|-----|----|----|----|----|----|----|----|----|-------|
| Inst / Para | A0                                                                                                       | RWR | ERD | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| TEOFF       | 0                                                                                                        | ↑   | 1   | 0  | 0  | 1  | 1  | 0  | 1  | 0  | 0  | (34h) |
| Parameter   | No Parameter                                                                                             |     |     |    |    |    |    |    |    |    |    |       |
| Description | -This command is used to turn OFF (Active Low) the Tearing Effect output signal from the TE signal line. |     |     |    |    |    |    |    |    |    |    |       |

## 8.1.18 TEON (35h): Tearing Effect Line On

| 35H         | TEON (Tearing Effect Line On)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |     |    |    |    |    |    |    |    |     |       |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----|----|----|----|----|----|----|-----|-------|
| Inst / Para | A0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | RWR | ERD | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0  | HEX   |
| TEON        | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ↑   | 1   | 0  | 0  | 1  | 1  | 0  | 1  | 0  | 1   | (35h) |
| Parameter   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ↑   | 1   | 0  | 0  | 0  | 0  | 0  | 0  | 0  | TEM |       |
| Description | <p>-This command is used to turn ON the Tearing Effect output signal from the TE signal line.</p> <p>-The Tearing Effect Line On has one parameter, which describes the mode of the Tearing Effect Output Line:</p> <p>-When TEM = '0': The Tearing Effect output line consists of V-Blanking information only</p> <p>Vertical time scale </p> <p>-When TEM = '1': The Tearing Effect output Line consists of both V-Blanking and H-Blanking information</p> |     |     |    |    |    |    |    |    |    |     |       |

|                             |                                                                                                                                                                                                                            |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             |  <p>Vertical time scale</p> <p>Note: During Sleep In Mode with Tearing Effect Line On, Tearing Effect Output pin will be active Low.</p> |
| Default Value<br>(D7 to D0) | 0x00h                                                                                                                                                                                                                      |

### 8.1.19 MADCTL (36h): Memory Data Access Control

| 36H                                                      | MADCTL (Memory Data Access Control)                                                                                                                                                                                                                                                                                                                                  |      |                      |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------------------|----|----|----|----|----|----|----|----|-------|-----|------|-------------|----|----|--------------------|----|----|----------------------|----|----|-------------------|----|----|-------------------|----|----|-----------------|
| Inst / Para                                              | A0                                                                                                                                                                                                                                                                                                                                                                   | RWR  | ERD                  | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
| MADCTL                                                   | 0                                                                                                                                                                                                                                                                                                                                                                    | ↑    | 1                    | 0  | 0  | 1  | 1  | 0  | 1  | 1  | 0  | (36h) |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
| Parameter                                                | 1                                                                                                                                                                                                                                                                                                                                                                    | ↑    | 1                    | MY | MX | MV | 0  | DO | GS | 0  | 0  |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
| Description                                              | -This command defines read/ write scanning direction of frame memory.                                                                                                                                                                                                                                                                                                |      |                      |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
|                                                          | <table><tr><th>Bit</th><th>NAME</th><th>DESCRIPTION</th></tr><tr><td>D7</td><td>MY</td><td>Page Address Order</td></tr><tr><td>D6</td><td>MX</td><td>Column Address Order</td></tr><tr><td>D5</td><td>MV</td><td>Page/Column Order</td></tr><tr><td>D3</td><td>DO</td><td>Data Output Order</td></tr><tr><td>D2</td><td>GS</td><td>Gate Scan Order</td></tr></table> |      |                      |    |    |    |    |    |    |    |    |       | Bit | NAME | DESCRIPTION | D7 | MY | Page Address Order | D6 | MX | Column Address Order | D5 | MV | Page/Column Order | D3 | DO | Data Output Order | D2 | GS | Gate Scan Order |
|                                                          | Bit                                                                                                                                                                                                                                                                                                                                                                  | NAME | DESCRIPTION          |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
|                                                          | D7                                                                                                                                                                                                                                                                                                                                                                   | MY   | Page Address Order   |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
|                                                          | D6                                                                                                                                                                                                                                                                                                                                                                   | MX   | Column Address Order |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
|                                                          | D5                                                                                                                                                                                                                                                                                                                                                                   | MV   | Page/Column Order    |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
|                                                          | D3                                                                                                                                                                                                                                                                                                                                                                   | DO   | Data Output Order    |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
|                                                          | D2                                                                                                                                                                                                                                                                                                                                                                   | GS   | Gate Scan Order      |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
|                                                          | -Bit Assignment                                                                                                                                                                                                                                                                                                                                                      |      |                      |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
|                                                          | <b>Bit D7- Page Address Order (MY)</b>                                                                                                                                                                                                                                                                                                                               |      |                      |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
|                                                          | "0" = Top to Bottom (When MADCTL D7="0").                                                                                                                                                                                                                                                                                                                            |      |                      |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
|                                                          | "1" = Bottom to Top (When MADCTL D7="1").                                                                                                                                                                                                                                                                                                                            |      |                      |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
|                                                          | <b>Bit D6- Column Address Order (MX)</b>                                                                                                                                                                                                                                                                                                                             |      |                      |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
|                                                          | "0" = Left to Right (When MADCTL D6="0").                                                                                                                                                                                                                                                                                                                            |      |                      |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
|                                                          | "1" = Right to Left (When MADCTL D6="1").                                                                                                                                                                                                                                                                                                                            |      |                      |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
| <b>Bit D5- Page/Column Order (MV)</b>                    |                                                                                                                                                                                                                                                                                                                                                                      |      |                      |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
| "0" = Column Direction Mode (When MADCTL D5="0").        |                                                                                                                                                                                                                                                                                                                                                                      |      |                      |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
| "1" = Page Direction Mode (When MADCTL D5="1")           |                                                                                                                                                                                                                                                                                                                                                                      |      |                      |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
| <b>Bit D3- Data Order (DO)</b>                           |                                                                                                                                                                                                                                                                                                                                                                      |      |                      |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
| "0" = Left to Right (When MADCTL D3="0" , using in MX=0) |                                                                                                                                                                                                                                                                                                                                                                      |      |                      |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
| "1" = Right to Left (When MADCTL D3="1" , using in MX=1) |                                                                                                                                                                                                                                                                                                                                                                      |      |                      |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
| <b>Bit D2- Gate Scan Order (GS)</b>                      |                                                                                                                                                                                                                                                                                                                                                                      |      |                      |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |
| "0" =Data refresh Top to Bottom (When MADCTL D2="0")     |                                                                                                                                                                                                                                                                                                                                                                      |      |                      |    |    |    |    |    |    |    |    |       |     |      |             |    |    |                    |    |    |                      |    |    |                   |    |    |                   |    |    |                 |

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | <p>"0" =Data refresh Bottom to Top (When MADCTL D2="1")</p> <p><b>Gate Scan Order (GS)</b></p> <div style="display: flex; justify-content: space-around;"> <div style="text-align: center;"> <p>Top-left (0, 0)</p> <p>GS="0"</p> </div> <div style="text-align: center;"> <p>Top-left (0, 0)</p> <p>GS="1"</p> </div> </div> <div style="display: flex; justify-content: space-around; margin-top: 20px;"> <div style="text-align: center;"> <p>Sent First</p> <p>Sent 2nd</p> <p>Sent 3rd</p> <p>...</p> <p>Sent Last</p> </div> <div style="text-align: center;"> <p>Top-left (0, 0)</p> <p>Display</p> </div> </div> |
| Default Value<br>(D7 to D0) | 0x00h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

### 8.1.20VSCSAD (37h): Vertical Scroll Start Address of RAM

| 37H                       | VSCSAD (Vertical Scroll Address of RAM) |     |     |      |      |      |      |      |      |      |      |       |
|---------------------------|-----------------------------------------|-----|-----|------|------|------|------|------|------|------|------|-------|
| Inst / Para               | A0                                      | RWR | ERD | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX   |
| VSCSAD                    | 0                                       | ↑   | 1   | 0    | 0    | 1    | 1    | 0    | 1    | 1    | 1    | (37h) |
| 1 <sup>st</sup> parameter | 1                                       | ↑   | 1   | 0    | 0    | 0    | 0    | 0    | 0    | 0    | VSP8 |       |
| 2 <sup>ND</sup> parameter | 1                                       | ↑   | 1   | VSP7 | VSP6 | VSP5 | VSP4 | VSP3 | VSP2 | VSP1 | VSP0 |       |

- The Vertical Scrolling Start Address command has two parameters which describe which line in the Frame Memory will be written as the first line after the last line on the display as illustrated below: .

Example:

When partial off (408 Duty) and VSP = '2' ( 0<=VSP[8:0] <=407 )

Available Scroll Area

Memory

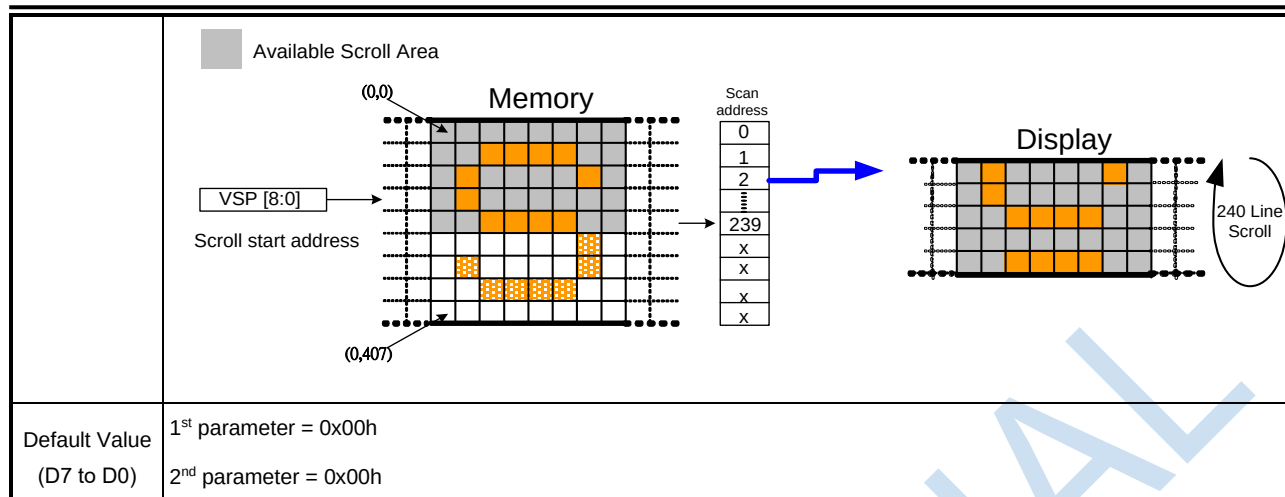
Scan address

|     |
|-----|
| 0   |
| 1   |
| 2   |
| 3   |
| ... |
| 406 |
| 407 |

Display

When partial on (240 Duty) and VSP = '2' ( 0<=VSP[8:0] <=239 )





## 8.1.21 HPM (38h): High Power Mode ON

| 38H         | HPM (High Power Mode ON)                                                                                                                                                                                                                 |     |     |    |    |    |    |    |    |    |    |       |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----|----|----|----|----|----|----|----|-------|
| Inst / Para | A0                                                                                                                                                                                                                                       | RWR | ERD | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| HPM         | 0                                                                                                                                                                                                                                        | ↑   | 1   | 0  | 0  | 1  | 1  | 1  | 0  | 0  | 0  | (38h) |
| Parameter   | No Parameter                                                                                                                                                                                                                             |     |     |    |    |    |    |    |    |    |    |       |
| Description | -This command is used to switch display drive mode to high power mode. In HPM, IC drivers LCD with frame rate (16 / 32) that set by <b>FRCTRL (0xB2h)</b> .<br>-Default setting of IC is HPM.<br>-After reset, it is in high power mode. |     |     |    |    |    |    |    |    |    |    |       |

## 8.1.22 LPM (39h): Low Power Mode ON

| 39H         | LPM (Low Power Mode ON)                                                                                                                                            |     |     |    |    |    |    |    |    |    |    |       |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----|----|----|----|----|----|----|----|-------|
| Inst / Para | A0                                                                                                                                                                 | RWR | ERD | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| LPM         | 0                                                                                                                                                                  | ↑   | 1   | 0  | 0  | 1  | 1  | 1  | 0  | 0  | 1  | (39h) |
| Parameter   | No Parameter                                                                                                                                                       |     |     |    |    |    |    |    |    |    |    |       |
| Description | -This command is used to switch display drive mode to low power mode. In LPM, IC drivers LCD with frame rate (1 to 8Hz) that set by <b>FRCTRL (0xB2h)</b> command. |     |     |    |    |    |    |    |    |    |    |       |

## 8.1.23 DTFORM (3Ah): Data Format Select

| 3AH         | DTFORM (Data Format Select)               |     |     |    |    |    |     |    |    |    |     |       |
|-------------|-------------------------------------------|-----|-----|----|----|----|-----|----|----|----|-----|-------|
| Inst / Para | A0                                        | RWR | ERD | D7 | D6 | D5 | D4  | D3 | D2 | D1 | D0  | HEX   |
| DTFORM      | 0                                         | ↑   | 1   | 0  | 0  | 1  | 1   | 1  | 0  | 1  | 0   | (3Ah) |
| Parameter   | 1                                         | ↑   | 1   | 0  | 0  | 0  | XDE | 0  | 0  | 0  | BPS |       |
| Description | -This command defines data format select. |     |     |    |    |    |     |    |    |    |     |       |

| Bit | NAME | DESCRIPTION            |
|-----|------|------------------------|
| D4  | XDE  | Data Up Down Switch    |
| D0  | BPS  | Bytes Per Pixel Select |

**Bit D4- Data Up Down Switch (XDE)**

"0" = Switch ON (Using with MY=1)

"1" = Switch OFF

**Bit D0- Bytes Per Pixel Select (BPS)**

"0" = 4 write operations for 24-bit data

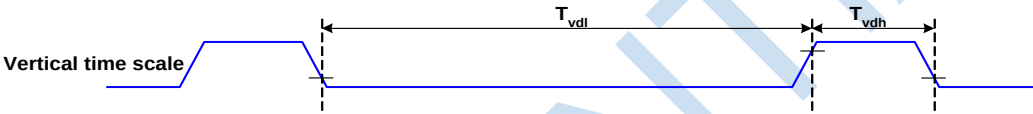
"1" = 3 write operations for 24-bit data

|                             |       |
|-----------------------------|-------|
| Default Value<br>(D7 to D0) | 0x00h |
|-----------------------------|-------|

### 8.1.24 WRMEMC (3Ch): Write Memory Continue

| 3CH                       | WRMEMC (Write Memory Continue)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |     |       |       |       |       |       |       |       |       |       |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Inst / Para               | A0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | RWR | ERD | D7    | D6    | D5    | D4    | D3    | D2    | D1    | D0    | HEX   |
| WRMEMC                    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ↑   | 1   | 0     | 0     | 1     | 1     | 1     | 1     | 0     | 0     | (3Ch) |
| 1 <sup>ST</sup> parameter | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ↑   | 1   | D1[7] | D1[6] | D1[5] | D1[4] | D1[3] | D1[2] | D1[1] | D1[0] |       |
| ⋮                         | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ↑   | 1   | Dx[7] | Dx[6] | Dx[5] | Dx[4] | Dx[3] | Dx[2] | Dx[1] | Dx[0] |       |
| N <sup>th</sup> parameter | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ↑   | 1   | Dn[7] | Dn[6] | Dn[5] | Dn[4] | Dn[3] | Dn[2] | Dn[1] | Dn[0] |       |
| Description               | -This command transfers image data from the host processor to the display module's frame memory continuing from the pixel location following the previous write memory continue or memory write command.                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |     |       |       |       |       |       |       |       |       |       |
|                           | -If MV=0:<br><br>Data is written continuing from the pixel location after the write range of the previous memory write or write memory continue. The column register is then incremented and pixels are written to the frame memory until the column register equals the end column (XE) value. The column register is then reset to XS and the page register is incremented. Pixels are written to the frame memory until the page register equals the end page (YE) value and the column register equals the XE value, or the host processor sends another command. If the number of pixels exceeds (XE-XS+1)*(YE-YS+1) the extra pixels are ignored. |     |     |       |       |       |       |       |       |       |       |       |
|                           | If MV=1:<br><br>Data is written continuing from the pixel location after the write range of the previous memory write or write memory continue. The page register is then incremented and pixels are written to the frame memory until the page register equals the end page (YE) value. The page register is then reset to YS and the column register is incremented. Pixels are written to the frame memory until the column register equals the end column (XE) value and the page register equals the YE value, or the host processor sends another command. If the number of pixels exceeds (XE-XS+1)*(YE-YS+1) the extra pixels are ignored.      |     |     |       |       |       |       |       |       |       |       |       |

## 8.1.25 TESCOAN (44h): Set Tear Scanline

| 44H                       | TESCAN (Set Tear Scanline)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |     |    |    |    |    |    |    |    |    |       |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----|----|----|----|----|----|----|----|-------|
| Inst / Para               | A0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | RWR | ERD | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| TESCAN                    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ↑   | 1   | 0  | 1  | 0  | 0  | 0  | 1  | 0  | 0  | (44h) |
| 1 <sup>st</sup> Parameter | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ↑   | 1   | 0  | 0  | 0  | 0  | 0  | 0  | 0  | N8 |       |
| 2 <sup>nd</sup> parameter | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ↑   | 1   | N7 | N6 | N5 | N4 | N3 | N2 | N1 | N0 |       |
| Description               | <p>-This command turns on the display module's Tearing Effect output signal on the TE signal line when the display module reaches line N. The TE signal is not affected by changing MV.</p> <p>-The tearing effect line on has one parameter that describes the tearing effect output line mode.</p> <p>-The tearing effect output line consist of V-blanking information only.</p>  <p>Note that set tear scanline with N=0 is equivalent to tearing effect line on with TEM=0.</p> <p>The tearing effect output line shall be active low when the display module is in sleep mode</p> |     |     |    |    |    |    |    |    |    |    |       |
| Default Value (D7 to D0)  | <p>1<sup>st</sup> parameter = 0x00h</p> <p>2<sup>nd</sup> parameter = 0x00h</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |     |    |    |    |    |    |    |    |    |       |

## 8.1.26 RDID1 (DAh): Read ID1

| DAH         | RDID1 (Read ID1)                                            |     |     |      |      |      |      |      |      |      |      |       |
|-------------|-------------------------------------------------------------|-----|-----|------|------|------|------|------|------|------|------|-------|
| Inst / Para | A0                                                          | RWR | ERD | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX   |
| RDID1       | 0                                                           | ↑   | 1   | 1    | 1    | 0    | 1    | 1    | 0    | 1    | 0    | (DAh) |
| Parameter   | 1                                                           | 1   | ↑   | ID17 | ID16 | ID15 | ID14 | ID13 | ID12 | ID11 | ID10 |       |
| Description | -This read byte is used to track the LCD module IC version. |     |     |      |      |      |      |      |      |      |      |       |

## 8.1.27 RDID2 (DBh): Read ID2

| DBH         | RDID2 (Read ID2)                                                   |     |     |      |      |      |      |      |      |      |      |       |
|-------------|--------------------------------------------------------------------|-----|-----|------|------|------|------|------|------|------|------|-------|
| Inst / Para | A0                                                                 | RWR | ERD | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX   |
| RDID2       | 0                                                                  | ↑   | 1   | 1    | 1    | 0    | 1    | 1    | 0    | 1    | 1    | (DBh) |
| Parameter   | 1                                                                  | 1   | ↑   | ID27 | ID26 | ID25 | ID24 | ID23 | ID22 | ID21 | ID20 |       |
| Description | -This read byte is used to track the LCD module/driver IC version. |     |     |      |      |      |      |      |      |      |      |       |

**8.1.28 RDID3 (DCh): Read ID3**

| DCH         | RDID3 (Read ID3)                                  |     |     |      |      |      |      |      |      |      |      |       |
|-------------|---------------------------------------------------|-----|-----|------|------|------|------|------|------|------|------|-------|
| Inst / Para | A0                                                | RWR | ERD | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX   |
| RDID3       | 0                                                 | ↑   | 1   | 1    | 1    | 0    | 1    | 1    | 1    | 0    | 0    | (DCh) |
| Parameter   | 1                                                 | 1   | ↑   | ID37 | ID36 | ID35 | ID34 | ID33 | ID32 | ID31 | ID30 |       |
| Description | -This read byte identifies the LCD module/driver. |     |     |      |      |      |      |      |      |      |      |       |

CONFIDENTIAL

## 8.2 Command Table2

| Instruction | A0 | R/W | ERD | D7       | D6            | D5        | D4        | D3       | D2            | D1       | D0       | Hex   | Function                                         |
|-------------|----|-----|-----|----------|---------------|-----------|-----------|----------|---------------|----------|----------|-------|--------------------------------------------------|
| 1. GTCON    | 0  | ↑   | 1   | 0        | 1             | 1         | 0         | 0        | 0             | 1        | 0        | (62h) | Gate<br>Timing<br>Control                        |
|             | 1  | ↑   | 1   | 0        | 0             | CMON      | CMODE     | GAB[3]   | GAB[2]        | GAB[1]   | GAB[0]   |       |                                                  |
|             | 1  | ↑   | 1   | 0        | 0             | CLKDIV[1] | CLKDIV[0] | FCMPA[3] | FCMPA[2]      | FCMPA[1] | FCMPA[0] |       |                                                  |
|             | 1  | ↑   | 1   | CMPWA[7] | CMPWA[6]      | CMPWA[5]  | CMPWA[4]  | CMPWA[3] | CMPWA[2]      | CMPWA[1] | CMPWA[0] |       |                                                  |
| 2. GATESET  | 0  | ↑   | 1   | 1        | 0             | 1         | 1         | 0        | 0             | 0        | 0        | (B0h) | Gate                                             |
|             | 1  | ↑   | 1   | 0        | GL[6]         | GL[5]     | GL[4]     | GL[3]    | GL[2]         | GL[1]    | GL[0]    |       | Setting                                          |
| 3. FSTCOM   | 0  | ↑   | 1   | 1        | 0             | 1         | 1         | 0        | 0             | 0        | 1        | (B1h) | First                                            |
|             | 1  | ↑   | 1   | 0        | 0             | 0         | 0         | 0        | 0             | 0        | FST[8]   |       | Gate                                             |
|             | 1  | ↑   | 1   | FST[7]   | FST[6]        | FST[5]    | FST[4]    | FST[3]   | FST[2]        | FST[1]   | FST[0]   |       | Setting                                          |
| 4. FRCTRL   | 0  | ↑   | 1   | 1        | 0             | 1         | 1         | 0        | 0             | 1        | 0        | (B2h) | FR                                               |
|             | 1  | ↑   | 1   | 0        | 0             | 0         | HFRA      | 0        | LFRA[2]       | LFRA[1]  | LFRA[0]  |       | Control                                          |
| 5. GTUPEQH  | 0  | ↑   | 1   | 1        | 0             | 1         | 1         | 0        | 0             | 1        | 1        | (B3h) | Update<br>Period<br>Gate EQ<br>Control<br>In HPM |
|             | 1  | ↑   | 1   | 1        | GEQEN         | 1         | 1         | 0        | 1             | 0        | 1        |       |                                                  |
|             | 1  | ↑   | 1   | 1        | 1             | 1         | 1         | 0        | 1             | 1        | 0        |       |                                                  |
|             | 1  | ↑   | 1   | 0        | GPCHPU0[2:0]  |           |           | 0        | GPCHPU1[2:0]  |          |          |       |                                                  |
|             | 1  | ↑   | 1   | 0        | GPCHPU2[2:0]  |           |           | 0        | GPCHPU3[2:0]  |          |          |       |                                                  |
|             | 1  | ↑   | 1   | 0        | GPCHPU4[2:0]  |           |           | 0        | GPCHPU5[2:0]  |          |          |       |                                                  |
|             | 1  | ↑   | 1   | 0        | GPCHPU6[2:0]  |           |           | 0        | GPCHPU7[2:0]  |          |          |       |                                                  |
|             | 1  | ↑   | 1   | 0        | GPCHPU8[2:0]  |           |           | 0        | GPCHPU9[2:0]  |          |          |       |                                                  |
|             | 1  | ↑   | 1   | 0        | GPCHPU10[2:0] |           |           | 0        | GPCHPU11[2:0] |          |          |       |                                                  |
|             | 1  | ↑   | 1   | 0        | GPCHPU12[2:0] |           |           | 0        | GPCHPU13[2:0] |          |          |       |                                                  |
|             | 1  | ↑   | 1   | 0        | GPCHPU14[2:0] |           |           | 0        | GPCHPU15[2:0] |          |          |       |                                                  |
| 6. GTUPEQL  | 0  | ↑   | 1   | 1        | 0             | 1         | 1         | 0        | 1             | 0        | 0        | (B4h) | Update<br>Period<br>Gate EQ<br>Control<br>In LPM |
|             | 1  | ↑   | 1   | 0        | GPCLPU0[2:0]  |           |           | 0        | GPCLPU1[2:0]  |          |          |       |                                                  |
|             | 1  | ↑   | 1   | 0        | GPCLPU2[2:0]  |           |           | 0        | GPCLPU3[2:0]  |          |          |       |                                                  |
|             | 1  | ↑   | 1   | 0        | GPCLPU4[2:0]  |           |           | 0        | GPCLPU5[2:0]  |          |          |       |                                                  |
|             | 1  | ↑   | 1   | 0        | GPCLPU6[2:0]  |           |           | 0        | GPCLPU7[2:0]  |          |          |       |                                                  |
|             | 1  | ↑   | 1   | 0        | GPCLPU8[2:0]  |           |           | 0        | GPCLPU9[2:0]  |          |          |       |                                                  |
|             | 1  | ↑   | 1   | 0        | GPCLPU10[2:0] |           |           | 0        | GPCLPU11[2:0] |          |          |       |                                                  |
|             | 1  | ↑   | 1   | 0        | GPCLPU12[2:0] |           |           | 0        | GPCLPU13[2:0] |          |          |       |                                                  |
|             | 1  | ↑   | 1   | 0        | GPCLPU14[2:0] |           |           | 0        | GPCLPU15[2:0] |          |          |       |                                                  |

| Instruction | A0 | R/W | ERD | D7        | D6        | D5        | D4        | D3        | D2        | D1        | D0        | Hex   | Function     |
|-------------|----|-----|-----|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-------|--------------|
| 7.SOUSEQ    | 0  | ↑   | 1   | 1         | 0         | 1         | 1         | 0         | 1         | 1         | 1         | (B7h) | Source EQ    |
|             | 0  | ↑   | 1   | 0         | 0         | 0         | SOUSEQ    | 0         | 0         | 1         | 1         |       | Enable       |
| 8.PNLSET    | 0  | ↑   | 1   | 1         | 0         | 1         | 1         | 1         | 0         | 0         | 0         | (B8h) | Panel        |
|             | 0  | ↑   | 1   | 0         | DOTINV[1] | DOTINV[0] | 0         | DPSCN[1]  | DPSCN[0]  | LAY[1]    | LAY[0]    |       | Setting      |
| 9.CLRAM     | 0  | ↑   | 1   | 1         | 0         | 1         | 1         | 1         | 0         | 1         | 1         | (BBh) | CLR          |
|             | 1  | ↑   | 1   | CLR       | 1         | 0         | 0         | 1         | 1         | 1         | 1         |       | RAM          |
| 10.GCTRL    | 0  | ↑   | 1   | 1         | 1         | 0         | 0         | 0         | 0         | 0         | 0         | (C0h) | Gate Control |
|             | 1  | ↑   | 1   | 0         | 0         | 0         | VGHA4     | VGHA3     | VGHA2     | VGHA1     | VGHA0     |       |              |
|             | 1  | ↑   | 1   | 0         | 0         | 0         | VGLA4     | VGLA3     | VGLA2     | VGLA1     | VGLA0     |       |              |
| 11.VSHPCTRL | 0  | ↑   | 1   | 1         | 1         | 0         | 0         | 0         | 0         | 0         | 1         | (C1h) | Source       |
|             | 1  | ↑   | 1   | VSHPA1[7] | VSHPA1[6] | VSHPA1[5] | VSHPA1[4] | VSHPA1[3] | VSHPA1[2] | VSHPA1[1] | VSHPA1[0] |       | High         |
|             | 1  | ↑   | 1   | VSHPA2[7] | VSHPA2[6] | VSHPA2[5] | VSHPA2[4] | VSHPA2[3] | VSHPA2[2] | VSHPA2[1] | VSHPA2[0] |       | Positive     |
|             | 1  | ↑   | 1   | VSHPA3[7] | VSHPA3[6] | VSHPA3[5] | VSHPA3[4] | VSHPA3[3] | VSHPA3[2] | VSHPA3[1] | VSHPA3[0] |       | Voltage      |
|             | 1  | ↑   | 1   | VSHPA4[7] | VSHPA4[6] | VSHPA4[5] | VSHPA4[4] | VSHPA4[3] | VSHPA4[2] | VSHPA4[1] | VSHPA4[0] |       | Control      |
| 12.VSLPCTRL | 0  | ↑   | 1   | 1         | 1         | 0         | 0         | 0         | 0         | 1         | 0         | (C2h) | Source       |
|             | 1  | ↑   | 1   | VSLPA1[7] | VSLPA1[6] | VSLPA1[5] | VSLPA1[4] | VSLPA1[3] | VSLPA1[2] | VSLPA1[1] | VSLPA1[0] |       | Low          |
|             | 1  | ↑   | 1   | VSLPA2[7] | VSLPA2[6] | VSLPA2[5] | VSLPA2[4] | VSLPA2[3] | VSLPA2[2] | VSLPA2[1] | VSLPA2[0] |       | Positive     |
|             | 1  | ↑   | 1   | VSLPA3[7] | VSLPA3[6] | VSLPA3[5] | VSLPA3[4] | VSLPA3[3] | VSLPA3[2] | VSLPA3[1] | VSLPA3[0] |       | Voltage      |
|             | 1  | ↑   | 1   | VSLPA4[7] | VSLPA4[6] | VSLPA4[5] | VSLPA4[4] | VSLPA4[3] | VSLPA4[2] | VSLPA4[1] | VSLPA4[0] |       | Control      |
| 13.VSHNCTRL | 0  | ↑   | 1   | 1         | 1         | 0         | 0         | 0         | 1         | 0         | 0         | (C4h) | Source       |
|             | 1  | ↑   | 1   | VSHNA1[7] | VSHNA1[6] | VSHNA1[5] | VSHNA1[4] | VSHNA1[3] | VSHNA1[2] | VSHNA1[1] | VSHNA1[0] |       | High         |
|             | 1  | ↑   | 1   | VSHNA2[7] | VSHNA2[6] | VSHNA2[5] | VSHNA2[4] | VSHNA2[3] | VSHNA2[2] | VSHNA2[1] | VSHNA2[0] |       | Negative     |
|             | 1  | ↑   | 1   | VSHNA3[7] | VSHNA3[6] | VSHNA3[5] | VSHNA3[4] | VSHNA3[3] | VSHNA3[2] | VSHNA3[1] | VSHNA3[0] |       | Voltage      |
|             | 1  | ↑   | 1   | VSHNA4[7] | VSHNA4[6] | VSHNA4[5] | VSHNA4[4] | VSHNA4[3] | VSHNA4[2] | VSHNA4[1] | VSHNA4[0] |       | Control      |
| 14.VSLNCTRL | 0  | ↑   | 1   | 1         | 1         | 0         | 0         | 0         | 1         | 0         | 1         | (C5h) | Source       |
|             | 1  | ↑   | 1   | VSLNA1[7] | VSLNA1[6] | VSLNA1[5] | VSLNA1[4] | VSLNA1[3] | VSLNA1[2] | VSLNA1[1] | VSLNA1[0] |       | Low          |
|             | 1  | ↑   | 1   | VSLNA2[7] | VSLNA2[6] | VSLNA2[5] | VSLNA2[4] | VSLNA2[3] | VSLNA2[2] | VSLNA2[1] | VSLNA2[0] |       | Negative     |
|             | 1  | ↑   | 1   | VSLNA3[7] | VSLNA3[6] | VSLNA3[5] | VSLNA3[4] | VSLNA3[3] | VSLNA3[2] | VSLNA3[1] | VSLNA3[0] |       | Voltage      |
|             | 1  | ↑   | 1   | VSLNA4[7] | VSLNA4[6] | VSLNA4[5] | VSLNA4[4] | VSLNA4[3] | VSLNA4[2] | VSLNA4[1] | VSLNA4[0] |       | Control      |

| Instruction   | A0 | R/W | ERD | D7              | D6    | D5     | D4    | D3    | D2      | D1           | D0    | Hex   | Function                |
|---------------|----|-----|-----|-----------------|-------|--------|-------|-------|---------|--------------|-------|-------|-------------------------|
| 15.VSHLSEL    | 0  | ↑   | 1   | 1               | 1     | 0      | 0     | 1     | 0       | 0            | 1     | (C9h) | Source                  |
|               | 1  | ↑   | 1   | 0               | 0     | 0      | 0     | 0     | 0       | VSHLSEL[1:0] |       |       | Voltage Select          |
| 16.ID1SET     | 0  | ↑   | 1   | 1               | 1     | 0      | 0     | 1     | 0       | 1            | 0     | (CAh) | ID1 Set                 |
|               | 1  | ↑   | 1   | ID17            | ID16  | ID15   | ID14  | ID13  | ID12    | ID11         | ID10  |       |                         |
| 17.ID2SET     | 0  | ↑   | 1   | 1               | 1     | 0      | 0     | 1     | 0       | 1            | 1     | (CBh) | ID2 Set                 |
|               | 1  | ↑   | 1   | ID27            | ID26  | ID25   | ID24  | ID23  | ID22    | ID21         | ID20  |       |                         |
| 18.ID3SET     | 0  | ↑   | 1   | 1               | 1     | 0      | 0     | 1     | 1       | 0            | 0     | (CCh) | ID3 Set                 |
|               | 1  | ↑   | 1   | ID37            | ID36  | ID35   | ID34  | ID33  | ID32    | ID31         | ID30  |       |                         |
| 19.AUTOPWR    | 0  | ↑   | 1   | 1               | 1     | 0      | 1     | 0     | 0       | 0            | 0     | (D0h) | Enable                  |
|               | 1  | ↑   | 1   | AUTOPWR         | 1     | 1      | 1     | 1     | 1       | 1            | 1     |       | Power Down              |
| 20.BSTEN      | 0  | ↑   | 1   | 1               | 1     | 0      | 1     | 0     | 0       | 0            | 1     | (D1h) | Booster                 |
|               | 1  | ↑   | 1   | 0               | 0     | 0      | 0     | 0     | 0       | 0            | BSTEN |       | Enable                  |
| 21.NVLOADCTRL | 0  | ↑   | 1   | 1               | 1     | 0      | 1     | 0     | 1       | 1            | 0     | (D6h) | NV Load Ctrl            |
|               | 1  | ↑   | 1   | 0               | 0     | 0      | 1     | 0     | VS_EN   | ID_EN        | 1     |       |                         |
|               | 1  | ↑   | 1   | 0               | 0     | 0      | 0     | 0     | NRDTIME | NRDCLP       | 0     |       |                         |
| 22.OSCSET     | 0  | ↑   | 1   | 1               | 1     | 0      | 1     | 1     | 0       | 0            | 0     | (D8h) | OSC Setting             |
|               | 1  | ↑   | 1   | OSCEN           | 0     | OSCSW2 | 0     | 0     | OSCSW1  | OSCSW0       | 0     |       |                         |
|               | 1  | ↑   | 1   | 1               | 1     | 1      | 0     | 1     | 0       | 0            | 1     |       |                         |
| 23.NVMRD      | 0  | ↑   | 1   | 1               | 1     | 1      | 0     | 1     | 0       | 0            | 1     | (E9h) | OTP                     |
|               | 1  | 1   | ↑   | DO[7]           | DO[6] | DO[5]  | DO[4] | DO[3] | DO[2]   | DO[1]        | DO[0] |       | Data Read               |
| 24.EXTBCTRL   | 0  | ↑   | 1   | 1               | 1     | 1      | 0     | 1     | 1       | 0            | 0     | (ECh) | EXTB                    |
|               | 1  | ↑   | 1   | EXTB_CTRL [7:0] |       |        |       |       |         |              |       |       | Control                 |
| 25.NVMCTRL1   | 0  | ↑   | 1   | 1               | 1     | 1      | 1     | 1     | 0       | 0            | 0     | (F8h) | NVM<br>WR/RD<br>Control |
|               | 1  | ↑   | 1   | 0               | 1     | 0      | 1     | 1     | 0       | 1            | 0     | (5Ah) |                         |
|               | 1  | ↑   | 1   | 0               | 1     | 1      | 0     | 1     | 0       | 0            | 1     | (69h) |                         |
|               | 1  | ↑   | 1   | 1               | 1     | 1      | 0     | 1     | 1       | 1            | 0     | (EEh) |                         |

| Instruction | A0 | R/W | ERD | D7     | D6     | D5     | D4     | D3     | D2     | D1     | D0     | Hex   | Function |
|-------------|----|-----|-----|--------|--------|--------|--------|--------|--------|--------|--------|-------|----------|
|             | 1  | ↑   | 1   | 0      | 0      | 0      | 0      | 0      | PROG   | RD     | 0      |       |          |
| 26.NVMCTRL2 | 0  | ↑   | 1   | 1      | 1      | 1      | 1      | 1      | 0      | 1      | 0      | (FAh) | NVM      |
|             | 1  | ↑   | 1   | 0      | 0      | PA[5]  | PA[4]  | PA[3]  | PA[2]  | PA[1]  | PA[0]  |       | Program  |
|             | 1  | ↑   | 1   | PIN[7] | PIN[6] | PIN[5] | PIN[4] | PIN[3] | PIN[2] | PIN[1] | PIN[0] |       | Setting  |
| 27.NVMRDEN  | 0  | ↑   | 1   | 1      | 1      | 1      | 1      | 1      | 0      | 1      | 1      | (FBh) | Enable   |
|             | 1  | ↑   | 1   | 0      | 1      | 0      | 0      | 1      | 0      | 1      | 0      | (4Ah) | NVM      |
|             | 1  | ↑   | 1   | 1      | 0      | 1      | 0      | 0      | 1      | 0      | 1      | (A5h) | Read     |
| 28.NVPROM   | 0  | ↑   | 1   | 1      | 1      | 1      | 1      | 1      | 1      | 0      | 0      | (FCh) | Enable   |
|             | 1  | ↑   | 1   | 0      | 0      | 1      | 0      | 1      | 0      | 0      | 1      | (29h) | NVM      |
|             | 1  | ↑   | 1   | 1      | 0      | 1      | 0      | 0      | 1      | 0      | 1      | (A5h) | Program  |



## 8.2.1 GTCON (62h): Gate Timing Control

| 62H                         | GTCON (Gate Timing Control)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |     |          |          |           |           |          |          |          |          |       |              |                         |    |             |    |             |    |             |    |             |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----|----------|----------|-----------|-----------|----------|----------|----------|----------|-------|--------------|-------------------------|----|-------------|----|-------------|----|-------------|----|-------------|
| Inst / Para                 | A0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | RWR                     | ERD | D7       | D6       | D5        | D4        | D3       | D2       | D1       | D0       | HEX   |              |                         |    |             |    |             |    |             |    |             |
| GTCON                       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ↑                       | 1   | 0        | 1        | 1         | 0         | 0        | 0        | 1        | 0        | (62h) |              |                         |    |             |    |             |    |             |    |             |
| 1 <sup>st</sup> parameter   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ↑                       | 1   | 0        | 0        | CMON      | CMODE     | GAB[3]   | GAB[2]   | GAB[1]   | GAB[0]   |       |              |                         |    |             |    |             |    |             |    |             |
| 2 <sup>nd</sup> parameter   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ↑                       | 1   | 0        | 0        | CLKDIV[1] | CLKDIV[0] | FCMPA[3] | FCMPA[2] | FCMPA[1] | FCMPA[0] |       |              |                         |    |             |    |             |    |             |    |             |
| 3 <sup>rd</sup> parameter   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ↑                       | 1   | CMPWA[7] | CMPWA[6] | CMPWA[5]  | CMPWA[4]  | CMPWA[3] | CMPWA[2] | CMPWA[1] | CMPWA[0] |       |              |                         |    |             |    |             |    |             |    |             |
| Description                 | <p>-This command is used to set gate timing during the compensation period in low power mode. The frame period is composed of display update period and compensation period.</p> <p>-As the frequency decreases, turning on this function and matching the appropriate waveform is more beneficial for slowing down the aging rate and phenomenon of the TFT components.</p> <p><b>CMON:</b> Compensation ON/OFF. “0”=&gt;OFF ; “1”=&gt;ON</p> <p><b>CMODE:</b> Compensation Mode. “0”=&gt;DC Mode ; “1”=&gt;AC Mode.</p> <p><b>GAB[3:0]:</b> GA and GB Frame Interval. GAB is set within the rage of 0 to 15 frames.</p> <p><b>CLKDIV[1:0]:</b> Compensation CLK Divide</p> <table><tr><th>CLKDIV [1:0]</th><th>Compensation CLK Divide</th></tr><tr><td>00</td><td>Divide by 1</td></tr><tr><td>01</td><td>Divide by 2</td></tr><tr><td>10</td><td>Divide by 4</td></tr><tr><td>11</td><td>Divide by 8</td></tr></table> <p><b>FCMPA[3:0]:</b> First Compensation Pulse to Data Upload End. Must be set within the rage of 0 to 15 (1 to 16 gate).</p> <p><b>CMPWA[7:0]:</b> Compensation Pulse Width. Must be set within the rage of 0 to 255.</p> |                         |     |          |          |           |           |          |          |          |          |       | CLKDIV [1:0] | Compensation CLK Divide | 00 | Divide by 1 | 01 | Divide by 2 | 10 | Divide by 4 | 11 | Divide by 8 |
|                             | CLKDIV [1:0]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Compensation CLK Divide |     |          |          |           |           |          |          |          |          |       |              |                         |    |             |    |             |    |             |    |             |
|                             | 00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Divide by 1             |     |          |          |           |           |          |          |          |          |       |              |                         |    |             |    |             |    |             |    |             |
|                             | 01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Divide by 2             |     |          |          |           |           |          |          |          |          |       |              |                         |    |             |    |             |    |             |    |             |
|                             | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Divide by 4             |     |          |          |           |           |          |          |          |          |       |              |                         |    |             |    |             |    |             |    |             |
|                             | 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Divide by 8             |     |          |          |           |           |          |          |          |          |       |              |                         |    |             |    |             |    |             |    |             |
|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                         |     |          |          |           |           |          |          |          |          |       |              |                         |    |             |    |             |    |             |    |             |
|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                         |     |          |          |           |           |          |          |          |          |       |              |                         |    |             |    |             |    |             |    |             |
|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                         |     |          |          |           |           |          |          |          |          |       |              |                         |    |             |    |             |    |             |    |             |
|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                         |     |          |          |           |           |          |          |          |          |       |              |                         |    |             |    |             |    |             |    |             |
|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                         |     |          |          |           |           |          |          |          |          |       |              |                         |    |             |    |             |    |             |    |             |
|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                         |     |          |          |           |           |          |          |          |          |       |              |                         |    |             |    |             |    |             |    |             |
| Default Value<br>(D7 to D0) | 1 <sup>st</sup> Parameter=0x32h<br>2 <sup>nd</sup> parameter=0x03h<br>3 <sup>rd</sup> parameter= 0x1Fh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                         |     |          |          |           |           |          |          |          |          |       |              |                         |    |             |    |             |    |             |    |             |

## 8.2.2 GATESET (B0h): Gate Line Setting

| B0H                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | GATESET (Gate Line Setting) |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------|-------|-------|-------|-------|---------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|---------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|-----|---|---|---|---|---|---|---|-----|---|---|---|---|---|---|---|-----|
| Inst / Para               | D/CX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | WRX                         | RDX   | D7    | D6    | D5    | D4    | D3                  | D2    | D1    | D0    | HEX   |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
| GATESET                   | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ↑                           | 1     | 1     | 0     | 1     | 1     | 0                   | 0     | 0     | 0     | (B0h) |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
| 1 <sup>st</sup> parameter | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ↑                           | 1     | 0     | GL[6] | GL[5] | GL[4] | GL[3]               | GL[2] | GL[1] | GL[0] |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
| Description               | <b>GL[6:0]</b> specifies the gate line of the module in 4-line basis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           | This command must be selected before sleep-out. Do not change this command while the display is turned on.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           | The relationship between the parameter GL[6:0] and the number of display lines is shown below.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           | <table><tr><th>GL[6]</th><th>GL[5]</th><th>GL[4]</th><th>GL[3]</th><th>GL[2]</th><th>GL[1]</th><th>GL[0]</th><th>Gate Line of Module</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>4</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>8</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>12</td></tr><tr><td>:</td><td>:</td><td>:</td><td>:</td><td>:</td><td>:</td><td>:</td><td>:</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>406</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>407</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>408</td></tr></table> |                             |       |       |       |       |       |                     |       |       |       |       | GL[6] | GL[5] | GL[4] | GL[3] | GL[2] | GL[1] | GL[0] | Gate Line of Module | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 4 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 8 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 12 | : | : | : | : | : | : | : | : | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 406 | 1 | 1 | 0 | 0 | 1 | 0 | 1 | 407 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 408 |
|                           | GL[6]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | GL[5]                       | GL[4] | GL[3] | GL[2] | GL[1] | GL[0] | Gate Line of Module |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0                           | 0     | 0     | 0     | 0     | 1     | 4                   |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0                           | 0     | 0     | 0     | 1     | 0     | 8                   |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0                           | 0     | 0     | 0     | 1     | 1     | 12                  |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
| :                         | :                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | :                           | :     | :     | :     | :     | :     |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
| 1                         | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0                           | 0     | 1     | 0     | 0     | 406   |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
| 1                         | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0                           | 0     | 1     | 0     | 1     | 407   |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
| 1                         | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0                           | 0     | 1     | 1     | 0     | 408   |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |
| </                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |       |       |       |                     |       |       |       |       |       |       |       |       |       |       |       |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |     |

## 8.2.3 FSTCOM (B1h): First Gate Setting

| B1H                       | FSTCOM (First Gate Setting)                                                                                 |             |             |             |             |             |             |             |             |                     |        |       |
|---------------------------|-------------------------------------------------------------------------------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|--------|-------|
| Inst / Para               | A0                                                                                                          | RWR         | ERD         | D7          | D6          | D5          | D4          | D3          | D2          | D1                  | D0     | HEX   |
| FSTCOM                    | 0                                                                                                           | ↑           | 1           | 1           | 0           | 1           | 1           | 0           | 0           | 0                   | 1      | (B1h) |
| 1 <sup>st</sup> parameter | 1                                                                                                           | ↑           | 1           | 0           | 0           | 0           | 0           | 0           | 0           | 0                   | FST[8] |       |
| 2 <sup>nd</sup> parameter | 1                                                                                                           | ↑           | 1           | FST[7]      | FST[6]      | FST[5]      | FST[4]      | FST[3]      | FST[2]      | FST[1]              | FST[0] |       |
| Description               | <b>FST[8:0]</b> specifies “the first output gate number – 1 “ that mapping to the RAM page address 0.       |             |             |             |             |             |             |             |             |                     |        |       |
|                           | Continuously increasing or decreasing the first gate setting results in vertical-scrolling in 4-line basis. |             |             |             |             |             |             |             |             |                     |        |       |
|                           | For detail setting value, please see the table as below.                                                    |             |             |             |             |             |             |             |             |                     |        |       |
|                           | <b>FST8</b>                                                                                                 | <b>FST7</b> | <b>FST6</b> | <b>FST5</b> | <b>FST4</b> | <b>FST3</b> | <b>FST2</b> | <b>FST1</b> | <b>FST0</b> | <b>Line address</b> |        |       |
|                           | 0                                                                                                           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 1                   |        |       |
|                           | 0                                                                                                           | 0           | 0           | 0           | 0           | 0           | 0           | 1           | 1           | 4                   |        |       |
|                           | 0                                                                                                           | 0           | 0           | 0           | 0           | 0           | 1           | 1           | 1           | 8                   |        |       |
|                           | :                                                                                                           | :           | :           | :           | :           | :           | :           | :           | :           | :                   |        |       |
|                           | 1                                                                                                           | 0           | 0           | 1           | 0           | 1           | 1           | 1           | 1           | 400                 |        |       |
|                           | 1                                                                                                           | 1           | 0           | 0           | 1           | 0           | 1           | 1           | 1           | 404                 |        |       |
| 1                         | 1                                                                                                           | 0           | 0           | 0           | 1           | 0           | 1           | 1           | 408         |                     |        |       |

|                             |                                                                                            |
|-----------------------------|--------------------------------------------------------------------------------------------|
|                             | <p>Example:</p> <p>If FST[8:0]=7h, Gate 8 would output the data of RAM page address 0.</p> |
| Default Value<br>(D7 to D0) | <p>1<sup>st</sup> Parameter=0x00h</p> <p>2<sup>nd</sup> parameter=0x00h</p>                |

## 8.2.4 FRCTRL (B2h): Frame Rate Control

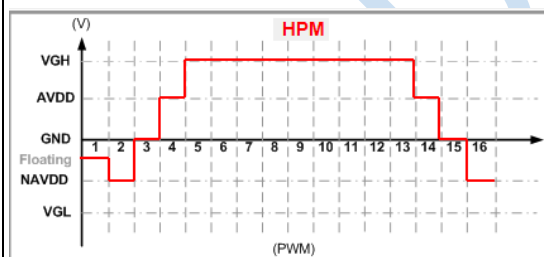
| B2H                         | FRCTRL (Frame Rate Control)                                                                          |     |                 |    |    |    |      |    |         |         |         |       |
|-----------------------------|------------------------------------------------------------------------------------------------------|-----|-----------------|----|----|----|------|----|---------|---------|---------|-------|
| Inst / Para                 | A0                                                                                                   | RWR | ERD             | D7 | D6 | D5 | D4   | D3 | D2      | D1      | D0      | HEX   |
| FRCTRL                      | 0                                                                                                    | ↑   | 1               | 1  | 0  | 1  | 1    | 0  | 0       | 1       | 0       | (B2h) |
| 1 <sup>st</sup> parameter   | 1                                                                                                    | ↑   | 1               | 0  | 0  | 0  | HFRA | 0  | LFRA[2] | LFRA[1] | LFRA[0] |       |
| Description                 | HFRA: Frame rate control in high power mode (HPM).                                                   |     |                 |    |    |    |      |    |         |         |         |       |
|                             | HFRA                                                                                                 |     | Frame rate (Hz) |    |    |    |      |    |         |         |         |       |
|                             | 0                                                                                                    |     | 16              |    |    |    |      |    |         |         |         |       |
|                             | 1                                                                                                    |     | 32              |    |    |    |      |    |         |         |         |       |
|                             | LFRA[2:0]: Frame rate control in low power mode (LPM)                                                |     |                 |    |    |    |      |    |         |         |         |       |
|                             | LFRA [2:0]                                                                                           |     | Frame rate (Hz) |    |    |    |      |    |         |         |         |       |
|                             | 0h                                                                                                   |     | 0.25            |    |    |    |      |    |         |         |         |       |
|                             | 1h                                                                                                   |     | 0.5             |    |    |    |      |    |         |         |         |       |
|                             | 2h                                                                                                   |     | 1               |    |    |    |      |    |         |         |         |       |
|                             | 3h                                                                                                   |     | 2               |    |    |    |      |    |         |         |         |       |
|                             | 4h                                                                                                   |     | 4               |    |    |    |      |    |         |         |         |       |
|                             | 5h                                                                                                   |     | 8               |    |    |    |      |    |         |         |         |       |
|                             | Note: The value of frame rate setting should be determined by the module loading and display effect. |     |                 |    |    |    |      |    |         |         |         |       |
| Default Value<br>(D7 to D0) | 0x12h                                                                                                |     |                 |    |    |    |      |    |         |         |         |       |

## 8.2.5 GTUPEQH (B3h): Update Period Gate EQ Control in HPM

| B3H                            | GTUPEQH (Update Period Gate EQ Control in HPM)                  |       |                 |    |               |    |    |    |               |    |    |       |
|--------------------------------|-----------------------------------------------------------------|-------|-----------------|----|---------------|----|----|----|---------------|----|----|-------|
| Inst / Para                    | A0                                                              | RWR   | ERD             | D7 | D6            | D5 | D4 | D3 | D2            | D1 | D0 | HEX   |
| GTUPEQH                        | 0                                                               | ↑     | 1               | 1  | 0             | 1  | 1  | 0  | 0             | 1  | 1  | (B3h) |
| 1 <sup>st</sup> parameter      | 1                                                               | ↑     | 1               | 1  | GEQEN         | 1  | 0  | 0  | 1             | 0  | 1  |       |
| 2 <sup>nd</sup> parameter      | 1                                                               | ↑     | 1               | 1  | 1             | 1  | 1  | 0  | 1             | 1  | 0  |       |
| 3 <sup>rd</sup> parameter      | 1                                                               | ↑     | 1               | 0  | GPCHPU0[2:0]  |    |    | 0  | GPCHPU1[2:0]  |    |    |       |
| 4 <sup>th</sup> parameter      | 1                                                               | ↑     | 1               | 0  | GPCHPU2[2:0]  |    |    | 0  | GPCHPU3[2:0]  |    |    |       |
| 5 <sup>th</sup> parameter      | 1                                                               | ↑     | 1               | 0  | GPCHPU4[2:0]  |    |    | 0  | GPCHPU5[2:0]  |    |    |       |
| 6 <sup>th</sup> parameter      | 1                                                               | ↑     | 1               | 0  | GPCHPU6[2:0]  |    |    | 0  | GPCHPU7[2:0]  |    |    |       |
| 7 <sup>th</sup> parameter      | 1                                                               | ↑     | 1               | 0  | GPCHPU8[2:0]  |    |    | 0  | GPCHPU9[2:0]  |    |    |       |
| 8 <sup>th</sup> parameter      | 1                                                               | ↑     | 1               | 0  | GPCHPU10[2:0] |    |    | 0  | GPCHPU11[2:0] |    |    |       |
| 9 <sup>th</sup> parameter      | 1                                                               | ↑     | 1               | 0  | GPCHPU12[2:0] |    |    | 0  | GPCHPU13[2:0] |    |    |       |
| 10 <sup>th</sup> parameter     | 1                                                               | ↑     | 1               | 0  | GPCHPU14[2:0] |    |    | 0  | GPCHPU15[2:0] |    |    |       |
| Description                    | -This command is used to set gate EQ waveform in update period. |       |                 |    |               |    |    |    |               |    |    |       |
|                                | GEQEN: Gate EQ Enable (HPM/LPM both)                            |       |                 |    |               |    |    |    |               |    |    |       |
|                                | GEQEN                                                           |       | Gate EQ Enable  |    |               |    |    |    |               |    |    |       |
|                                | 0                                                               |       | Disable         |    |               |    |    |    |               |    |    |       |
|                                | 1                                                               |       | Enable          |    |               |    |    |    |               |    |    |       |
|                                | GPCHPU0~ GPCHPU15: Gate EQ Control in HPM at 0~15 PWM           |       |                 |    |               |    |    |    |               |    |    |       |
|                                | Gate EQ of HPM update period includes 16 PWMs to achieve..      |       |                 |    |               |    |    |    |               |    |    |       |
|                                | GPCHPU0~15[2:0]                                                 |       | Gate EQ Voltage |    |               |    |    |    |               |    |    |       |
|                                | 000                                                             |       | Floating        |    |               |    |    |    |               |    |    |       |
|                                | 001                                                             |       | VGL             |    |               |    |    |    |               |    |    |       |
| 100                            |                                                                 | GND   |                 |    |               |    |    |    |               |    |    |       |
| 101                            |                                                                 | NAVDD |                 |    |               |    |    |    |               |    |    |       |
| 110                            |                                                                 | AVDD  |                 |    |               |    |    |    |               |    |    |       |
| 111                            |                                                                 | VGH   |                 |    |               |    |    |    |               |    |    |       |
| Default HPM Update EQ Setting. |                                                                 |       |                 |    |               |    |    |    |               |    |    |       |

| HPM PWM  | Default Parameter | Voltage  |
|----------|-------------------|----------|
| GPCHPU0  | 000               | Floating |
| GPCHPU1  | 101               | NAVDD    |
| GPCHPU2  | 100               | GND      |
| GPCHPU3  | 110               | AVDD     |
| GPCHPU4  | 111               | VGH      |
| GPCHPU5  | 111               | VGH      |
| GPCHPU6  | 111               | VGH      |
| GPCHPU7  | 111               | VGH      |
| GPCHPU8  | 111               | VGH      |
| GPCHPU9  | 111               | VGH      |
| GPCHPU10 | 111               | VGH      |
| GPCHPU11 | 111               | VGH      |
| GPCHPU12 | 111               | VGH      |
| GPCHPU13 | 110               | AVDD     |
| GPCHPU14 | 100               | GND      |
| GPCHPU15 | 101               | NAVDD    |

Default HPM Update EQ Waveform as below

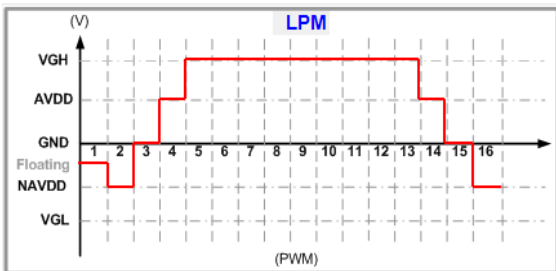


Default Value  
(D7 to D0)

1<sup>st</sup> Parameter=0xE5h  
 2<sup>nd</sup> parameter=0xF6h  
 3<sup>rd</sup> parameter=0x05h  
 4<sup>th</sup> parameter=0x46h  
 5<sup>th</sup> parameter=0x77h  
 6<sup>th</sup> parameter=0x77h  
 7<sup>rd</sup> parameter= 0x77h  
 8<sup>th</sup> parameter=0x77h  
 9<sup>th</sup> parameter=0x76h  
 10<sup>th</sup> parameter=0x45h

## 8.2.6 GTUPEQL (B4h): Update Period Gate EQ Control in LPM

| B4H                       | GTUPEQL (Update Period Gate EQ Control in LPM)                                                                            |     |                          |      |                        |    |    |    |               |    |    |       |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------|-----|--------------------------|------|------------------------|----|----|----|---------------|----|----|-------|
| Inst / Para               | A0                                                                                                                        | RWR | ERD                      | D7   | D6                     | D5 | D4 | D3 | D2            | D1 | D0 | HEX   |
| GTUPEQL                   | 0                                                                                                                         | ↑   | 1                        | 1    | 0                      | 1  | 1  | 0  | 1             | 0  | 0  | (B4h) |
| 1 <sup>st</sup> parameter | 1                                                                                                                         | ↑   | 1                        | 0    | GPCLPU0[2:0]           |    |    | 0  | GPCLPU1[2:0]  |    |    |       |
| 2 <sup>nd</sup> parameter | 1                                                                                                                         | ↑   | 1                        | 0    | GPCLPU2[2:0]           |    |    | 0  | GPCLPU3[2:0]  |    |    |       |
| 3 <sup>rd</sup> parameter | 1                                                                                                                         | ↑   | 1                        | 0    | GPCLPU4[2:0]           |    |    | 0  | GPCLPU5[2:0]  |    |    |       |
| 4 <sup>th</sup> parameter | 1                                                                                                                         | ↑   | 1                        | 0    | GPCLPU6[2:0]           |    |    | 0  | GPCLPU7[2:0]  |    |    |       |
| 5 <sup>th</sup> parameter | 1                                                                                                                         | ↑   | 1                        | 0    | GPCLPU8[2:0]           |    |    | 0  | GPCLPU9[2:0]  |    |    |       |
| 6 <sup>th</sup> parameter | 1                                                                                                                         | ↑   | 1                        | 0    | GPCLPU10[2:0]          |    |    | 0  | GPCLPU11[2:0] |    |    |       |
| 7 <sup>th</sup> parameter | 1                                                                                                                         | ↑   | 1                        | 0    | GPCLPU12[2:0]          |    |    | 0  | GPCLPU13[2:0] |    |    |       |
| 8 <sup>th</sup> parameter | 1                                                                                                                         | ↑   | 1                        | 0    | GPCLPU14[2:0]          |    |    | 0  | GPCLPU15[2:0] |    |    |       |
| Description               | <b>GPCLPU0~ GPCLPU15: Gate EQ Control in LPM at 0~15 PWM</b><br>Gate EQ of LPM update period includes 16 PWMs to achieve. |     |                          |      |                        |    |    |    |               |    |    |       |
|                           | <b>GPCLPU0~15[2:0]</b>                                                                                                    |     |                          |      | <b>Gate EQ Voltage</b> |    |    |    |               |    |    |       |
|                           | 000                                                                                                                       |     |                          |      | Floating               |    |    |    |               |    |    |       |
|                           | 001                                                                                                                       |     |                          |      | VGL                    |    |    |    |               |    |    |       |
|                           | 100                                                                                                                       |     |                          |      | GND                    |    |    |    |               |    |    |       |
|                           | 101                                                                                                                       |     |                          |      | NAVDD                  |    |    |    |               |    |    |       |
|                           | 110                                                                                                                       |     |                          |      | AVDD                   |    |    |    |               |    |    |       |
|                           | 111                                                                                                                       |     |                          |      | VGH                    |    |    |    |               |    |    |       |
|                           | Default HPM Update EQ Setting.                                                                                            |     |                          |      |                        |    |    |    |               |    |    |       |
|                           | <b>LPM PWM</b>                                                                                                            |     | <b>Default Parameter</b> |      | <b>Voltage</b>         |    |    |    |               |    |    |       |
|                           | GPCLPU0                                                                                                                   |     | 000                      |      | Floating               |    |    |    |               |    |    |       |
|                           | GPCLPU1                                                                                                                   |     | 101                      |      | NAVDD                  |    |    |    |               |    |    |       |
| GPCLPU2                   |                                                                                                                           | 100 |                          | GND  |                        |    |    |    |               |    |    |       |
| GPCLPU3                   |                                                                                                                           | 110 |                          | AVDD |                        |    |    |    |               |    |    |       |
| GPCLPU4                   |                                                                                                                           | 111 |                          | VGH  |                        |    |    |    |               |    |    |       |
| GPCLPU5                   |                                                                                                                           | 111 |                          | VGH  |                        |    |    |    |               |    |    |       |
| GPCLPU6                   |                                                                                                                           | 111 |                          | VGH  |                        |    |    |    |               |    |    |       |
| GPCLPU7                   |                                                                                                                           | 111 |                          | VGH  |                        |    |    |    |               |    |    |       |
| GPCLPU8                   |                                                                                                                           | 111 |                          | VGH  |                        |    |    |    |               |    |    |       |
| GPCLPU9                   |                                                                                                                           | 111 |                          | VGH  |                        |    |    |    |               |    |    |       |
| GPCLPU10                  |                                                                                                                           | 111 |                          | VGH  |                        |    |    |    |               |    |    |       |

|                                                                                   |                                                                                                                                                                                                                                                                                       |     |       |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|
|                                                                                   | <b>GPCLPU11</b>                                                                                                                                                                                                                                                                       | 111 | VGH   |
|                                                                                   | <b>GPCLPU12</b>                                                                                                                                                                                                                                                                       | 111 | VGH   |
|                                                                                   | <b>GPCLPU13</b>                                                                                                                                                                                                                                                                       | 110 | AVDD  |
|                                                                                   | <b>GPCLPU14</b>                                                                                                                                                                                                                                                                       | 100 | GND   |
|                                                                                   | <b>GPCLPU15</b>                                                                                                                                                                                                                                                                       | 101 | NAVDD |
| Default LPM Update EQ Waveform as below                                           |                                                                                                                                                                                                                                                                                       |     |       |
|  |                                                                                                                                                                                                                                                                                       |     |       |
| Default Value<br>(D7 to D0)                                                       | 1 <sup>st</sup> Parameter=0x05h<br>2 <sup>nd</sup> parameter=0x46h<br>3 <sup>rd</sup> parameter=0x77h<br>4 <sup>th</sup> parameter=0x77h<br>5 <sup>th</sup> parameter=0x77h<br>6 <sup>th</sup> parameter=0x77h<br>7 <sup>rd</sup> parameter= 0x76h<br>8 <sup>th</sup> parameter=0x45h |     |       |

### 8.2.7 SOUEQ (B7h): Source EQ Enable

| B7H                         | SOUEQ (Source EQ Enable)                                          |     |     |    |    |    |       |    |    |    |    |       |
|-----------------------------|-------------------------------------------------------------------|-----|-----|----|----|----|-------|----|----|----|----|-------|
| Inst / Para                 | A0                                                                | RWR | ERD | D7 | D6 | D5 | D4    | D3 | D2 | D1 | D0 | HEX   |
| SOUEQ                       | 0                                                                 | ↑   | 1   | 1  | 0  | 1  | 1     | 0  | 1  | 1  | 1  | (B7h) |
| 1 <sup>st</sup> parameter   | 1                                                                 | ↑   | 1   | 0  | 0  | 0  | SOUEQ | 0  | 0  | 1  | 1  |       |
| Description                 | <b>SOUEQ</b><br>"0": Disable Source EQ.<br>"1": Enable Source EQ. |     |     |    |    |    |       |    |    |    |    |       |
| Default Value<br>(D7 to D0) | 0x13h                                                             |     |     |    |    |    |       |    |    |    |    |       |

## 8.2.8 PNLSET (B8h): Panel Setting

| B8H                           | PNLSET (Panel Setting)                                       |                    |                      |    |           |           |    |          |          |        |        |       |
|-------------------------------|--------------------------------------------------------------|--------------------|----------------------|----|-----------|-----------|----|----------|----------|--------|--------|-------|
| Inst / Para                   | A0                                                           | RWR                | ERD                  | D7 | D6        | D5        | D4 | D3       | D2       | D1     | D0     | HEX   |
| PNLSET                        | 0                                                            | ↑                  | 1                    | 1  | 0         | 1         | 1  | 1        | 0        | 0      | 0      | (B8h) |
| Parameter                     | 1                                                            | ↑                  | 1                    | 0  | DOTINV[1] | DOTINV[0] | 0  | DPSCN[1] | DPSCN[0] | LAY[1] | LAY[0] |       |
| Description                   | -This command is used to set the gate scan and panel layout. |                    |                      |    |           |           |    |          |          |        |        |       |
|                               | DOTINV[1:0]:                                                 |                    |                      |    |           |           |    |          |          |        |        |       |
|                               | DOTINV [1:0]                                                 |                    | Column/Dot Inversion |    |           |           |    |          |          |        |        |       |
|                               | 00                                                           |                    | Column inversion     |    |           |           |    |          |          |        |        |       |
|                               | 01                                                           |                    | 1-Dot inversion      |    |           |           |    |          |          |        |        |       |
|                               | 10                                                           |                    | 2-Dot inversion      |    |           |           |    |          |          |        |        |       |
|                               | DPSCN[1:0]: Display Gate Scan Mode Select                    |                    |                      |    |           |           |    |          |          |        |        |       |
|                               | DPSCN [1:0]                                                  |                    | Gate Scan Mode       |    |           |           |    |          |          |        |        |       |
|                               | 00                                                           |                    | Two Line Interval    |    |           |           |    |          |          |        |        |       |
|                               | 01                                                           |                    | One Line Interval    |    |           |           |    |          |          |        |        |       |
| 10                            |                                                              | Frame interval     |                      |    |           |           |    |          |          |        |        |       |
| LAY[1:0]: Panel Layout Select |                                                              |                    |                      |    |           |           |    |          |          |        |        |       |
| LAY [1:0]                     |                                                              | Panel Layout       |                      |    |           |           |    |          |          |        |        |       |
| 00                            |                                                              | Two Line Interlace |                      |    |           |           |    |          |          |        |        |       |
| 01                            |                                                              | One Line Interlace |                      |    |           |           |    |          |          |        |        |       |
| 10                            |                                                              | Non Interlace      |                      |    |           |           |    |          |          |        |        |       |
| Default Value (D7 to D0)      | 0x0Ah                                                        |                    |                      |    |           |           |    |          |          |        |        |       |

## 8.2.9 CLRAM (BBh): Enable Clear RAM

| BBH         | CLRAM (Enable Clear RAM )                                                                                                                                                |     |     |     |    |    |    |    |    |    |    |       |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|----|----|----|----|----|----|----|-------|
| Inst / Para | A0                                                                                                                                                                       | RWR | ERD | D7  | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| CLRAM       | 0                                                                                                                                                                        | ↑   | 1   | 1   | 0  | 1  | 1  | 1  | 0  | 1  | 1  | (BBh) |
| Parameter   | 1                                                                                                                                                                        | ↑   | 1   | CLR | 1  | 0  | 0  | 1  | 1  | 1  | 1  |       |
| Description | <p>-This command is used to clear RAM data to 0x00. Ensure to enable clear RAM in Display OFF mode.</p> <p>CLR=0 ; Disable Clear RAM</p> <p>CLR=1 ; Enable Clear RAM</p> |     |     |     |    |    |    |    |    |    |    |       |



## 8.2.10 GCTRL (C0h): Gate Voltage Control

| C0H                             | GCTRL (Gate Voltage Control)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----|----|----|----|-------------------------|---------|-------|-------|-------|-------|-----|-----|-----|-----|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|----------------|-----|------|-----|------|-----|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|-----------|---------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|-----------------|-----|-------|-----|-------|-----|-------|-----|-------|-----|-------|-----|-------|-----|-------|-----|-------|-----|-------|-----|-------|
| Inst / Para                     | A0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | RWR            | ERD | D7 | D6 | D5 | D4                      | D3      | D2    | D1    | D0    | HEX   |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| GCTRL                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | ↑              | 1   | 1  | 1  | 0  | 0                       | 0       | 0     | 0     | 0     | (C0h) |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| 1 <sup>st</sup> parameter       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | ↑              | 1   | 0  | 0  | 0  | VGHA4                   | VGHA3   | VGHA2 | VGHA1 | VGHA0 |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| 2 <sup>nd</sup> parameter       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | ↑              | 1   | 0  | 0  | 0  | VGLA4                   | VGLA3   | VGLA2 | VGLA1 | VGLA0 |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| Description                     | -This command is used to adjust gate voltage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | VGHA[4:0]: VGH Setting.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                |     |    |    |    | VGLA[4:0]: VGL Setting. |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | <table><tr><th>VGHA[4:0]</th><th>VGH (V)</th></tr><tr><td>00h</td><td>8.0</td></tr><tr><td>01h</td><td>8.5</td></tr><tr><td>02h</td><td>9.0</td></tr><tr><td>03h</td><td>9.5</td></tr><tr><td>04h</td><td>10.0</td></tr><tr><td>05h</td><td>10.5</td></tr><tr><td>06h</td><td>11.0</td></tr><tr><td>07h</td><td>11.5</td></tr><tr><td>08h</td><td>12.0</td></tr><tr><td>09h</td><td>12.5</td></tr><tr><td>0Ah</td><td>13.0</td></tr><tr><td>0Bh</td><td>13.5</td></tr><tr><td>0Ch</td><td>14.0</td></tr><tr><td>0Dh</td><td>14.5</td></tr><tr><td>0Eh</td><td>15.0 (default)</td></tr><tr><td>0Fh</td><td>15.5</td></tr><tr><td>10h</td><td>16.0</td></tr><tr><td>11h</td><td>16.5</td></tr></table> |                |     |    |    |    | VGHA[4:0]               | VGH (V) | 00h   | 8.0   | 01h   | 8.5   | 02h | 9.0 | 03h | 9.5 | 04h | 10.0 | 05h | 10.5 | 06h | 11.0 | 07h | 11.5 | 08h | 12.0 | 09h | 12.5 | 0Ah | 13.0 | 0Bh | 13.5 | 0Ch | 14.0 | 0Dh | 14.5 | 0Eh | 15.0 (default) | 0Fh | 15.5 | 10h | 16.0 | 11h | 16.5 | <table><tr><th>VGLA[4:0]</th><th>VGL (V)</th></tr><tr><td>02h</td><td>-6.0</td></tr><tr><td>03h</td><td>-6.5</td></tr><tr><td>04h</td><td>-7.0</td></tr><tr><td>05h</td><td>-7.5</td></tr><tr><td>06h</td><td>-8.0</td></tr><tr><td>07h</td><td>-8.5</td></tr><tr><td>08h</td><td>-9.0</td></tr><tr><td>09h</td><td>-9.5</td></tr><tr><td>0Ah</td><td>-10.0 (default)</td></tr><tr><td>0Bh</td><td>-10.5</td></tr><tr><td>0Ch</td><td>-11.0</td></tr><tr><td>0Dh</td><td>-11.5</td></tr><tr><td>0Eh</td><td>-12.0</td></tr><tr><td>0Fh</td><td>-12.5</td></tr><tr><td>10h</td><td>-13.0</td></tr><tr><td>11h</td><td>-13.5</td></tr><tr><td>12h</td><td>-14.0</td></tr><tr><td>13h</td><td>-14.5</td></tr><tr><td>14h</td><td>-15.0</td></tr></table> |  |  |  |  |  | VGLA[4:0] | VGL (V) | 02h | -6.0 | 03h | -6.5 | 04h | -7.0 | 05h | -7.5 | 06h | -8.0 | 07h | -8.5 | 08h | -9.0 | 09h | -9.5 | 0Ah | -10.0 (default) | 0Bh | -10.5 | 0Ch | -11.0 | 0Dh | -11.5 | 0Eh | -12.0 | 0Fh | -12.5 | 10h | -13.0 | 11h | -13.5 | 12h | -14.0 | 13h | -14.5 | 14h | -15.0 |
|                                 | VGHA[4:0]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | VGH (V)        |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 00h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 8.0            |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 01h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 8.5            |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 02h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 9.0            |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 03h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 9.5            |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 04h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 10.0           |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 05h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 10.5           |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 06h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 11.0           |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 07h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 11.5           |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 08h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 12.0           |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 09h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 12.5           |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 0Ah                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 13.0           |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 0Bh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 13.5           |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 0Ch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 14.0           |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 0Dh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 14.5           |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 0Eh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 15.0 (default) |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 0Fh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 15.5           |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 10h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 16.0           |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 11h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 16.5           |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | VGLA[4:0]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | VGL (V)        |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 02h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | -6.0           |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 03h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | -6.5           |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 04h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | -7.0           |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
|                                 | 05h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | -7.5           |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| 06h                             | -8.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| 07h                             | -8.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| 08h                             | -9.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| 09h                             | -9.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| 0Ah                             | -10.0 (default)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| 0Bh                             | -10.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| 0Ch                             | -11.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| 0Dh                             | -11.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| 0Eh                             | -12.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| 0Fh                             | -12.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| 10h                             | -13.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| 11h                             | -13.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| 12h                             | -14.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| 13h                             | -14.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| 14h                             | -15.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| Default Value                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| (D7 to D0)                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| 1 <sup>st</sup> Parameter=0x0Eh |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |
| 2 <sup>nd</sup> parameter=0x0Ah |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |     |    |    |    |                         |         |       |       |       |       |     |     |     |     |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                |     |      |     |      |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |           |         |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |                 |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |     |       |

## 8.2.11 VSHPCTRL (C1h): Source High Positive Voltage Control

| C1H                       | VSHPCTRL (Source High Positive Voltage Control )                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |                |           |           |       |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|----------------|-----------|-----------|-------|
| Inst / Para               | A0                                                                                                                                                                                                                                                                                                                                                                                                                                                | RWR       | ERD       | D7        | D6        | D5        | D4        | D3        | D2             | D1        | D0        | HEX   |
| VSHPCTRL                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ↑         | 1         | 1         | 1         | 0         | 0         | 0         | 0              | 0         | 1         | (C1h) |
| 1 <sup>st</sup> parameter | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ↑         | 1         | VSHPA1[7] | VSHPA1[6] | VSHPA1[5] | VSHPA1[4] | VSHPA1[3] | VSHPA1[2]      | VSHPA1[1] | VSHPA1[0] |       |
| 2 <sup>nd</sup> parameter | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ↑         | 1         | VSHPA2[7] | VSHPA2[6] | VSHPA2[5] | VSHPA2[4] | VSHPA2[3] | VSHPA2[2]      | VSHPA2[1] | VSHPA2[0] |       |
| 3 <sup>rd</sup> parameter | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ↑         | 1         | VSHPA3[7] | VSHPA3[6] | VSHPA3[5] | VSHPA3[4] | VSHPA3[3] | VSHPA3[2]      | VSHPA3[1] | VSHPA3[0] |       |
| 4 <sup>th</sup> parameter | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ↑         | 1         | VSHPA4[7] | VSHPA4[6] | VSHPA4[5] | VSHPA4[4] | VSHPA4[3] | VSHPA4[2]      | VSHPA4[1] | VSHPA4[0] |       |
| Description               | -This command is used to adjust source high positive voltage in different mode.<br>The calculation of VSHPAx is as shown below:<br><b>VSHPAx= 3.7+0.02x VSHPAx[7:0]</b><br>VSHPA1[7:0] : Source High Positive Voltage Mode1<br>VSHPA2[7:0] : Source High Positive Voltage Mode2<br>VSHPA3[7:0] : Source High Positive Voltage Mode3<br>VSHPA4[7:0] : Source High Positive Voltage Mode4<br>The suggestion of usable VSHPAx voltage is shown below |           |           |           |           |           |           |           |                |           |           |       |
|                           | VSHPAx[7]                                                                                                                                                                                                                                                                                                                                                                                                                                         | VSHPAx[6] | VSHPAx[5] | VSHPAx[4] | VSHPAx[3] | VSHPAx[2] | VSHPAx[1] | VSHPAx[0] | VSHPAx(V)      |           |           |       |
|                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0         | 0         | 0         | 0         | 0         | 0         | 0         | 3.70           |           |           |       |
|                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0         | 0         | 0         | 0         | 0         | 0         | 1         | 3.72           |           |           |       |
|                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0         | 0         | 0         | 0         | 0         | 1         | 0         | 3.74           |           |           |       |
|                           | :                                                                                                                                                                                                                                                                                                                                                                                                                                                 | :         | :         | :         | :         | :         | :         | :         | :              |           |           |       |
|                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1         | 0         | 0         | 0         | 0         | 0         | 1         | 5.00 (default) |           |           |       |
|                           | :                                                                                                                                                                                                                                                                                                                                                                                                                                                 | :         | :         | :         | :         | :         | :         | :         | :              |           |           |       |
|                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1         | 1         | 1         | 0         | 0         | 0         | 1         | 5.96           |           |           |       |
|                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1         | 1         | 1         | 0         | 0         | 1         | 0         | 5.98           |           |           |       |
| 0                         | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1         | 1         | 0         | 0         | 1         | 1         | 6.00      |                |           |           |       |
| Default Value (D7 to D0)  | 1 <sup>st</sup> Parameter=0x41h<br>2 <sup>nd</sup> parameter=0x41h<br>3 <sup>rd</sup> parameter=0x41h<br>4 <sup>th</sup> parameter=0x41h                                                                                                                                                                                                                                                                                                          |           |           |           |           |           |           |           |                |           |           |       |

## 8.2.12 VSLPCTRL (C2h): Source Low Positive Voltage Control

| C2H                         | VSLPCTRL (Source Low Positive Voltage Control)                                 |           |           |           |           |           |           |                |           |           |           |       |
|-----------------------------|--------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|----------------|-----------|-----------|-----------|-------|
| Inst / Para                 | A0                                                                             | RWR       | ERD       | D7        | D6        | D5        | D4        | D3             | D2        | D1        | D0        | HEX   |
| VSLPCTRL                    | 0                                                                              | ↑         | 1         | 1         | 1         | 0         | 0         | 0              | 0         | 1         | 0         | (C2h) |
| 1 <sup>st</sup> parameter   | 1                                                                              | ↑         | 1         | VSLPA1[7] | VSLPA1[6] | VSLPA1[5] | VSLPA1[4] | VSLPA1[3]      | VSLPA1[2] | VSLPA1[1] | VSLPA1[0] |       |
| 2 <sup>nd</sup> parameter   | 1                                                                              | ↑         | 1         | VSLPA2[7] | VSLPA2[6] | VSLPA2[5] | VSLPA2[4] | VSLPA2[3]      | VSLPA2[2] | VSLPA2[1] | VSLPA2[0] |       |
| 3 <sup>rd</sup> parameter   | 1                                                                              | ↑         | 1         | VSLPA3[7] | VSLPA3[6] | VSLPA3[5] | VSLPA3[4] | VSLPA3[3]      | VSLPA3[2] | VSLPA3[1] | VSLPA3[0] |       |
| 4 <sup>th</sup> parameter   | 1                                                                              | ↑         | 1         | VSLPA4[7] | VSLPA4[6] | VSLPA4[5] | VSLPA4[4] | VSLPA4[3]      | VSLPA4[2] | VSLPA4[1] | VSLPA4[0] |       |
| Description                 | -This command is used to adjust source low positive voltage in different mode. |           |           |           |           |           |           |                |           |           |           |       |
|                             | The calculation of VSLPAx is as shown below:                                   |           |           |           |           |           |           |                |           |           |           |       |
|                             | <b>VSLPAx= 0.02x VSLPAx [7:0]</b>                                              |           |           |           |           |           |           |                |           |           |           |       |
|                             | VSLPA1 [7:0] : Source Low Positive Voltage Mode1                               |           |           |           |           |           |           |                |           |           |           |       |
|                             | VSLPA2 [7:0] : Source Low Positive Voltage Mode2                               |           |           |           |           |           |           |                |           |           |           |       |
|                             | VSLPA3 [7:0] : Source Low Positive Voltage Mode3                               |           |           |           |           |           |           |                |           |           |           |       |
|                             | VSLPA4 [7:0] : Source Low Positive Voltage Mode4                               |           |           |           |           |           |           |                |           |           |           |       |
|                             | The suggestion of usable VSLPAx voltage is shown below                         |           |           |           |           |           |           |                |           |           |           |       |
|                             | VSLPAx[7]                                                                      | VSLPAx[6] | VSLPAx[5] | VSLPAx[4] | VSLPAx[3] | VSLPAx[2] | VSLPAx[1] | VSLPAx[0]      | VSLPAx(V) |           |           |       |
|                             | 0                                                                              | 0         | 0         | 0         | 0         | 0         | 0         | 0              | 0         |           |           |       |
|                             | 0                                                                              | 0         | 0         | 0         | 0         | 0         | 0         | 1              | 0.02      |           |           |       |
|                             | 0                                                                              | 0         | 0         | 0         | 0         | 0         | 1         | 0              | 0.04      |           |           |       |
| :                           | :                                                                              | :         | :         | :         | :         | :         | :         | :              |           |           |           |       |
| 0                           | 1                                                                              | 1         | 0         | 0         | 0         | 1         | 0         | 1.00 (default) |           |           |           |       |
| :                           | :                                                                              | :         | :         | :         | :         | :         | :         | :              |           |           |           |       |
| 0                           | 1                                                                              | 1         | 0         | 0         | 0         | 0         | 1         | 1.96           |           |           |           |       |
| 0                           | 1                                                                              | 1         | 0         | 0         | 0         | 1         | 0         | 1.98           |           |           |           |       |
| 0                           | 1                                                                              | 1         | 0         | 0         | 0         | 1         | 1         | 2.00           |           |           |           |       |
| Default Value<br>(D7 to D0) | 1 <sup>st</sup> Parameter=0x32h                                                |           |           |           |           |           |           |                |           |           |           |       |
|                             | 2 <sup>nd</sup> parameter=0x32h                                                |           |           |           |           |           |           |                |           |           |           |       |
|                             | 3 <sup>rd</sup> parameter=0x32h                                                |           |           |           |           |           |           |                |           |           |           |       |
|                             | 4 <sup>th</sup> parameter=0x32h                                                |           |           |           |           |           |           |                |           |           |           |       |

## 8.2.13 VSHNCTRL (C4h): Source High Negative Voltage Control

| C4H                       | VSHNCTRL (Source High Negative Voltage Control )                                                                                                                                                                                                                                                                                                                                                                                                                   |           |           |           |           |           |           |           |                 |           |           |       |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------------|-----------|-----------|-------|
| Inst / Para               | A0                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | RWR       | ERD       | D7        | D6        | D5        | D4        | D3        | D2              | D1        | D0        | HEX   |
| VSHNCTRL                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ↑         | 1         | 1         | 1         | 0         | 0         | 0         | 1               | 0         | 0         | (C4h) |
| 1 <sup>st</sup> parameter | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ↑         | 1         | VSHNA1[7] | VSHNA1[6] | VSHNA1[5] | VSHNA1[4] | VSHNA1[3] | VSHNA1[2]       | VSHNA1[1] | VSHNA1[0] |       |
| 2 <sup>nd</sup> parameter | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ↑         | 1         | VSHNA2[7] | VSHNA2[6] | VSHNA2[5] | VSHNA2[4] | VSHNA2[3] | VSHNA2[2]       | VSHNA2[1] | VSHNA2[0] |       |
| 3 <sup>rd</sup> parameter | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ↑         | 1         | VSHNA3[7] | VSHNA3[6] | VSHNA3[5] | VSHNA3[4] | VSHNA3[3] | VSHNA3[2]       | VSHNA3[1] | VSHNA3[0] |       |
| 4 <sup>th</sup> parameter | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ↑         | 1         | VSHNA4[7] | VSHNA4[6] | VSHNA4[5] | VSHNA4[4] | VSHNA4[3] | VSHNA4[2]       | VSHNA4[1] | VSHNA4[0] |       |
| Description               | -This command is used to adjust source high negative voltage in different mode.<br><br>The calculation of VSHNAx is as shown below:<br><br><b>VSHNAx= -2.5-0.02x VSHNAx[7:0]</b><br><br>VSHNA1[7:0] : Source High Negative Voltage Mode1<br>VSHNA2[7:0] : Source High Negative Voltage Mode2<br>VSHNA3[7:0] : Source High Negative Voltage Mode3<br>VSHNA4[7:0] : Source High Negative Voltage Mode4<br><br>The suggestion of usable VSHNAx voltage is shown below |           |           |           |           |           |           |           |                 |           |           |       |
|                           | VSHNAx[7]                                                                                                                                                                                                                                                                                                                                                                                                                                                          | VSHNAx[6] | VSHNAx[5] | VSHNAx[4] | VSHNAx[3] | VSHNAx[2] | VSHNAx[1] | VSHNAx[0] | VSHNAx(V)       |           |           |       |
|                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0         | 0         | 0         | 0         | 0         | 0         | 0         | -2.50           |           |           |       |
|                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0         | 0         | 0         | 0         | 0         | 0         | 1         | -2.52           |           |           |       |
|                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0         | 0         | 0         | 0         | 0         | 1         | 0         | -2.54           |           |           |       |
|                           | :                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | :         | :         | :         | :         | :         | :         | :         | :               |           |           |       |
|                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1         | 0         | 0         | 0         | 1         | 1         | 0         | -3.90 (default) |           |           |       |
|                           | :                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | :         | :         | :         | :         | :         | :         | :         | :               |           |           |       |
|                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1         | 1         | 1         | 1         | 0         | 1         | 1         | -4.96           |           |           |       |
|                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1         | 1         | 1         | 1         | 1         | 0         | 0         | -4.98           |           |           |       |
|                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1         | 1         | 1         | 1         | 1         | 0         | 1         | -5.00           |           |           |       |
| Default Value (D7 to D0)  | 1 <sup>st</sup> Parameter=0x46h                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |           |           |           |           |           |           |                 |           |           |       |
|                           | 2 <sup>nd</sup> parameter=0x46h                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |           |           |           |           |           |           |                 |           |           |       |
|                           | 3 <sup>rd</sup> parameter=0x46h                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |           |           |           |           |           |           |                 |           |           |       |
|                           | 4 <sup>th</sup> parameter=0x46h                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |           |           |           |           |           |           |                 |           |           |       |

## 8.2.14 VSLNCTRL (C5h): Source Low Negative Voltage Control

| C5H                         | VSLNCTRL (Source Low Positive Voltage Control)                                                                                                                                                                                                                                                                                                                                                                                                                                     |           |           |           |           |           |           |           |           |           |           |                 |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------------|
| Inst / Para                 | A0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | RWR       | ERD       | D7        | D6        | D5        | D4        | D3        | D2        | D1        | D0        | HEX             |
| VSLNCTRL                    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ↑         | 1         | 1         | 1         | 0         | 0         | 0         | 1         | 0         | 1         | (C5h)           |
| 1 <sup>st</sup> parameter   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ↑         | 1         | VSLNA1[7] | VSLNA1[6] | VSLNA1[5] | VSLNA1[4] | VSLNA1[3] | VSLNA1[2] | VSLNA1[1] | VSLNA1[0] |                 |
| 2 <sup>nd</sup> parameter   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ↑         | 1         | VSLNA2[7] | VSLNA2[6] | VSLNA2[5] | VSLNA2[4] | VSLNA2[3] | VSLNA2[2] | VSLNA2[1] | VSLNA2[0] |                 |
| 3 <sup>rd</sup> parameter   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ↑         | 1         | VSLNA3[7] | VSLNA3[6] | VSLNA3[5] | VSLNA3[4] | VSLNA3[3] | VSLNA3[2] | VSLNA3[1] | VSLNA3[0] |                 |
| 4 <sup>th</sup> parameter   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ↑         | 1         | VSLNA4[7] | VSLNA4[6] | VSLNA4[5] | VSLNA4[4] | VSLNA4[3] | VSLNA4[2] | VSLNA4[1] | VSLNA4[0] |                 |
| Description                 | <p>-This command is used to adjust source low positive voltage in different mode.</p> <p>The calculation of VSLNAx is as shown below:</p> <p><b>VSLNAx=1- 0.02x VSLNAx [7:0]</b></p> <p>VSLNA1 [7:0] : Source Low Negative Voltage Mode1</p> <p>VSLNA2 [7:0] : Source Low Negative Voltage Mode2</p> <p>VSLNA3 [7:0] : Source Low Negative Voltage Mode3</p> <p>VSLNA4 [7:0] : Source Low Negative Voltage Mode4</p> <p>The suggestion of usable VSLNAx voltage is shown below</p> |           |           |           |           |           |           |           |           |           |           |                 |
|                             | VSLNAx[7]                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | VSLNAx[6] | VSLNAx[5] | VSLNAx[4] | VSLNAx[3] | VSLNAx[2] | VSLNAx[1] | VSLNAx[0] | VSLNAx(V) |           |           |                 |
|                             | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0         | 0         | 0         | 0         | 0         | 0         | 0         | 0         | 0         | 0         | 1.00            |
|                             | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0         | 0         | 0         | 0         | 0         | 0         | 0         | 0         | 1         | 0         | 0.98            |
|                             | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0         | 0         | 0         | 0         | 0         | 0         | 1         | 0         | 0         | 0         | 0.96            |
|                             | :                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | :         | :         | :         | :         | :         | :         | :         | :         | :         | :         | :               |
|                             | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1         | 0         | 0         | 0         | 1         | 1         | 0         | 0         | 0         | 0         | -0.40 (default) |
|                             | :                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | :         | :         | :         | :         | :         | :         | :         | :         | :         | :         | :               |
|                             | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0         | 0         | 0         | 1         | 0         | 1         | 0         | 0         | 0         | 0         | -1.76           |
|                             | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0         | 0         | 0         | 1         | 0         | 1         | 1         | 0         | 0         | 0         | -1.78           |
|                             | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0         | 0         | 0         | 1         | 1         | 0         | 0         | 0         | 0         | 0         | -1.80           |
| Default Value<br>(D7 to D0) | <p>1<sup>st</sup> Parameter=0x46h</p> <p>2<sup>nd</sup> parameter=0x46h</p> <p>3<sup>rd</sup> parameter=0x46h</p> <p>4<sup>th</sup> parameter=0x46h</p>                                                                                                                                                                                                                                                                                                                            |           |           |           |           |           |           |           |           |           |           |                 |

## 8.2.15 VSHLSEL (C9h): Source Voltage Select

| C9H                         | VSHLSEL (Source Voltage Select)                 |     |     |                                |    |    |    |    |    |              |    |       |
|-----------------------------|-------------------------------------------------|-----|-----|--------------------------------|----|----|----|----|----|--------------|----|-------|
| Inst / Para                 | A0                                              | RWR | ERD | D7                             | D6 | D5 | D4 | D3 | D2 | D1           | D0 | HEX   |
| VSHLSEL                     | 0                                               | ↑   | 1   | 1                              | 1  | 0  | 0  | 1  | 0  | 0            | 1  | (C9h) |
| Parameter                   | 1                                               | ↑   | 1   | 0                              | 0  | 0  | 0  | 0  | 0  | VSHLSEL[1:0] |    |       |
| Description                 | -This command is used to switch source voltage. |     |     |                                |    |    |    |    |    |              |    |       |
|                             | VSHLSEL [1:0]                                   |     |     | Source Voltage                 |    |    |    |    |    |              |    |       |
|                             | 00                                              |     |     | VSHPA1/ VSLPA1/ VSHNA1/ VSLNA1 |    |    |    |    |    |              |    |       |
|                             | 01                                              |     |     | VSHPA2/ VSLPA2/ VSHNA2/ VSLNA2 |    |    |    |    |    |              |    |       |
|                             | 10                                              |     |     | VSHPA3/ VSLPA3/ VSHNA3/ VSLNA3 |    |    |    |    |    |              |    |       |
|                             | 11                                              |     |     | VSHPA4/ VSLPA4/ VSHNA4/ VSLNA4 |    |    |    |    |    |              |    |       |
| Default Value<br>(D7 to D0) | Parameter=0x00h                                 |     |     |                                |    |    |    |    |    |              |    |       |

## 8.2.16 ID1SET (CAh): ID1 Setting

| CAH                      | ID1SET (ID1 Code Setting) |     |     |      |      |      |      |      |      |      |      |       |
|--------------------------|---------------------------|-----|-----|------|------|------|------|------|------|------|------|-------|
| Inst / Para              | A0                        | RWR | ERD | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX   |
| ID1SET                   | 0                         | ↑   | 1   | 1    | 1    | 0    | 0    | 1    | 0    | 1    | 0    | (CAh) |
| Parameter                | 1                         | ↑   | 1   | ID17 | ID16 | ID15 | ID14 | ID13 | ID12 | ID11 | ID10 |       |
| Description              | ID1[7:0]: ID1 Setting.    |     |     |      |      |      |      |      |      |      |      |       |
| Default Value (D7 to D0) | Parameter=0x00h           |     |     |      |      |      |      |      |      |      |      |       |

## 8.2.17 ID2SET (CBh): ID2 Setting

| CBH                      | ID2SET (ID2 Code Setting) |     |     |      |      |      |      |      |      |      |      |       |
|--------------------------|---------------------------|-----|-----|------|------|------|------|------|------|------|------|-------|
| Inst / Para              | A0                        | RWR | ERD | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX   |
| ID2SET                   | 0                         | ↑   | 1   | 1    | 1    | 0    | 0    | 1    | 0    | 1    | 1    | (CBh) |
| Parameter                | 1                         | ↑   | 1   | ID27 | ID26 | ID25 | ID24 | ID23 | ID22 | ID21 | ID20 |       |
| Description              | ID2[7:0]: ID2 Setting.    |     |     |      |      |      |      |      |      |      |      |       |
| Default Value (D7 to D0) | Parameter=0x00h           |     |     |      |      |      |      |      |      |      |      |       |

## 8.2.18 ID3SET (CCh): ID3 Setting

| CCH                         | ID3SET (ID3 Code Setting) |     |     |      |      |      |      |      |      |      |      |       |
|-----------------------------|---------------------------|-----|-----|------|------|------|------|------|------|------|------|-------|
| Inst / Para                 | A0                        | RWR | ERD | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX   |
| ID3SET                      | 0                         | ↑   | 1   | 1    | 1    | 0    | 0    | 1    | 1    | 0    | 0    | (CCh) |
| Parameter                   | 1                         | ↑   | 1   | ID37 | ID36 | ID35 | ID34 | ID33 | ID32 | ID31 | ID30 |       |
| Description                 | ID3[7:0]: ID3 Setting.    |     |     |      |      |      |      |      |      |      |      |       |
| Default Value<br>(D7 to D0) | Parameter=0x00h           |     |     |      |      |      |      |      |      |      |      |       |

## 8.2.19 AUTOPWRCTRL (D0h): Enable Auto Power Down

| D0H                         | AUTOPWRCTRL (Enable Auto Power Down)                                                                                                       |     |     |         |    |    |    |    |    |    |    |       |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|---------|----|----|----|----|----|----|----|-------|
| Inst / Para                 | A0                                                                                                                                         | RWR | ERD | D7      | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| AUTOPWRCTRL                 | 0                                                                                                                                          | ↑   | 1   | 1       | 1  | 0  | 1  | 0  | 0  | 0  | 0  | (D0h) |
| Parameter                   | 1                                                                                                                                          | ↑   | 1   | AUTOPWR | 1  | 1  | 1  | 1  | 1  | 1  | 1  |       |
| Description                 | -This command is used to enable/disable auto power down<br><b>AUTOPWR:</b><br>"0": Disable Auto Power down.<br>"1": Enable Auto Power down |     |     |         |    |    |    |    |    |    |    |       |
| Default Value<br>(D7 to D0) | Parameter=0xFFh                                                                                                                            |     |     |         |    |    |    |    |    |    |    |       |

## 8.2.20 BSTEN (D1h): Booster Enable

| D1H                         | BSTEN (Booster Enable)                                       |     |     |    |    |    |    |    |    |    |       |       |
|-----------------------------|--------------------------------------------------------------|-----|-----|----|----|----|----|----|----|----|-------|-------|
| Inst / Para                 | A0                                                           | RWR | ERD | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0    | HEX   |
| BSTEN                       | 0                                                            | ↑   | 1   | 1  | 1  | 0  | 1  | 0  | 0  | 0  | 1     | (D1h) |
| Parameter                   | 1                                                            | ↑   | 1   | 0  | 0  | 0  | 0  | 0  | 0  | 0  | BSTEN |       |
| Description                 | <b>BSTEN:</b><br>"0": Disable Booster<br>"1": Enable Booster |     |     |    |    |    |    |    |    |    |       |       |
| Default Value<br>(D7 to D0) | Parameter=0x00h                                              |     |     |    |    |    |    |    |    |    |       |       |

## 8.2.21 NVMLoadCTRL (D6h): NVM Load Control

| D6H                         | NVMLoadCTRL (NVM Load Control)                                               |                               |     |    |    |    |    |                                                                                                |         |        |    |       |  |
|-----------------------------|------------------------------------------------------------------------------|-------------------------------|-----|----|----|----|----|------------------------------------------------------------------------------------------------|---------|--------|----|-------|--|
| Inst / Para                 | A0                                                                           | RWR                           | ERD | D7 | D6 | D5 | D4 | D3                                                                                             | D2      | D1     | D0 | HEX   |  |
| NVMLoadCTRL                 | 0                                                                            | ↑                             | 1   | 1  | 1  | 0  | 1  | 0                                                                                              | 1       | 1      | 0  | (D6h) |  |
| 1 <sup>st</sup> parameter   | 1                                                                            | ↑                             | 1   | 0  | 0  | 0  | 1  | 0                                                                                              | VS_EN   | ID_EN  | 1  |       |  |
| 2 <sup>nd</sup> parameter   | 1                                                                            | ↑                             | 1   | 0  | 0  | 0  | 0  | 0                                                                                              | NRDTIME | NRDSLP | 0  |       |  |
| Description                 | -This command is used to control the items load by NVM and NVM Load control. |                               |     |    |    |    |    |                                                                                                |         |        |    |       |  |
|                             | Bit                                                                          | NVM Load Item                 |     |    |    |    |    | Value                                                                                          |         |        |    |       |  |
|                             | VS_EN                                                                        | Source High/Low Voltage       |     |    |    |    |    | '0' = Disable Source High/Low Voltage Load<br>'1' = Enable Source High/Low Voltage Load        |         |        |    |       |  |
|                             | ID_EN                                                                        | ID1/ID2/ID3                   |     |    |    |    |    | '0' = Disable ID1/ID/ID3 Load<br>'1' = Enable ID1/ID/ID3 Load                                  |         |        |    |       |  |
|                             | NRDTIME                                                                      | NVM Load Control by Timer     |     |    |    |    |    | "0": NVM load will not be trigger by Timer.<br>"1": NVM load will be trigger by Timer          |         |        |    |       |  |
|                             | NRDSLP                                                                       | NVM Load Control by Sleep-out |     |    |    |    |    | "0": NVM load will not be trigger by sleep Out.<br>"1": NVM load will be trigger by sleep Out. |         |        |    |       |  |
| Default Value<br>(D7 to D0) | 1 <sup>st</sup> Parameter=0x17h<br>2 <sup>nd</sup> parameter=0x00h           |                               |     |    |    |    |    |                                                                                                |         |        |    |       |  |

## 8.2.22 OSCSET (D8h): OSC Setting

| D8H                         | OSCSET (OSC Setting)                                               |     |     |       |             |    |    |    |    |    |    |       |
|-----------------------------|--------------------------------------------------------------------|-----|-----|-------|-------------|----|----|----|----|----|----|-------|
| Inst / Para                 | A0                                                                 | RWR | ERD | D7    | D6          | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| OSCEN                       | 0                                                                  | ↑   | 1   | 1     | 1           | 0  | 1  | 1  | 0  | 0  | 0  | (D8h) |
| 1 <sup>st</sup> parameter   | 1                                                                  | ↑   | 1   | OSCEN | 0           | 1  | 0  | 0  | 1  | 1  | 0  |       |
| 2 <sup>nd</sup> parameter   | 1                                                                  | ↑   | 1   | 1     | 1           | 1  | 0  | 1  | 0  | 0  | 1  |       |
| Description                 | OSCEN (OSC Enable/Disable)                                         |     |     |       |             |    |    |    |    |    |    |       |
|                             | OSCEN                                                              |     |     |       | OSC Enable  |    |    |    |    |    |    |       |
|                             | 0                                                                  |     |     |       | Disable OSC |    |    |    |    |    |    |       |
|                             | 1                                                                  |     |     |       | Enable OSC  |    |    |    |    |    |    |       |
| Default Value<br>(D7 to D0) | 1 <sup>st</sup> Parameter=0x26h<br>2 <sup>nd</sup> parameter=0xE9h |     |     |       |             |    |    |    |    |    |    |       |



## 8.2.23 NVMRD (E9h): NVM Data Read

| E9H         | NVMRD (NVM Data Read)                                      |     |     |       |       |       |       |       |       |       |       |       |
|-------------|------------------------------------------------------------|-----|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Inst / Para | A0                                                         | RWR | ERD | D7    | D6    | D5    | D4    | D3    | D2    | D1    | D0    | HEX   |
| NVMRD       | 0                                                          | ↑   | 1   | 1     | 1     | 1     | 0     | 1     | 0     | 0     | 1     | (E9h) |
| Parameter   | 1                                                          | ↑   | 1   | DO[7] | DO[6] | DO[5] | DO[4] | DO[3] | DO[2] | DO[1] | DO[0] |       |
| Description | -This command is used to read register that load from NVM. |     |     |       |       |       |       |       |       |       |       |       |

## 8.2.24 EXTBCtrl (ECh): EXTB Control

| ECH                         | EXTBCTRL (EXTB Control)                                              |     |     |                |    |    |    |    |    |    |    |       |
|-----------------------------|----------------------------------------------------------------------|-----|-----|----------------|----|----|----|----|----|----|----|-------|
| Inst / Para                 | A0                                                                   | RWR | ERD | D7             | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| EXTBCTRL                    | 0                                                                    | ↑   | 1   | 1              | 1  | 1  | 0  | 1  | 1  | 0  | 0  | (ECh) |
| Parameter                   | 1                                                                    | ↑   | 1   | EXTB_CTRL[7:0] |    |    |    |    |    |    |    |       |
| Description                 | <b>EXTB_CTRL[7:0]:</b><br>"5Ah": EXTB=H (VDD)<br>"A5h": EXTB=L (VSS) |     |     |                |    |    |    |    |    |    |    |       |
| Default Value<br>(D7 to D0) | Parameter=0x5Ah                                                      |     |     |                |    |    |    |    |    |    |    |       |

## 8.2.25 NVMCTRL1 (F8h): NVM WR/RD Control

| F8H                         | NVMCTRL1 (NVM WR/RD Control)                                                                          |     |                                                       |    |    |    |    |    |      |    |    |       |
|-----------------------------|-------------------------------------------------------------------------------------------------------|-----|-------------------------------------------------------|----|----|----|----|----|------|----|----|-------|
| Inst / Para                 | A0                                                                                                    | RWR | ERD                                                   | D7 | D6 | D5 | D4 | D3 | D2   | D1 | D0 | HEX   |
| NVMCTRL1                    | 0                                                                                                     | ↑   | 1                                                     | 1  | 1  | 1  | 1  | 1  | 0    | 0  | 0  | (F8h) |
| 1 <sup>st</sup> parameter   | 1                                                                                                     | ↑   | 1                                                     | 0  | 1  | 0  | 1  | 1  | 0    | 1  | 0  | (5Ah) |
| 2 <sup>nd</sup> parameter   | 1                                                                                                     | ↑   | 1                                                     | 0  | 1  | 1  | 0  | 1  | 0    | 0  | 1  | (69h) |
| 3 <sup>rd</sup> parameter   | 1                                                                                                     | ↑   | 1                                                     | 1  | 1  | 1  | 0  | 1  | 1    | 1  | 0  | (EEh) |
| 4 <sup>th</sup> parameter   | 1                                                                                                     | ↑   | 1                                                     | 0  | 0  | 0  | 0  | 0  | PROG | RD | 0  |       |
| Description                 | -NVM WR/RD mode select                                                                                |     |                                                       |    |    |    |    |    |      |    |    |       |
|                             | Bit                                                                                                   |     | Value                                                 |    |    |    |    |    |      |    |    |       |
|                             | PROG                                                                                                  |     | '1' =Enable Program Mode<br>'0' =Disable Program Mode |    |    |    |    |    |      |    |    |       |
|                             | RD                                                                                                    |     | '1' = Enable Read Mode<br>'0' = Disable Read Mode     |    |    |    |    |    |      |    |    |       |
| Default Value<br>(D7 to D0) | 1 <sup>st</sup> Parameter=0x5Ah<br>2 <sup>nd</sup> parameter=0x69h<br>3 <sup>rd</sup> parameter=0xEEh |     |                                                       |    |    |    |    |    |      |    |    |       |

4<sup>th</sup> parameter=0x00h**8.2.26 NVCTRL2 (FAh): NVM Program Setting**

| FAH                         | NVCTRL2 (NVM Program Setting)                                      |     |     |        |        |        |        |        |        |        |        |       |
|-----------------------------|--------------------------------------------------------------------|-----|-----|--------|--------|--------|--------|--------|--------|--------|--------|-------|
| Inst / Para                 | A0                                                                 | RWR | ERD | D7     | D6     | D5     | D4     | D3     | D2     | D1     | D0     | HEX   |
| NVCTRL2                     | 0                                                                  | ↑   | 1   | 1      | 1      | 1      | 1      | 1      | 0      | 1      | 0      | (FAh) |
| 1 <sup>st</sup> parameter   | 1                                                                  | ↑   | 1   | 0      | 0      | PA[5]  | PA[4]  | PA[3]  | PA[2]  | PA[1]  | PA[0]  |       |
| 2 <sup>nd</sup> parameter   | 1                                                                  | ↑   | 1   | PIN[7] | PIN[6] | PIN[5] | PIN[4] | PIN[3] | PIN[2] | PIN[1] | PIN[0] |       |
| Description                 | -NVM program register address and data setting.                    |     |     |        |        |        |        |        |        |        |        |       |
| Default Value<br>(D7 to D0) | 1 <sup>st</sup> Parameter=0x00h<br>2 <sup>nd</sup> parameter=0x00h |     |     |        |        |        |        |        |        |        |        |       |

**8.2.27 NVMRDEN (FBh): NVM Read Enable**

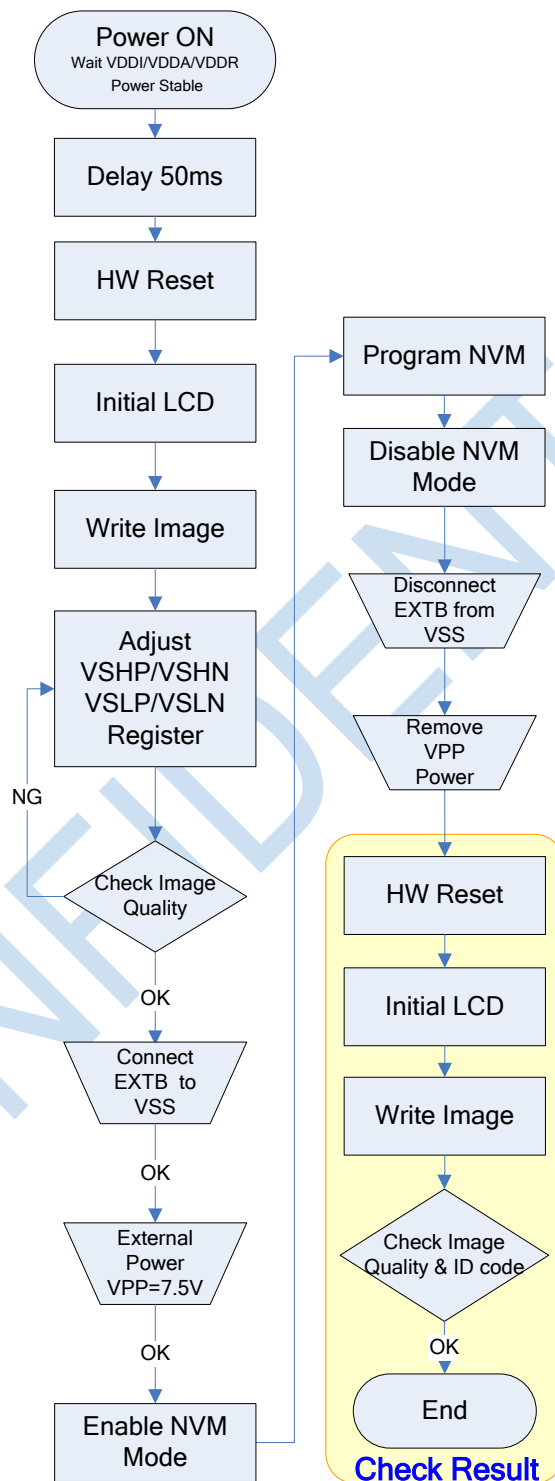
| FBH                       | NVMRDEN (NVM Read Enable) |     |     |    |    |    |    |    |    |    |    |       |
|---------------------------|---------------------------|-----|-----|----|----|----|----|----|----|----|----|-------|
| Inst / Para               | A0                        | RWR | ERD | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| NVMRDEN                   | 0                         | ↑   | 1   | 1  | 1  | 1  | 1  | 1  | 0  | 1  | 1  | (FBh) |
| 1 <sup>st</sup> parameter | 1                         | ↑   | 1   | 0  | 1  | 0  | 0  | 1  | 0  | 1  | 0  | (4Ah) |
| 2 <sup>nd</sup> parameter | 1                         | ↑   | 1   | 1  | 0  | 1  | 0  | 0  | 1  | 0  | 1  | (A5h) |
| Description               | -Enable NVM read mode.    |     |     |    |    |    |    |    |    |    |    |       |

**8.2.28 NVMPROM (FCh): NVM Program Enable**

| FCH                       | NVMPROM (NVM Program Enable) |     |     |    |    |    |    |    |    |    |    |       |
|---------------------------|------------------------------|-----|-----|----|----|----|----|----|----|----|----|-------|
| Inst / Para               | A0                           | RWR | ERD | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |
| NVMPROM                   | 0                            | ↑   | 1   | 1  | 1  | 1  | 1  | 1  | 1  | 0  | 0  | (FCh) |
| 1 <sup>st</sup> parameter | 1                            | ↑   | 1   | 0  | 0  | 1  | 0  | 1  | 0  | 0  | 1  | (29h) |
| 2 <sup>nd</sup> parameter | 1                            | ↑   | 1   | 1  | 0  | 1  | 0  | 0  | 1  | 0  | 1  | (A5h) |
| Description               | -Enable NVM program mode.    |     |     |    |    |    |    |    |    |    |    |       |

## 9 NVM PROGRAMMING FLOW

NVM Programming Flow



## 10 ABSOLUTE MAXIMUM RATING

| Item                           | Symbol  | Rating            | Unit |
|--------------------------------|---------|-------------------|------|
| I/O Power Supply Voltage       | VDDI    | - 0.3 ~ +4.0      | V    |
| Analog Power Supply Voltage    | VDDA    | - 0.3 ~ +4.0      | V    |
| Reference Power Supply Voltage | VDDR    | - 0.3 ~ +4.0      | V    |
| LCD Driver Supply Voltage      | AVDD    | - 0.3 ~ +6.4      | V    |
|                                | VSH-VSL | -0.3 ~ +6.2       | V    |
|                                | VGH-VGL | -0.3 ~ +33.0      | V    |
| Logic Input Voltage Range      | VIN     | -0.3 ~ VDDI + 0.5 | V    |
| Logic Output Voltage Range     | VO      | -0.3 ~ VDDI + 0.5 | V    |
| Operating Temperature Range    | TOPR    | -30 ~ +85         | °C   |
| Storage Temperature Range      | TSTG    | -40 ~ +125        | °C   |

Table 2 Maximum Ratings

Note: If one of the above items is exceeded its maximum limitation momentarily, the quality of the product may be degraded. Absolute maximum limitation, therefore, specify the values exceeding which the product may be physically damaged. Be sure to use the product within the recommend range.

## 11 DC CHARACTERISTICS

| Parameter                             | Symbol            | Condition          | Specification |      |         | Unit | Related Pins |
|---------------------------------------|-------------------|--------------------|---------------|------|---------|------|--------------|
|                                       |                   |                    | MIN.          | TYP. | MAX.    |      |              |
| Power & Operation Voltage             |                   |                    |               |      |         |      |              |
| Analog Power Supply<br>(Normal Mode)  | VDDA/VDDR         | Analog Power       | 2.55          |      | 3.6     | V    |              |
| Digital Power Supply<br>(Normal Mode) | VDDI              | I/O Supply Voltage | 1.65          |      | 3.6     | V    |              |
| Power Supply<br>(1.8V Mode)           | VDDA/VDDR<br>VDDI | Single Power       | 1.7           | 1.8  | 1.9     | V    |              |
| Gate Driver High Voltage              | VGH               |                    | 8.0           |      | 16.5    | V    | Note 1       |
| Gate Driver Low Voltage               | VGL               |                    | -15.0         |      | -6.0    | V    |              |
| Gate Driver Supply Voltage            |                   | VGH-VGL            | 14.0          |      | 31.5    | V    | Note 2       |
| Input / Output                        |                   |                    |               |      |         |      |              |
| Logic-High Input Voltage              | VIH               |                    | 0.7VDDI       |      | VDDI    | V    |              |
| Logic-Low Input Voltage               | VIL               |                    | VSS           |      | 0.3VDDI | V    |              |
| Logic-High Output Voltage             | VOH               | IOH = -<br>1.0mA   | 0.8VDDI       |      | VDDI    | V    |              |
| Logic-Low Output Voltage              | VOL               | IOL =<br>+1.0mA    | VSS           |      | 0.2VDDI | V    |              |
| Input Leakage Current                 | IIL               | IOH = -<br>1.0mA   | -0.1          |      | +0.1    | uA   |              |
| VCOM Voltage                          |                   |                    |               |      |         |      |              |
| VCOM amplitude                        | VCOM              |                    |               | VSS  |         | V    |              |
| Source Driver                         |                   |                    |               |      |         |      |              |
| Source High Voltage<br>(Positive)     | VSHP              |                    | 3.7           |      | 6.0     | V    |              |
| Source Low Voltage<br>(Positive)      | VSLP              |                    | 0             |      | 2.0     | V    |              |
| Source High Voltage<br>(Negative)     | VSHN              |                    | -5.0          |      | -2.5    | V    |              |
| Source Low Voltage<br>(Negative)      | VSLN              |                    | -1.8          |      | 1.0     | V    |              |

Table 3 Basic DC Characteristics

*Notes:*

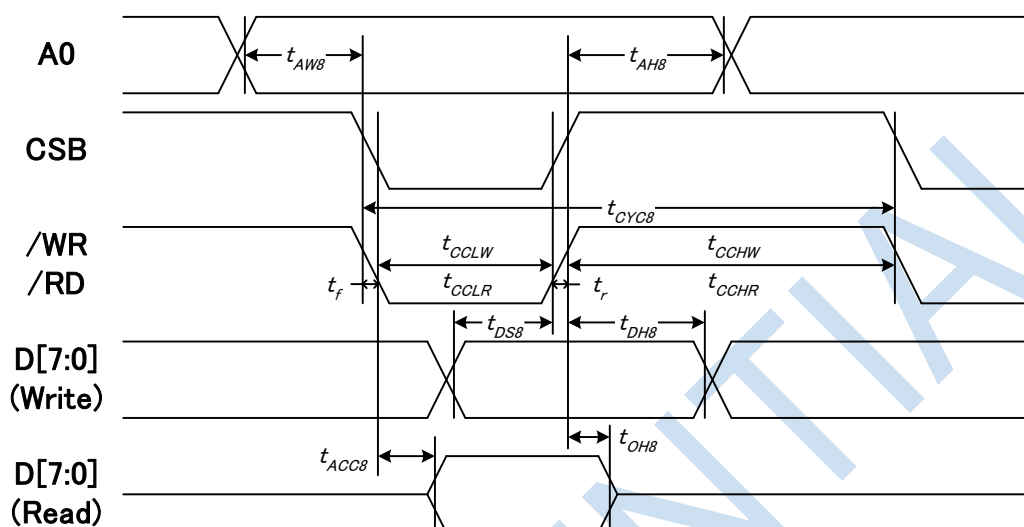
- 1. When evaluating the maximum and minimum of  $V_{GH}$ ,  $V_{DD}=2.8V$ .*
- 2. The maximum value of  $|V_{GH}-V_{GL}|$  can not over 32.0V.*

CONFIDENTIAL

## 12 AC CHARACTERISTICS

### 12.1 Interface Timing

#### 12.1.1 System Bus Timing for Parallel 8080 MCU Interface



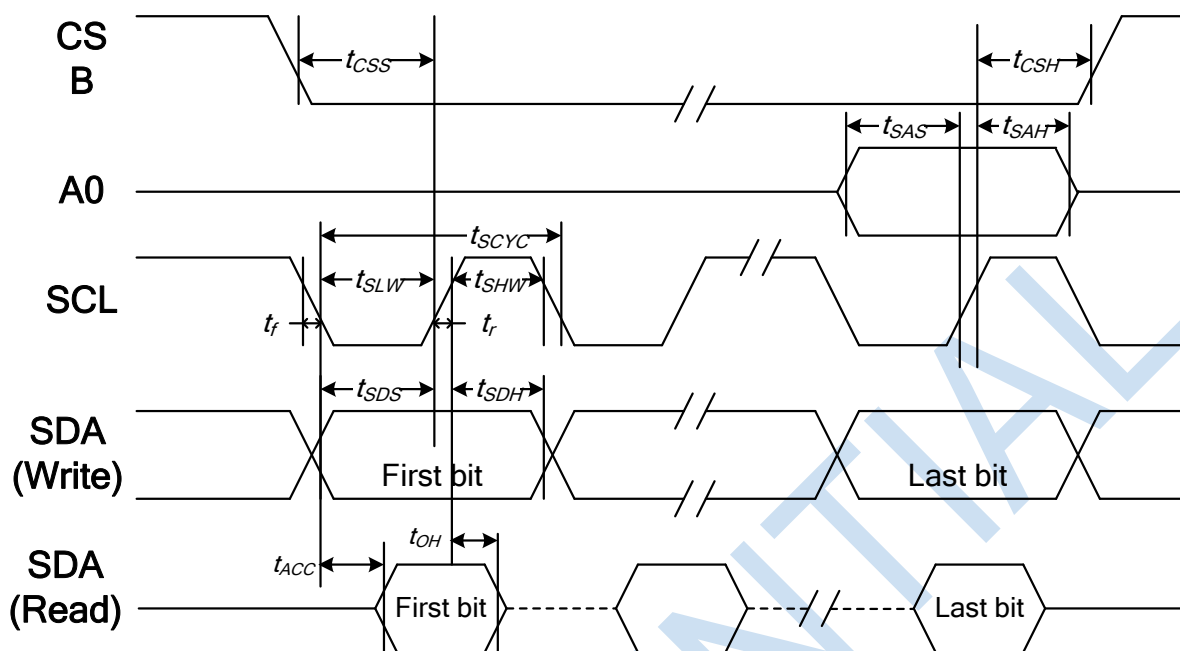
VDDI = 1.8~3.3V, Ta = 25°C

| Item                      | Signal | Symbol     | Condition  | Min. | Max. | Unit |
|---------------------------|--------|------------|------------|------|------|------|
| Address setup time        | A0     | $t_{AW8}$  |            | 20   | —    | ns   |
| Address hold time         |        | $t_{AH8}$  |            | 0    | —    |      |
| System cycle time (WRITE) | /WR    | $t_{CYC8}$ |            | 160  | —    |      |
| /WR L pulse width (WRITE) |        | $t_{CCLW}$ |            | 70   | —    |      |
| /WR H pulse width (WRITE) |        | $t_{CCHW}$ |            | 70   | —    |      |
| System cycle time (READ)  | RD     | $t_{CYC8}$ |            | 400  | —    |      |
| /RD L pulse width (READ)  |        | $t_{CCLR}$ |            | 180  | —    |      |
| /RD H pulse width (READ)  |        | $t_{CCHR}$ |            | 180  | —    |      |
| WRITE Data setup time     | D[7:0] | $t_{DS8}$  |            | 15   | —    |      |
| WRITE Data hold time      |        | $t_{DH8}$  |            | 15   | —    |      |
| READ access time          |        | $t_{ACC8}$ | CL = 30 pF | —    | 100  |      |
| READ Output disable time  |        | $t_{OH8}$  | CL = 30 pF | 10   | 110  |      |

Note:

- The input signal rise time and fall time ( $t_r$ ,  $t_f$ ) is specified at 15 ns or less. When the system cycle time is extremely fast,  $(t_r + t_f) \leq (t_{CYC8} - t_{CCLW} - t_{CCHW})$  for  $(t_r + t_f) \leq (t_{CYC8} - t_{CCLR} - t_{CCHR})$  are specified.
- All timing is specified using 20% and 80% of VDDI as the reference.

## 12.1.2 System Bus Timing for 4SPI MCU Interface



VDDI = 1.8~3.3V, Ta = 25°C

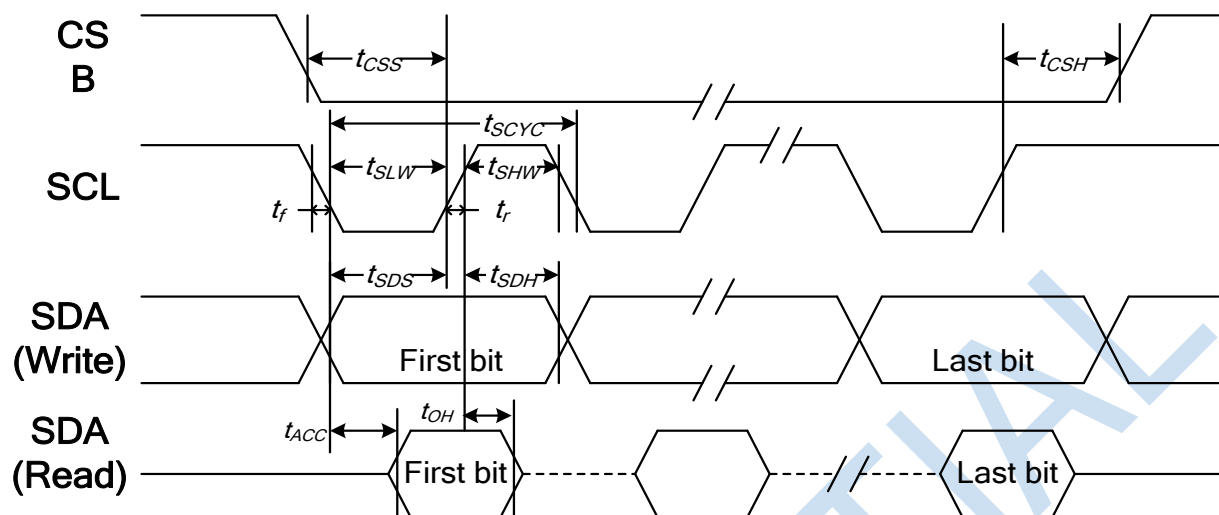
| Item                          | Signal      | Symbol | Condition          | Min. | Max. | Unit |
|-------------------------------|-------------|--------|--------------------|------|------|------|
| Serial clock period (Write)   | SCL         | tSCYC  |                    | 30   | —    | ns   |
| Serial clock period (Read)    |             |        |                    | 150  |      |      |
| SCLK "H" pulse width (Write)  |             | tSHW   |                    | 15   | —    |      |
| SCLK "H" pulse width (Read)   |             |        |                    | 60   |      |      |
| SCLK "L" pulse width (Write)  |             | tSLW   |                    | 15   | —    |      |
| SCLK "L" pulse width (Read)   |             |        |                    | 60   |      |      |
| Address setup time            | A0          | tSAS   |                    | 10   | —    |      |
| Address hold time             |             | tSAH   |                    | 10   | —    |      |
| Data setup time               | SDA (Write) | tSDS   |                    | 10   | —    |      |
| Data hold time                |             | tSDH   |                    | 10   | —    |      |
| Read data access time         | SDA (Read)  | tACC   | For maximum CL=30p | 10   | 50   |      |
| Read data output disable time |             | tOH    | For minimum CL=8p  | 15   | 50   |      |
| CSB-SCLK time                 | CSB         | tCSS   |                    | 10   | —    |      |
| CSB-SCLK time                 |             | tCSH   |                    | 10   | —    |      |

Note:

1. The input signal rise and fall time ( $t_r$ ,  $t_f$ ) are specified at 15 ns or less.
2. All timing is specified using 20% and 80% of VDDI as the standard.



## 12.1.3 System Bus Timing for 3SPI MCU Interface



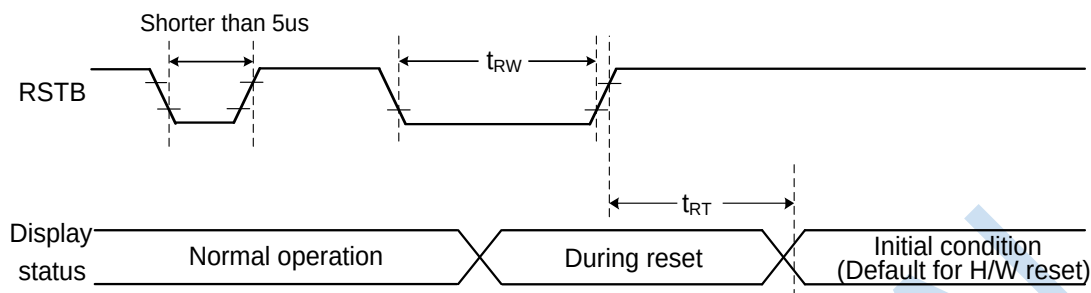
VDDI = 1.8~3.3V, Ta = 25°C

| Item                         | Signal      | Symbol | Condition          | Min. | Max. | Unit |
|------------------------------|-------------|--------|--------------------|------|------|------|
| Serial clock period (Write)  | SCL         | tSCYC  |                    | 30   | —    | ns   |
| Serial clock period (Read)   |             |        |                    | 150  |      |      |
| SCLK "H" pulse width (Write) |             | tSHW   |                    | 15   | —    |      |
| SCLK "H" pulse width (Read)  |             |        |                    | 60   |      |      |
| SCLK "L" pulse width (Write) |             | tSLW   |                    | 15   | —    |      |
| SCLK "L" pulse width (Read)  |             |        |                    | 60   |      |      |
| Data setup time              | SDA (Write) | tSDS   |                    | 10   | —    |      |
| Data hold time               |             | tSDH   |                    | 10   | —    |      |
| Data setup time              | SDA (Read)  | tACC   | For maximum CL=30p | 10   | 50   |      |
| Data hold time               |             | tOH    | For minimum CL=8p  | 15   | 50   |      |
| CSB-SCLK time                | CSB         | tCSS   |                    | 10   | —    |      |
| CSB-SCLK time                |             | tCSH   |                    | 10   | —    |      |

Note:

1. The input signal rise and fall time ( $t_r$ ,  $t_f$ ) are specified at 15 ns or less.
2. All timing is specified using 20% and 80% of VDDI as the standard.

## 12.1.4 Reset Timing

VDDI = 1.8~3.3V,  $T_a = 25^\circ\text{C}$ 

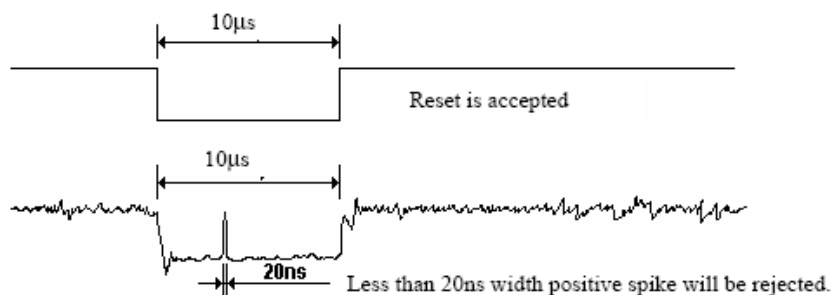
| Item                  | Symbol   | Condition                    | Min. | Max. | Unit |
|-----------------------|----------|------------------------------|------|------|------|
| Reset "L" pulse width | $t_{RW}$ |                              | 1    | —    | ms   |
| Reset cancel          | $t_{RT}$ | Note1, 5 (sleep-in mode)     | —    | 5    | ms   |
|                       |          | Note1, 6, 7 (sleep-out mode) | —    | 120  | ms   |

## Notes:

- The reset cancel includes also required time for loading ID bytes, VSource setting and other settings from NVM (or similar device) to registers. This loading is done every time when there is HW reset cancel time ( $t_{RT}$ ) within 5 ms after a rising edge of RSTB.
- Spike due to an electrostatic discharge on RSTB line does not cause irregular system reset according to the table below:

| RSTB Pulse          | Action         |
|---------------------|----------------|
| Shorter than 5us    | Reset Rejected |
| Longer than 9us     | Reset          |
| Between 5us and 9us | Reset starts   |

- During the Resetting period, the display will be blanked (The display is entering blanking sequence, which maximum time is 120 ms, when Reset Starts in Sleep Out –mode. The display remains the blank state in Sleep In –mode.) and then return to Default condition for Hardware Reset.
- Spike Rejection also applies during a valid reset pulse as shown below:



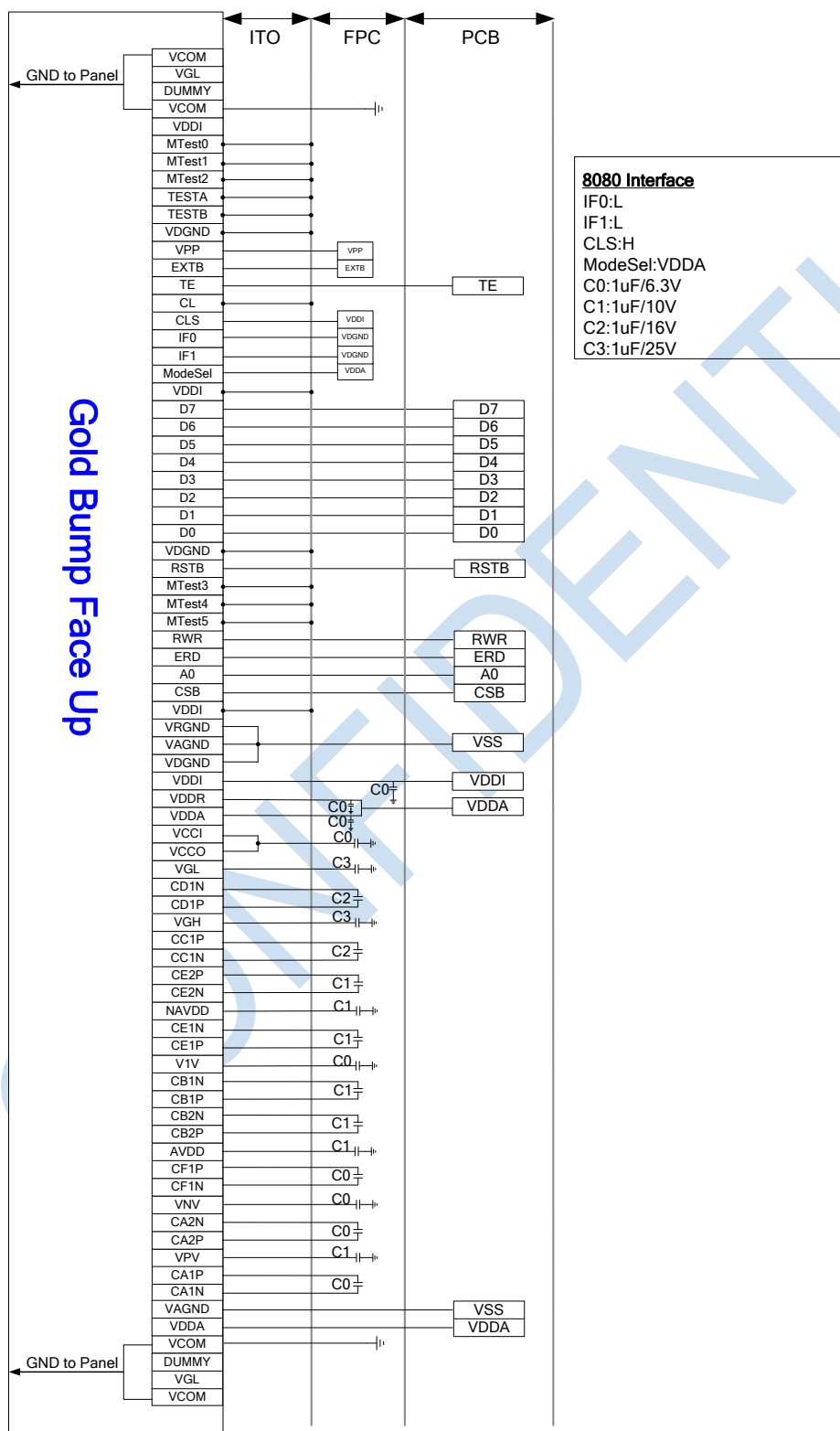
5. *When Reset applied during Sleep In Mode.*
6. *When Reset applied during Sleep Out Mode.*
7. *It is necessary to wait 5msec after releasing RSTB before sending commands. Also Sleep Out command cannot be sent for 120msec.*

CONFIDENTIAL

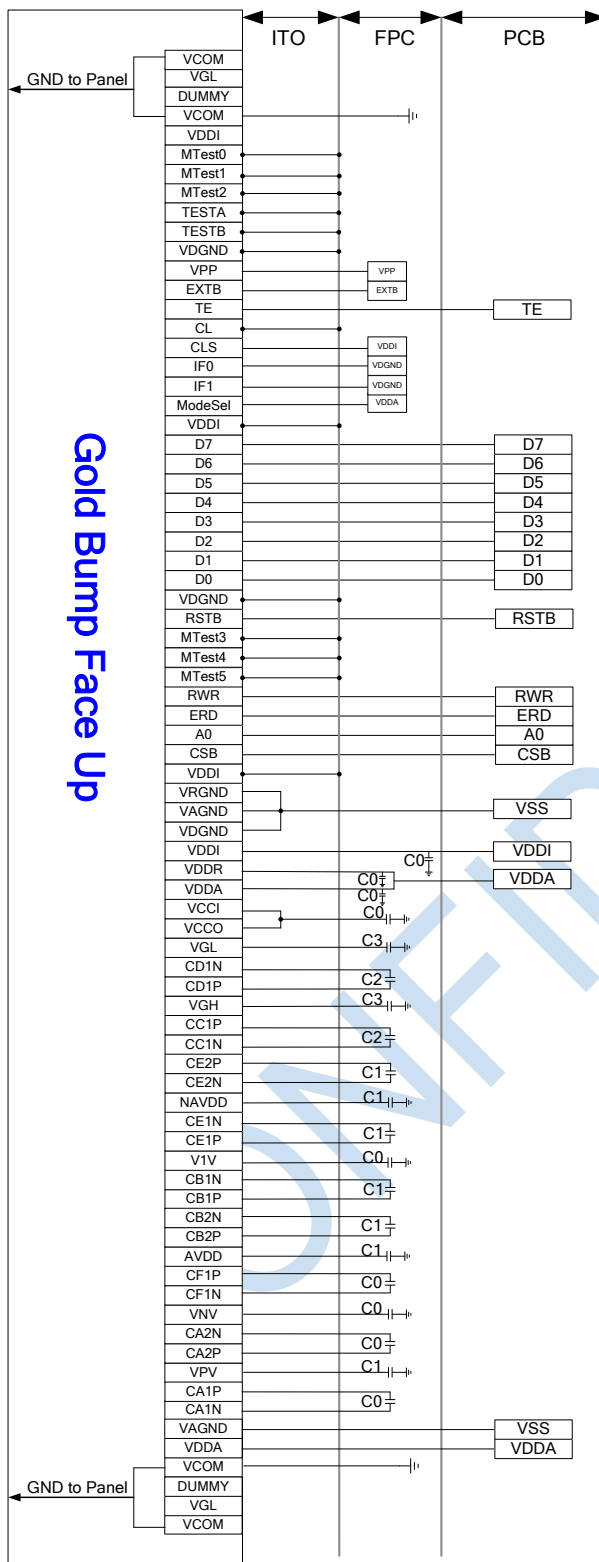
## 13 APPLICATION NOTE

## 13.1 Parallel 8080 Interface

## 13.1.1 VDDA=3.3V

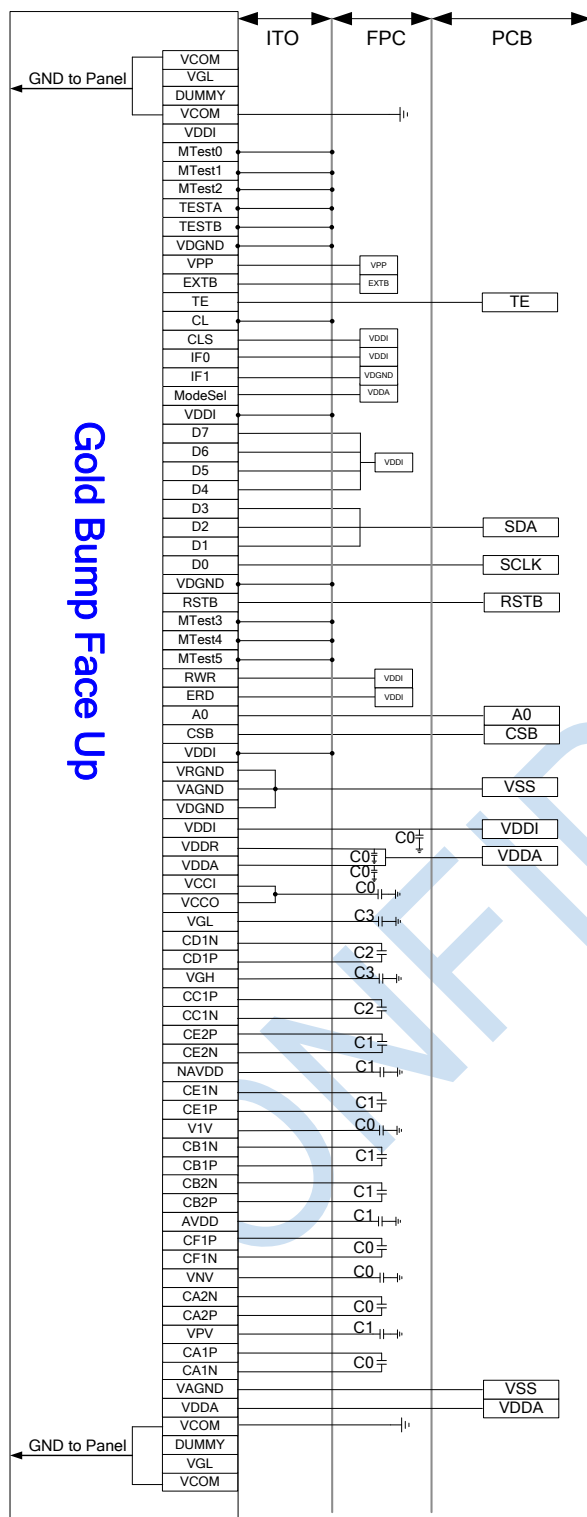


## 13.1.2 VDDI=VDDA=1.8V

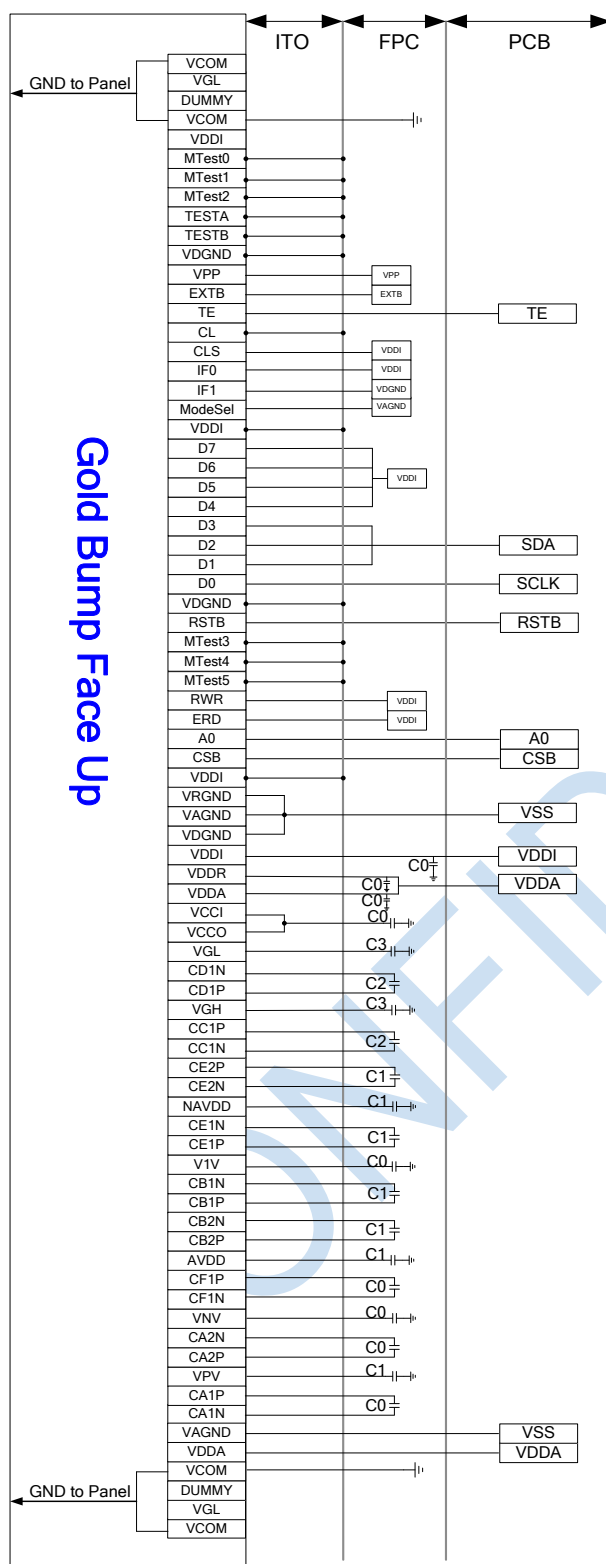


## 13.2 Single 4-line SPI Interface

### 13.2.1 VDDA=3.3V

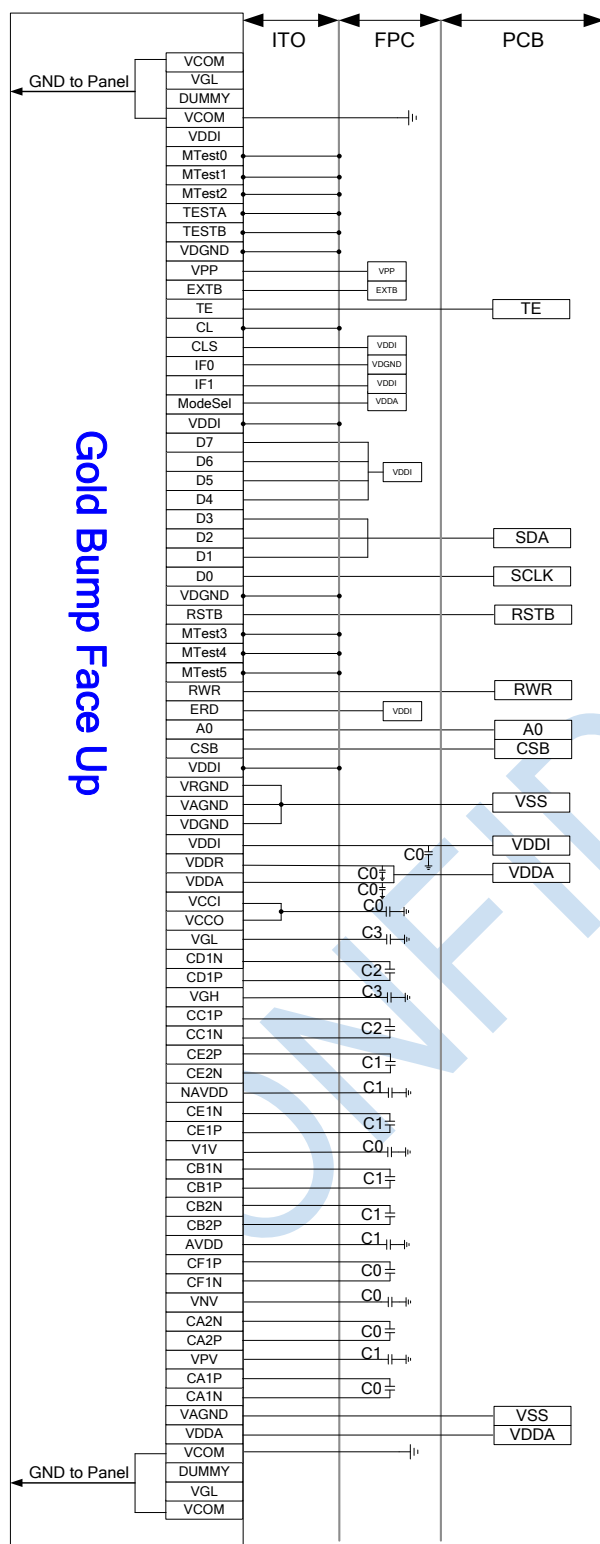


## 13.2.2 VDDI=VDDA=1.8V



## 13.3 Dual 4-line SPI Interface

### 13.3.1 VDDA=3.3V

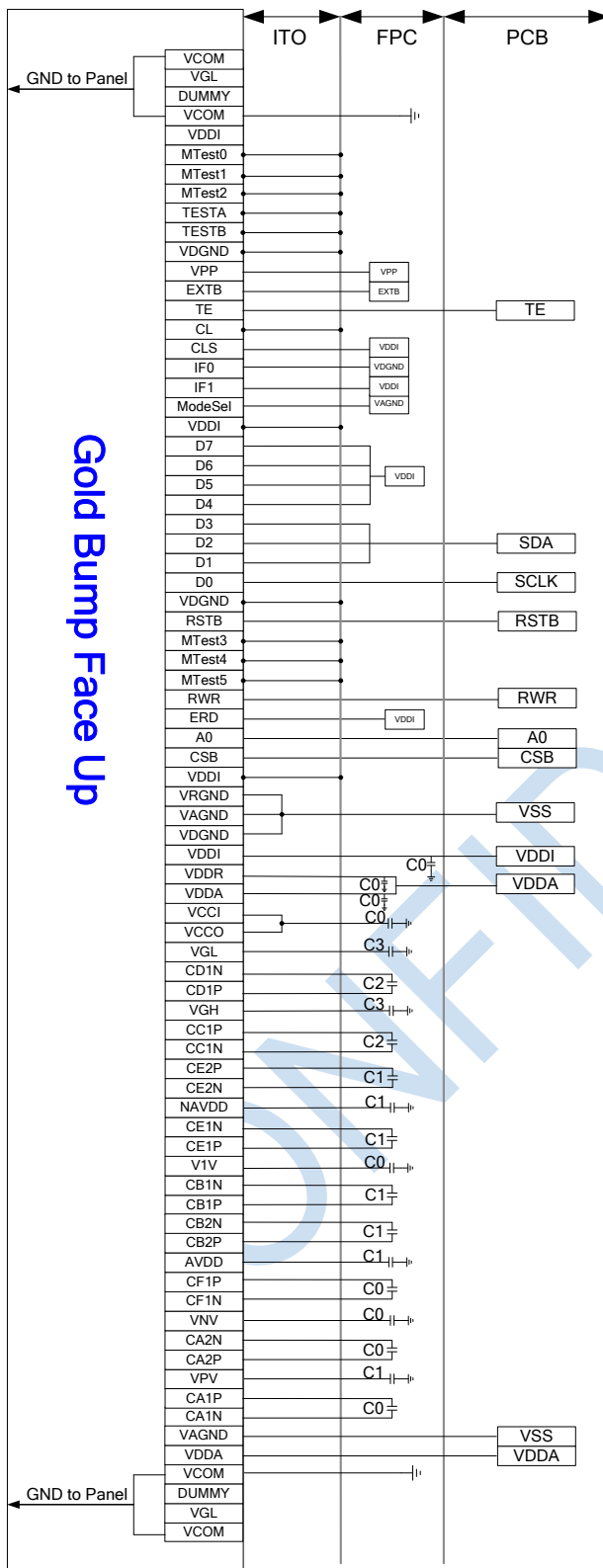


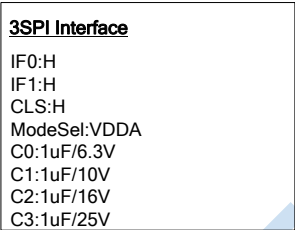
#### 4SPI Interface

IF0:L  
IF1:H  
CLS:H  
ModeSel:VDDA  
C0:1uF/6.3V  
C1:1uF/10V  
C2:1uF/16V  
C3:1uF/25V

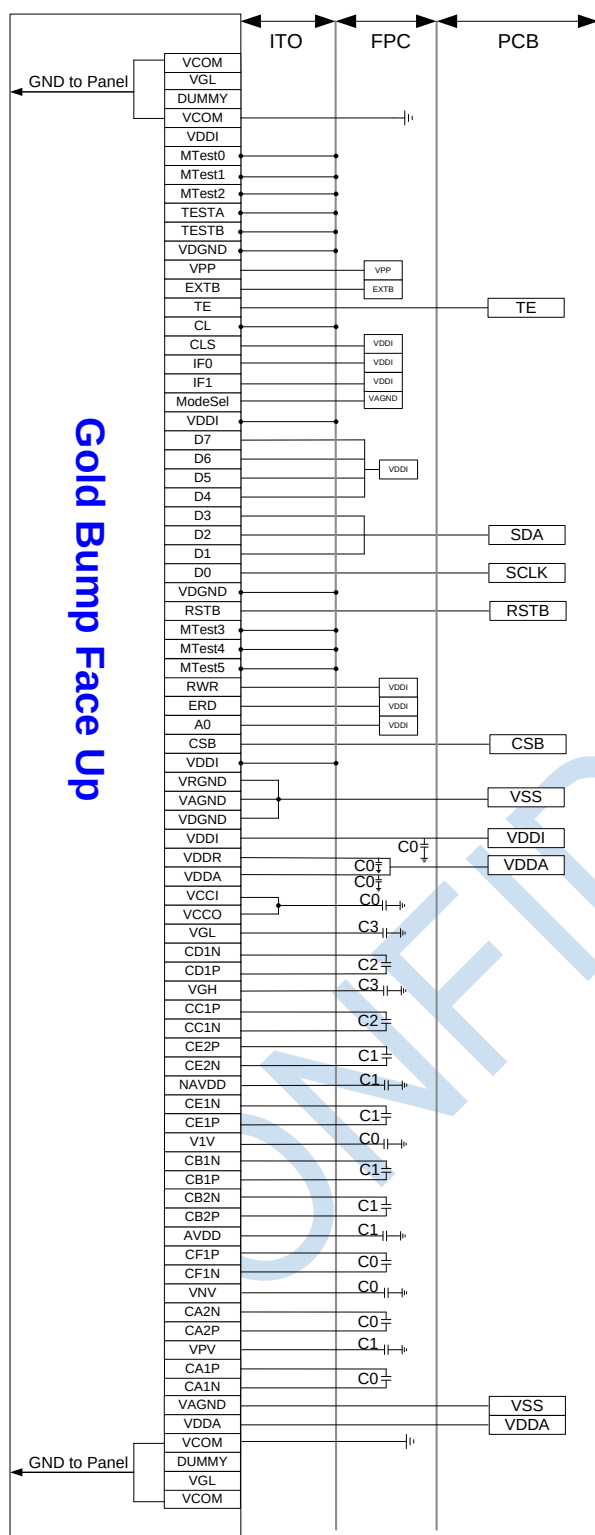


## 13.3.2 VDDI=VDDA=1.8V





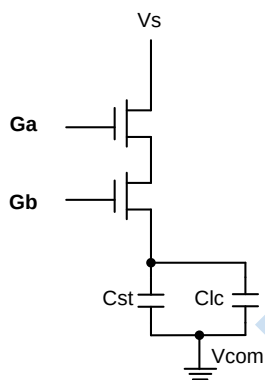
## 13.4.2 VDDI=VDDA=1.8V



## 14 PANEL LAYOUT

### 14.1 TFT Structure

The display module will operate at very low frame frequency to conserve power, so the recommended panel structure is two transistors for one line to control gate scan sequence. The benefits of this panel layout are to reduce leakage and enhance panel reliability.



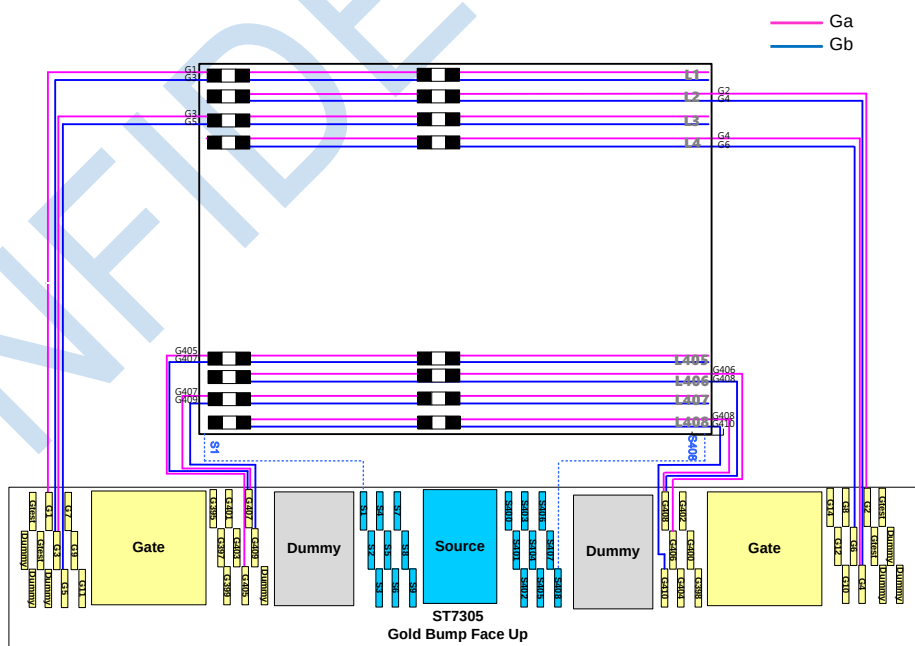
### 14.2 Gate Layout

#### 14.2.1 1-line interlace

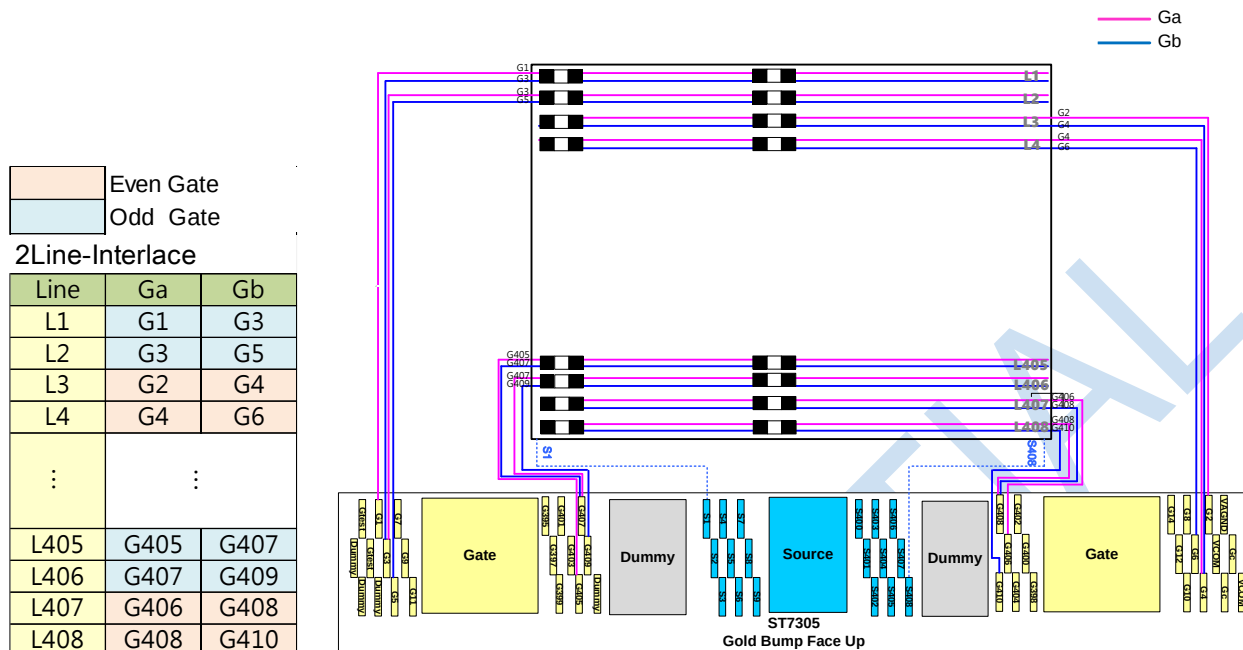
|  |           |
|--|-----------|
|  | Even Gate |
|  | Odd Gate  |

#### 1Line-Interlace

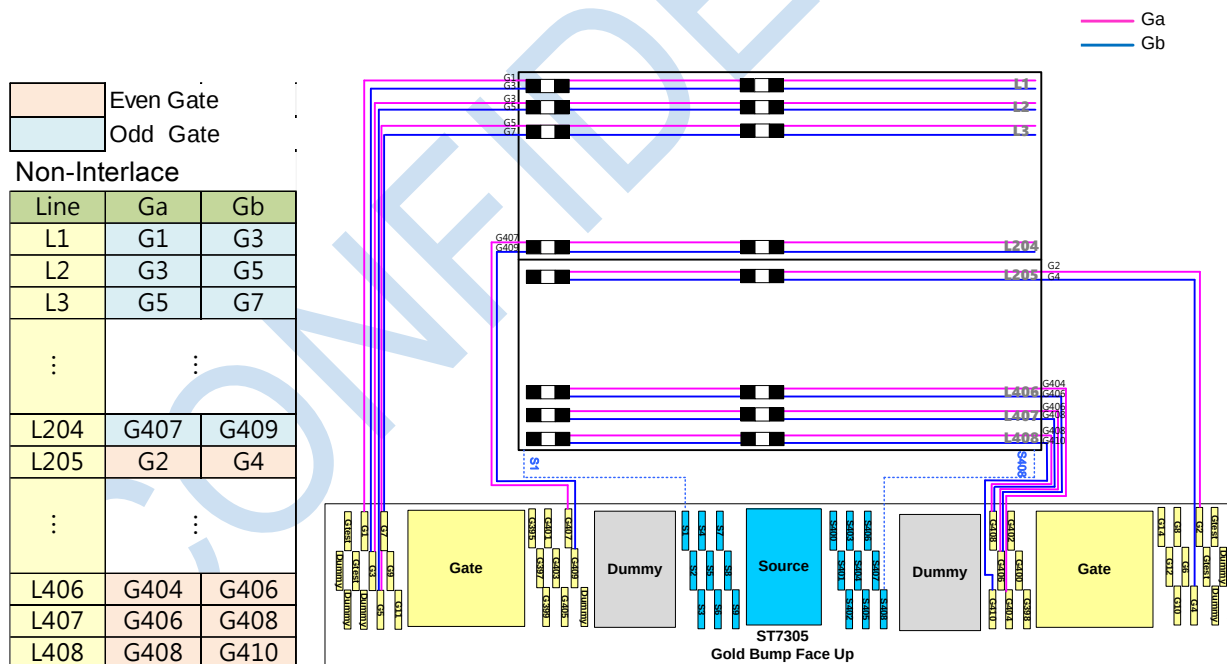
| Line | Ga   | Gb   |
|------|------|------|
| L1   | G1   | G3   |
| L2   | G2   | G4   |
| L3   | G3   | G5   |
| L4   | G4   | G6   |
| ⋮    | ⋮    | ⋮    |
| L405 | G405 | G407 |
| L406 | G406 | G408 |
| L407 | G407 | G409 |
| L408 | G408 | G410 |



## 14.2.2 2-line interlace



## 14.2.3 Non interlace



**15 REVISION HISTORY**

| Version | Date    | Description                                                                                                                                           |
|---------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| V0.1    | 2021/03 | <ul style="list-style-type: none"><li>● Preliminary.</li></ul>                                                                                        |
| V0.2    | 2021/04 | <ul style="list-style-type: none"><li>● Modify Source PAD Arrangement</li></ul>                                                                       |
| V0.3    | 2021/09 | <ul style="list-style-type: none"><li>● Modify Command B2h/BBh/C0h/C1h/C4h</li><li>● Remove 2Eh/3Eh</li><li>● Modify DC CHARACTERISTICS</li></ul>     |
| V0.4    | 2022/06 | <ul style="list-style-type: none"><li>● Add "7.8 Dual TFT Waveform"</li><li>● Modify Command 0x62h/0xB2/0xD8h</li><li>● Add 14.PANEL LAYOUT</li></ul> |
| V0.5    | 2023/02 | <ul style="list-style-type: none"><li>● Modify resolution to 408H x 408V</li><li>● Remove B9h/C8h</li></ul>                                           |